

Standard Design and Construction Guidelines for Microtunneling

This document uses both the
International System of Units (SI)
and customary units

American Society of Civil Engineers

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This standard has been prepared in accordance with recognized engineering principles and should not be used without the user's competent knowledge for a given application. The publication of this standard by ASCE is not intended to warrant that the information contained therein is suitable for any general or specific use, and ASCE takes no position respecting the validity of patent rights. The user is advised that the determination of patent rights or risk of infringement is entirely his or her own responsibility.

A complete list of currently available standards is available in the ASCE Library (<http://ascelibrary.org/page/books/s-standards>).

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PREFACE

These design and construction guidelines for microtunneling have been created by a group of engineers, owners, contractors, suppliers, and manufacturers working over the past several years. The changes to these microtunneling guidelines take into account many of the advances that have occurred since their original publication in 2001. A number of new sections have been added. Many of the existing sections have been updated because of changes in technology as well as changes in construction that make microtunneling a more acceptable means of construction.

Sections have also been expanded and modified so that those reviewing the guidelines can best understand what is needed in detail in order to undertake a microtunneling project. In addition, the methods for preparing construction contract documents,

including drawings, technical specifications, and contractual specifications, have been updated. The qualifications of contractors who construct a microtunneling project as well as those of the engineers who design them have been updated.

The committee that worked on these guidelines acknowledges that there have been significant improvements in best practices and technology since 2001. No document, including this one, can encompass all of the issues on a particular microtunneling project. In addition, improvements in best practices and technology continue to evolve so quickly that consideration of these guidelines in connection with any project must take into account not only the specific characteristics of the particular project but also further improvements in best practices and technology.

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PART I GENERAL

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CHAPTER 1

SCOPE

- 1.1 This standard guideline covers planning, design, pipe materials, and construction of microtunneling. The term *microtunneling* is defined as a trenchless construction method for installing pipelines. The North American definition of microtunneling describes a method but does not impose size limitations on that method; therefore, a tunnel may be considered a microtunnel if all of the following are used during construction:
- *Remote control:* The microtunneling boring machine (MTBM) is operated from a control panel, normally located on the surface. The system simultaneously installs pipe as the ground is excavated and removed. Personnel entry is not required for routine operation.
 - *Guidance:* The guidance system usually consists of a laser beam projected onto a target in the MTBM. With this system, the MTBM is capable of installing gravity sewers or other types of pipelines to the required tolerance for line and grade.
 - *Pipe jacking:* A pipeline is constructed by consecutively pushing pipes and the MTBM through the ground, using a jacking system for thrust.
 - *Continuous support:* Continuous pressure is provided to the face of the excavation to balance groundwater and earth pressures.
- 1.2 Part II of this standard guideline presents planning aspects of microtunneling.
- 1.3 Part III of this standard guideline presents design aspects of microtunneling.
- 1.4 Part IV of this standard guideline presents pipe material aspects of microtunneling.
- 1.5 Part V of this standard guideline presents construction aspects of microtunneling.
- 1.6 This standard guideline may be used as a reference by owners, engineers, contractors, and construction managers. This document is to be used as a guideline only for development of the trenchless project, including engineering, drawings, specifications, submittal requirements, and work plans. Each project has its own set of unique challenges and conditions. All sections of this guideline may not be applicable to all projects. The engineer should use a standard of care and good judgment in selecting which items are appropriate and relevant to the specific project after a thorough review of the project site conditions. In no case should this document be used as the project specification.
- 1.7 The guidelines in this document are intended for use by owners, engineers, and contractors familiar with the installation and pipe characteristics that affect the structural behavior of pipe for jacking installations and the significance of the installation requirements associated with the construction method. Before applying the guidelines given in Parts II, III, IV, and V, the user should review the guidance and requirements given in this standard practice and the commentary.
- 1.8 The values of dimensions and quantities are expressed in SI (metric) units, which are to be regarded as standard. SI unit values are converted to English unit values, which are presented in parentheses following the SI units. The use of SI units is in accordance with ASTM Practice E380. English units expressed in parentheses are supplied for information only and are not a part of this standard practice.
- Some of the applicable standards referenced may have a double designation (Axxx/AxxxM) or separate inch-pound (English) and SI (metric) units.
- 1.9 The commentary in the applicable sections of this guideline is provided for background and clarification.
- 1.10 Microtunneling is a highly specialized and technical form of construction. The engineer and the contractor should have experience that covers the size, length, ground conditions, complexities, and risks of a particular project. Qualifications of the engineer designing the microtunnel, the engineering firm responsible for the design, and key personnel should be checked and verified before the design and implementation of the project to ensure a proper microtunnel design and to ensure that performance and project goals are met. Engineering firms without the requisite microtunneling qualifications should hire qualified staff or a microtunneling subconsultant knowledgeable in the field. The contractor and its key personnel should be prequalified before bidding or before award of the microtunneling project. More details on suggested contractor qualifications are discussed in Sections 15.4 and 21.2.

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CHAPTER 2

REFERENCE DOCUMENTS

2.1 AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

- ASCE 27-00: *Standard Practice for Direct Design of Precast Concrete Pipe for Jacking in Trenchless Construction*
- ASCE/CI 38-02: *Standard Guidelines for the Collection and Depiction of Existing Subsurface Utility Data*
- ASCE/AIME: *Geotechnical Baseline Reports for Construction: Suggested Guidelines*, 2007
- Manual of Practice No. 46: *Pipeline Route Selection for Rural and Cross-Country Pipelines* (Out of print)
- Manual of Practice No. 60/WEF FD-5: *Gravity Sanitary Sewer Design and Construction, Second Edition*
- Manual of Practice No. 77: *Design and Construction of Urban Stormwater Management Systems* (Out of print)
- Manual of Practice No. 89: *Pipeline Crossings* (Out of print)
- Manual of Practice No. 106: *Horizontal Auger Boring Projects*
- Manual of Practice No. 108: *Pipeline Design for Installation by Horizontal Directional Drilling*
- Manual of Practice No. 112: *Pipe Bursting Projects*
- Manual of Practice No. 115: *Pipe Ramming Projects*
- Manual of Practice No. 120: *Trenchless Renewal of Culverts and Storm Sewers*
- Manual of Practice No. 125: *Pipelines for Water Conveyance and Drainage*

Commentary. The manuals are available for purchase through the ASCE website. Check the website at www.asce.org/bookstore for the latest versions and editions.

2.2 AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

2.2.1 Standards for Soil, Rock, and Groundwater Sampling and Testing

- C295: *Standard Guide for Petrographic Examination of Aggregates for Concrete*
- D421: *Standard Practice for Dry Preparation of Soil Samples for Particle-Size Analysis and Determination of Soil Constants*
- D422: *Standard Test Method for Particle-Size Analysis of Soils*
- D653: *Standard Terminology Relating to Soil, Rock, and Contained Fluids*
- D698: *Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft³ (600 kN-m/m³))*
- D854: *Standard Test Methods for Specific Gravity of Soil Solids by Water Pycnometer*

- D1140: *Standard Test Methods for Amount of Material in Soils Finer Than No. 200 (75- μ m) Sieve*
- D1556: *Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method*
- D1557: *Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft³ (2,700 kN-m/m³))*
- D1587: *Standard Practice for Thin-Walled Tube Sampling of Soils for Geotechnical Purposes*
- D2216: *Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass*
- D2217: *Standard Practice for Wet Preparation of Soil Samples for Particle Size Analysis and Determination of Soil Constants*
- D2435: *Standard Test Methods for One-Dimensional Consolidation Properties of Soils Using Incremental Loading*
- D2487: *Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)*
- D2488: *Standard Practice for Description and Identification of Soils (Visual Manual Procedure)*
- D2850: *Standard Test Method for Unconsolidated-Undrained Triaxial Compression Test on Cohesive Soils*
- D3080: *Standard Test Method for Direct Shear Test of Soils under Consolidated Drained Conditions*
- D3550: *Standard Practice for Thick Wall, Ring-Lined, Split Barrel, Drive Sampling of Soils*
- D3967: *Standard Test Method for Splitting Tensile Strength of Intact Rock Core Specimens*
- D4221: *Standard Test Method for Dispersive Characteristics of Clay Soil by Double Hydrometer*
- D4318: *Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils*
- D4546: *Standard Test Methods for One-Dimensional Swell or Collapse of Soils*
- D4643: *Standard Test Method for Determination of Water (Moisture) Content of Soil by Microwave Oven Heating*
- D4767: *Standard Test Method for Consolidated Undrained Triaxial Compression Test for Cohesive Soils*
- D5434: *Standard Guide for Field Logging of Subsurface Exploration of Soil and Rock*
- D5873: *Standard Test Method for Determination of Rock Hardness by Rebound Hammer Method*
- D6032: *Standard Test Method for Determining Rock Quality Designation (RQD) of Rock Core*
- D6938: *Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)*
- D7012: *Standard Test Methods for Compressive Strength and Elastic Moduli of Intact Rock Core Specimens under Varying States of Stress and Temperatures*

D7625: *Standard Test Method for Laboratory Determination of Abrasiveness of Rock Using the CERCHAR Method*

2.2.2 Standards for Concrete Pipes

- C76: *Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe*
C361: *Standard Specification for Reinforced Concrete Low-Head Pressure Pipe*
C506: *Standard Specification for Reinforced Concrete Arch Culvert, Storm Drain, and Sewer Pipe*
C507: *Standard Specification for Reinforced Concrete Elliptical Culvert, Storm Drain, and Sewer Pipe*
C655: *Standard Specification for Reinforced Concrete D-Load Culvert, Storm Drain, and Sewer Pipe*
C822: *Standard Terminology Relating to Concrete Pipe and Related Products*
C1417: *Standard Specification for Manufacture of Reinforced Concrete Sewer, Storm Drain, and Culvert Pipe for Direct Design*

2.2.3 Standards for Ductile Iron Pipes

- A377: *Standard Index of Specifications for Ductile-Iron Pressure Pipe*
A674: *Standard Practice for Polyethylene Encasement for Ductile Iron Pipe for Water or Other Liquids*
A716: *Standard Specification for Ductile Iron Culvert Pipe*
A746: *Standard Specification for Ductile Iron Gravity Sewer Pipe*

2.2.4 Standards for Fiberglass Pipes

- D3262: *Standard Specification for "Fiberglass" (Glass-Fiber-Reinforced Thermosetting-Resin) Sewer Pipe*
D3517: *Standard Specification for "Fiberglass" (Glass-Fiber-Reinforced Thermosetting-Resin) Pressure Pipe*
D3567: *Standard Practice for Determining Dimensions of Fiberglass (Glass-Fiber-Reinforced Thermosetting Resin) Pipe and Fittings*
D3681: *Standard Test Method for Chemical Resistance of Fiberglass (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe in a Deflected Condition*
D3754: *Standard Specification for Fiberglass (Glass-Fiber-Reinforced Thermosetting-Resin) Sewer and Industrial Pressure Pipe*
D5365: *Standard Test Method for Long-Term Ring-Bending Strain of Fiberglass (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe*

2.2.5 Standards for Polymer Concrete Pipes

- D6783: *Standard Specification for Polymer Concrete Pipe*

2.2.6 Standards for Steel Pipes

- A36: *Standard Specification for Carbon Structural Steel*
A139: *Standard Specification for Electric-Fusion (ARC) Welded Steel Pipe (NPS 4 and Over)*
A515: *Standard Specification for Pressure Vessel Plates Carbon Steel for Intermediate and Higher Temperature Service*
A572: *Standard Specification for High-Strength Low Alloy Columbium-Vanadium Structural Steel*
A999: *Standard Specification for General Requirements for Alloy and Stainless Steel Pipe*

2.2.7 Standards for Vitrified Clay Pipe

- C12: *Standard Practice for Installing Vitrified Clay Pipe Lines*
C301: *Standard Test Methods for Vitrified Clay Pipe*

- C700: *Standard Specification for Vitrified Clay Pipe, Extra Strength, Standard Strength, and Perforated*
C828: *Standard Test Method for Low-Pressure Air Test of Vitrified Clay Pipe Lines*

C1091: *Standard Test Method for Hydrostatic Infiltration Testing of Vitrified Clay Pipe Lines*

C1208: *Standard Specification for Vitrified Clay Pipe and Joints for Use in Microtunneling, Sliplining, Pipe Bursting, and Tunnels*

2.2.8 Standards for Slurries, Lubricants, and Other Additives

- D3910: *Standard Practices for Design, Testing, and Construction of Slurry Seal*
D4380: *Standard Test Method for Density of Bentonitic Slurries*
D4381: *Standard Test Method for Sand Content by Volume of Bentonitic Slurries*
D6910: *Standard Test Method for Marsh Funnel Viscosity of Clay Construction Slurries*

2.2.9 Other Standards

IEEE/ASTM SI 10: *American National Standard for Metric Practice*

Commentary. The ASTM standards are available for purchase through the ASTM website. Check the website at www.astm.org for the latest versions and editions.

2.3 AMERICAN WATER WORKS ASSOCIATION (AWWA)

- C150/A21.50: *Thickness Design of Ductile-Iron Pipe*
C151/A21.51: *Ductile Iron Pipe, Centrifugally Cast, for Water or Other Liquids*
C200: *Steel Water Pipe, 6 Inch (150 mm) and Larger*
C300: *Reinforced Concrete Pressure Pipe, Steel-Cylinder Type*
C302: *Reinforced Concrete Pressure Pipe, Noncylinder Type*
C950: *Fiberglass Pressure Pipe*
M11: *Steel Pipe: Guide for Design and Installation*

Commentary. The AWWA standards are available for purchase through the AWWA website. Check the website at www.awwa.org for the latest versions and editions.

2.4 AMERICAN PETROLEUM INSTITUTE (API)

- API 2B: *Specification for the Fabrication of Structural Steel Pipe*
API 5L: *Specification for Line Pipe*
API 13A: *Specification for Drilling Fluid Materials*
API 13G: *Recommended Practice Standard Procedure for Drilling Mud Report Form*
API 29: *Suggested Procedures for Laboratory Evaluation Drilling Mud Materials*
API 1102: *Steel Pipelines Crossing Railroads and Highways*
API 1104: *Welding of Pipelines and Related Facilities*

Commentary. The API standards are available for purchase through the API website. Check the website at www.api.org for the latest versions and editions.

Note that standards developed by the American Petroleum Institute for drilling slurries allow the use of petroleum

by-products, which may not be appropriate for microtunneling applications because the fluids may contaminate the ground.

2.5 EUROPEAN STANDARDS

ATV-A 125: *Pipe Jacking*

ATV-A 127: *Static Calculation of Drains and Sewers*

ATV-A 161: *Structural Calculation of Driven Pipes*

BS 5911-1: *Concrete Pipes and Ancillary Concrete Products*

DIN EN 12889: *Trenchless Construction and Testing of Drains and Sewers*

DIN EN 14636-1: *Plastics Piping Systems for Non-Pressure Drainage and Sewerage: Polyester Resin Concrete (PRC); Part 1: Pipes and Fittings with Flexible Joints*

DIN 54815-1: *Pipe Made of Filled Polyester Resin Molding Materials; Part 1: Dimensions, Material, and Marking*

DIN 54815-2: *Pipe Made of Filled Polyester Resin Molding Materials; Part 2: Requirements, Testing*

DWA-A 125E: *Pipe Jacking and Related Techniques*

EN 295-7: *Requirements for Vitrified Clay Pipes and Joints for Pipe Jacking*

Commentary. The European standards are available for purchase through various websites. Check the Internet for the latest versions and editions. Note that the European standards are typically not used in North America but may be useful resources.

2.6 NORTH AMERICAN SOCIETY FOR TRENCHLESS TECHNOLOGY (NASTT)

Horizontal Directional Drilling (HDD) Good Practices Guidelines

Pipe Bursting Good Practices Guideline Manual

Commentary. The NASTT manuals are available for purchase through the NASTT website. Check the website at www.nastt.org for the latest versions and editions.

2.7 OTHER STANDARDS

ANSI/AWS D1.1/D1.1M: *Structural Welding Code—Steel Common Ground Alliance Best Practices*

NSF/ANSI International 60: *Drinking Water Treatment Chemicals—Health Effects*

NSF/ANSI International 61: *Drinking Water System Components—Health Effects*

29CFR 1926: *Code of Federal Regulations, Title 29—Labor, Part 1926—Safety and Health Regulations for Construction*

Commentary. The standards are available through the associated websites. Check the Internet for the latest versions and editions.

Check the state website where the project is located to verify the enforcement agency for 29CFR 1926 because some states have more restrictive regulations.

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CHAPTER 3 DEFINITIONS

adapter ring: In microtunneling, a fabricated ring, usually made from steel, that is mounted on the first pipe of the pipe string. This ring is intended to transfer the jacking force from the microtunnel boring machine's (MTBM's) bearing area to the jacking pipe and to create a waterproof seal between the machine and the spigot of the first joint.

advance rate: Forward progress over a given period of time; includes penetration rate, makeup time, downtime, and idle time; typically measured in feet or meters per shift.

annular space: The theoretical volume between the gauge cut and the outside radius of the jacking pipe times the length of the installation, equal to annulus $\times$ length of tunnel.

annular thickness: The radial distance between the excavated radius created by the gauge cutter and the outside radius of the jacking pipe. The radial distance is equal to the overcut (gauge cut) plus the shield cut.

annulus: The theoretical area between the gauge cut and the pipe outside diameter (OD) equal to $(\pi \times (\text{gauge cut OD}^2 - \text{pipe OD}^2)/4)$. It is the combined area created by the overcut and the shield cut.

auger boring: A technique for forming a bore from a jacking or drive shaft to a receiving shaft by means of a rotating auger with cutting tools. The casings are jacked forward sequentially in a cyclic process while the auger is turned. Spoils are removed back to the drive shaft by helically wound auger flights rotating in the steel casing. The equipment may have limited guidance and steering capability.

auger MTBM: A type of microtunnel boring machine that uses auger flights to remove the spoils through a separate, smaller casing placed through the jacking pipe.

backfill grouting: Grout injected into the void space between the carrier pipe and casing after the drive is completed.

belly pans: Shims or other steel plates added to the bottom of the MTBM to adjust the machine's location within the excavation. Also known as *belly plates*.

Berlin construction method: A sewer collection system developed in Berlin, Germany, whereby sewer laterals are brought to a manhole and then dropped into the sewer interceptor. This method of connecting laterals to sewer is more compatible with trenchless installations.

boulder: Per ASTM D653, a rock fragment, usually rounded by weathering or abrasion, with a dimension of 12 in. (305 mm) or more.

buried objects: Items in the ground that can affect MTBM advancement and line and grade control. Buried objects that prevent continued advancement along the design path, which includes tolerances, are called *obstructions*.

caisson: A watertight chamber used in shaft construction under water. The ground within the chamber is excavated, allowing the chamber to sink under its own weight into the ground.

carbon footprint: The amount of carbon dioxide generated and caused by a construction project, including any traffic effects.

carrier pipe: Pipe for conveyance of water, gas, sewage, or other products and services. Some carrier pipe can be installed by direct jacking (i.e., one-pass microtunneling).

cased bore: A bore in which a pipe, usually a steel casing, is inserted simultaneously with the boring operation.

casing: A jacked pipe in a two pass system that supports a bore. The casing is not a carrier pipe.

cemented soils: Soils in which the particles are held together by a naturally occurring or human-made chemical agent that changes the soils' physical and mechanical properties.

charge line: See **feed line**.

closed-face: Term referring to one mode of operation of a traditional tunnel boring machine (TBM).

cobble: Per ASTM D653, a rock fragment, usually rounded or semirounded, with dimensions between 3 and 12 in. (75 and 305 mm).

cohesionless soils (noncohesive soils): Earth materials containing less than 20 percent soil particles passing the No. 200 sieve. Any free-running soil, such as sand or gravel, whose strength depends on friction between particles.

cohesive soils: Earth materials containing 20 percent or more soil particles passing the No. 200 sieve.

compression ring: A ring fitted between the end-bearing area of the leading pipe bell and the trailing pipe spigot to help distribute jacking forces more uniformly. A compression ring is attached to the trailing end of each pipe and is compressed between the pipe sections during jacking. The compression ring compensates for steering corrections, pipe misalignment, and pipe dimensional tolerances. Also referred to as a *packer*.

contact grouting: Grout injected into the theoretical space between the jacking pipe and the ground after the drive is completed.

contaminated plume: The underground trace of an environmental pollutant.

contract documents: Documents prepared by the owner and project engineer for bidding and for awarding a project; they can include bid forms, general conditions, special conditions, technical specifications, drawings, geotechnical data reports, and geotechnical baseline reports.

control console: An electronic unit typically inside a container located on the ground surface that controls the operation of the microtunneling machine. As the machine operator drives the tunnel from the control console, electronic information is transmitted to the control console from the MTBM. This information includes head position, steering angle, jacking force, penetration rates, machine-face torque, slurry feed and return flow rates and pressures, and laser position. Some control consoles are equipped with a computer that tracks and

stores the data and allows real-time analysis of the tunnel drive.

controls: Part of the microtunneling system that allows synchronized excavation, removal of spoils, and jacking of pipe to balance forward movement with excavation so that ground settlement and heave are managed.

crossing: Pipeline installation in which the primary purpose is to provide one or more passages beneath a waterway, road, or other surface obstruction.

crown: The highest point of the pipe or tunnel.

cut and cover: See **open-cut**.

cutability: The capacity of rock to be spalled, ripped, chipped, and broken into smaller pieces to allow the advance of the MTBM. Key properties affecting cutability include the frequency of discontinuities, unconfined compressive strength, tensile strength, hardness, and abrasiveness of the rock.

cutter chamber access: See **face access**.

cutter wheel: Any rotating tool or system of tools on a common support that excavates at the face of a bore.

cuttings: See **spoils**.

delay time: Lost time associated with the owner's issues.

design specification: A document that establishes specific requirements the contractor must use, including means and methods. A design specification creates an implied warranty that if the contractor uses the specified means and methods along with industry-accepted good practices, the constructed product will meet the specification requirements.

desktop study: A compilation and review of available site, geotechnical, and other information pertinent to a project. Typically performed at the outset of a project, before field explorations and detailed design.

dewatering: The act of removing groundwater or lowering the groundwater elevation, using a system of wells and pumps.

differential cut: The spiral, screwlike pattern created on the excavated surface as the gauge cutter is advanced while being rotated around the bored periphery of the cutter wheel.

direct-jacked pipe: Pipe installed in the ground that serves as the casing and carrier pipe.

discharge line: See **return line**.

downtime: Time lost when the MTBM is unable to operate; generally associated with equipment failure.

drilling fluid: Water that may contain additives, including bentonite, polymers, soda ash, surfactants, or other materials, to enhance stability and excavatability. If the drilling fluid contains additives and is designed to have specific properties, it is considered to be an *engineered drilling fluid*, whether specified by the engineer or selected by the contractor. Some additives may increase the effectiveness of the drilling fluid and reduce adhesion of the spoils (cuttings). The drilling fluid is used in a closed-loop system for transporting spoils and for counterbalancing earth and groundwater pressures during microtunneling.

drive: Designation of the pipe installed from a jacking shaft to a reception shaft.

drive shaft: See **jacking shaft**.

earth piercing: The use of a tool that comprises a percussive hammer within a suitable, generally torpedo-shaped casing. The hammer may be pneumatic or hydraulic. The term is usually associated with nonsteered devices.

effective stresses: In a saturated mass of soil or rock, the total stress P minus pore water pressure, U_w .

emergency recovery shaft: See **rescue shaft**.

engineered drilling fluid: See **drilling fluid**.

entrance seal: See **launch seal**.

entry ring: See **launch seal**.

EPB machine: Earth pressure balance tunneling machine, by which mechanical pressure is applied to the material at the face and controlled to provide the correct counterbalance to earth and groundwater pressures in order to prevent heave or subsidence. The term is not applicable to microtunneling systems for which the primary counterbalance of earth and groundwater pressures is supplied by pressurized slurry.

exit seal: Same as a launch seal except for the retrieval of the machine at the reception shaft. Used in high groundwater and unstable soils to prevent loss of ground.

exit shaft: See **reception shaft**.

eye: The opening in the shaft wall through which the MTBM is launched.

face: The location where excavation is taking place.

face access: Access to the location where excavation is taking place, typically through the cutting chamber. MTBM configuration may only allow limited personnel access to the back of the cutter wheel from within the pipe string. An air lock may also be required in the rear of the MTBM to maintain air pressure for counterbalancing earth and groundwater pressures.

face pressure: Earth and groundwater pressures applied against the cross-sectional area of the microtunnel face.

factor of safety: The ratio of the structural capacity of a system to expected or actual loads on the system.

feed line: Pipeline that transports drilling fluid from the slurry separation plant to the face of the MTBM. Also known as a *charge line*.

filter cake: A thin layer of clay or polymer from the slurry at the face and perimeter of the formation being excavated. The filter cake is formed through filtrate loss.

filtrate: A liquid that has passed through a filter.

finer: Portion of a soil finer than a No. 200 (75 μ m) U.S. standard sieve.

frac-out: See **inadvertent return** and **hydrofracture**.

gauge cut: The outermost projection of the excavation, measured as a diameter. The gauge cut is subject to cutter wear and to the differential cut.

gauge cutter: The cutter that creates the gauge cut and is subject to wear.

gel: Per ASTM D653, the condition of a liquid that has begun to exhibit measurable shear strength.

gel time: The measured time interval between the point when the slurry mix is in a fluid state and the point when it begins to form a gel.

gravel: Per ASTM D653, rounded or semirounded particles of rock that will pass a 3-in. (76.2-mm) sieve and be retained on a No. 4 (4.75-mm) U.S. standard sieve.

ground cover: Distance between the ground surface and the crown of the pipe.

groundwater: All subsurface water, as distinct from water on the surface.

grouting: The process of filling voids or of modifying or improving ground conditions. Grouting materials may be cementitious, chemical, or composed of other mixtures.

guidance system: System that locates the actual position of the MTBM relative to the laser or other device. The laser or other device should be referenced to the design line and grade.

guided boring method: Another term for the *pilot tube method*, in which an auger boring power unit is used with a pilot tube's guidance and steering system.

heave: Measurable upward movement of the ground or structure as the result of the excavation process.

horizontal directional drilling (HDD): A surface-launched trenchless technology for the installation of pipes, conduits,

and cables. HDD creates a pilot bore along the design pathway and reams the pilot bore in one or more passes to a diameter suitable for the product, which is pulled into the prepared bore in the final step of the process.

HQ: A diamond core wire line tube drill bit size where the hole diameter is 96 mm and the core diameter is 63.5 mm.

hydrofracture: A special form of inadvertent return in which drilling fluid reaches the surface or waterway. Also called a *frac-out*.

idle time: Lost time when the MTBM is not in operation; not including makeup time or downtime.

in-line microtunneling: A method of replacing an existing pipe by filling the pipe with flowable fill to prevent fluid loss and then excavating the entire pipe and surrounding ground with an MTBM for a new installation.

inadvertent return: The loss of drilling fluid, including slurry and lubrication, from the slurry or lubrication system. A special form of inadvertent return, where the fluid exceeds the strength and confining pressure of the ground and reaches the surface or waterway, is called a *hydrofracture* or *frac-out*.

inclinometer: (1) A geotechnical instrument used to measure horizontal deflection transverse to an installed vertical pipe casing by passing a probe through a casing (via two sets of grooves in the casing). (2) An instrument used within the MTBM to measure pitch and roll.

inlet line: See **feed line**.

interjack pipes: Pipes specially designed for use with an intermediate jacking station.

intermediate jacking station (IJS): A fabricated steel cylinder fitted with hydraulic jacks, which is incorporated into a pipeline between two specially fabricated pipe segments. Its function is to provide additional thrust in order to overcome skin friction and distribute the jacking forces over the pipe string on long drives.

invert: The lowest point of the pipe, tunnel, or shaft.

jacking force: The total force required to overcome the face pressure component and the frictional resistance component along the pipe to allow forward movement of the MTBM and the trailing pipe string.

jacking frame: A structural component that houses the hydraulic cylinders used to propel the microtunneling machine and pipeline. The jacking frame cradles the MTBM and jacking pipes, and serves to distribute the thrust load to the pipeline and the reaction load to the shaft wall or thrust block.

jacking loads: See **jacking force**.

jacking pipes: Pipes designed to be installed using pipe jacking techniques.

jacking shaft: Excavation from which trenchless technology equipment is launched for the installation of a pipeline. The jacking shaft may incorporate a thrust wall to spread reaction loads to the ground and an entry ring to control inflows of groundwater and soil at the portal.

jacking shield: A fabricated steel cylinder from within which the excavation is carried out either by hand or machine. An articulated steering section and steering jacks are incorporated into the shield to allow it to be adjusted to control line and grade. The term is associated with traditional tunneling.

Jersey barrier: See **K-rail**.

K-rail: A modular concrete or plastic barrier used to separate lanes of traffic. Also known as *Jersey barrier*.

laser: A device commonly incorporated into the guidance system and used to track alignment and grade during the tunneling or jacking operation.

launch seal: A mechanical seal, usually consisting of a rubber flange that is mounted to the wall of the jacking shaft. The flange seal is distended by the MTBM as the machine passes through the shaft. The seal is intended to prevent the flow of groundwater, soils, slurry, and lubrication into the shaft during tunneling operations.

lubrication: The act of injecting a fluid, normally bentonite and/or polymers, to reduce the skin friction and jacking forces on the jacking pipe during installation. The fluid fills the annular space.

makeup time: The time spent to add a pipe segment and make connections.

Marsh funnel viscosity: A measure of the flow rate of a fluid through a calibrated funnel. A Marsh funnel is typically used to measure drilling fluid and slurry viscosity, in seconds per quart (sec/qt) of fluid. The Marsh funnel viscosity of water is 26 sec/qt.

maximum allowable jacking force: The largest jacking force that can be applied to the jacked pipe, allowing for an appropriate factor of safety.

maximum anticipated jacking force: The largest anticipated jacking force required to advance jacking pipe and machine along a drive.

microtunneling: See Section 1.1.

mixed face: An interface within the excavated tunnel zone between two geological units that have a significant contrast in engineering properties (e.g., rock overlain by soft ground, or very soft, low-strength soil overlain or underlain by a very stiff, high-strength soil).

mixed ground: The excavated tunnel zone, without a defined interface, that contains soil and coarse fractions of gravels, cobbles, and boulders (i.e., silt with boulders). This term is not currently defined by ASTM or USCS.

mixed reach: A significant change in ground type, characteristics, or behavior along the excavated tunnel between the jacking and receiving shafts. This term is not currently defined by ASTM or USCS.

muck: Spoils or removal of same.

N-value: The number of blows per foot required to drive a standard penetration test (SPT) soil sampler into the ground during geotechnical exploration. The *N*-value is often used to estimate in situ consistency of clays and silts and relative density of sands.

noncohesive soils: See **cohesionless soils**.

NQ: A diamond core wire line tube drill bit size where the hole diameter is 75.7 mm and the core diameter is 47.6 mm.

obstruction: Any buried object that lies completely or partially within the cross section of the microtunnel and that impedes continued forward progress along the design path and within allowable tolerances.

one-pass: A procedure by which the carrier pipe is directly jacked into the ground without a casing.

open-cut: A method by which access is gained to the required underground level for installation, maintenance, or inspection of a pipe, conduit, or cable. The excavation is then backfilled, and the surface is reinstated.

open-face: Term referring to one mode of operation of a traditional TBM.

overcut: The theoretical difference between the radial measurement of the gauge cut and the MTBM shield; equal to $(\text{gauge cut OD} - \text{MTBM OD})/2$. Actual overcut is reduced as the gauge cutter is worn and because of the differential cut.

over excavation: A process of excavating more material than the theoretical volume of the tunnel based on diameter and

advanced distance. Over excavation can lead to ground settlement and the formation of voids.

packer: See **compression ring**.

penetration rate: Instantaneous excavation distance per time while the MTBM is operating, typically measured in inches per minute or millimeters per minute.

perched water: An accumulation of groundwater that is above the regional groundwater table in an unsaturated zone.

performance specification: A document that establishes the performance criteria the constructed product must meet. A performance specification leaves the selection of means and methods to the contractor.

piercing tools: See **earth piercing**.

piezometer: A specialized monitoring well that provides discrete screens and underground seals to produce a measurement of the groundwater pressure at specific intervals below ground. The groundwater pressure may be measured using open standpipes or vibrating wire transducers.

pilot tube method: A multistage method of accurately installing a product pipe to line and grade by use of a guided pilot tube followed by upsizing to install the product pipe. The pilot tube method does not fit within the definition of microtunneling.

pipe brake: A mechanical device designed to prevent the MTBM and pipe string from moving back into the shaft.

pipe eating: See **in-line microtunneling**.

pipe jacking: A system of using hydraulic jacking from a drive shaft to directly install pipes behind a shield machine so that they form a continuous string in the ground.

pipe lubricant: See **lubrication**.

pipe ramming: A trenchless installation method whereby a percussive hammer is attached, via an adapter, to an open-end casing, which is then driven through the ground. To create an open casing, the spoils within the casing are removed after the drive is completed, or periodically during the drive.

pipe string: The succession of joined individual pipes being used to advance and support the excavation.

pitch: The upward or downward angle of the MTBM, measured from a theoretical horizontal plane passing through the longitudinal axis of the MTBM.

plastic viscosity: A measure of the internal resistance to fluid flow, expressed as the tangential shear stress in excess of the yield stress divided by the resulting rate of shear.

plowing: A condition where the MTBM is being steered in one direction yet continues along an undesired direction.

pore water pressure: The pressure of the groundwater held within a soil or rock: in gaps (pores) between particles.

potholing: Small, carefully dug, nonmechanized excavation, used to locate a utility or other subsurface feature.

product pipe: See **carrier pipe**.

push ring adapter: Mechanical component mounted on the thrust ring to prevent the thrust ring from coming into contact with and damaging the pipe collar.

receiving shaft: See **reception shaft**.

reception shaft: Excavation into which the microtunneling equipment is driven and recovered.

rescue shaft: An unplanned additional shaft required to remove obstacles/obstructions and/or retrieve or repair the MTBM. The rescue shaft may need to function as a jacking shaft to complete the drive.

return line: Pipeline that transports slurry from the face of the MTBM to the slurry separation plant. Also known as a *discharge line*.

rider: A shallow collection line that runs along the microtunneled trunk sewer to a drop manhole.

rock: Per ASTM D653 and ISRM, any naturally formed aggregate of mineral matter occurring in large masses or fragments.

roll: The angle of rotation about the theoretical longitudinal centerline of the MTBM, measured from the 12 o'clock position.

settlement: Measurable downward movement of the ground or of an overlying utility or other structure as the result of excavation or dewatering.

settlement point or marker: An instrument installed at ground surface or driven into the ground to monitor vertical ground deformations before, during, and after construction.

shield cut: The theoretical difference between the radial measurement of the MTBM shield and the jacking pipe equal to $(\text{MTBM OD} - \text{pipe OD})/2$.

skinning: The act of fitting a steel cylinder over the MTBM shield body to increase the diameter of an existing MTBM with the use of an appropriately upsized cutter wheel. Also referred to as *upsizing*.

slurry: A mixture of drilling fluid and spoils.

slurry chamber: A chamber in which excavated material is mixed with slurry for transport through the return line to the separation plant at the surface. The slurry chamber is located behind the cutter wheel of a slurry microtunneling machine.

slurry lines: Parallel hoses or pipes that transport spoils and slurry from the face of a slurry microtunneling machine through the return line to the ground surface for separation, and then return the slurry to the face for reuse through the feed line (the *feed line* is also known as a *charge line*).

slurry separation: A process in which excavated material is separated from the circulation slurry. Mechanical separation is typical, although gravity separation in pits or tanks is possible with coarse-grained soils.

spacers: Mechanical structures used to transfer the jacking force from the jacking thrust ring to the pipe and to accommodate lengths of pipe that are longer than the stroke length of the jacks.

specials: Pipe sections immediately ahead of and behind the IJS. Specials have ends that are specifically manufactured to physically accommodate the IJS.

spoils: Earth, rock, and other materials excavated during the installation process. Also referred to as *cuttings*.

sump pump: A device placed at the base of a shaft or in an MTBM to collect and remove fluids incidental to the construction process and resulting from shaft leakage and to prevent the excavation equipment from flooding.

surface water: Any water encountered or collected at ground level.

thrust block: An engineered structure, located between the jacking frame and the shaft wall, that distributes the jacking force developed by the hydraulic jacking frame over a large surface area to the ground behind the back wall of the shaft.

thrust ring: A fabricated ring that is mounted on the face of the jacking frame. It is intended to transfer the jacking force from the jacking frame to the thrust-bearing area of the pipe section being jacked.

tooling: Ground-engaging elements of the cutter wheel designed to excavate and penetrate the ground.

top hat: A steel can with a faceplate bolted to a reception shaft wall to receive the arriving MTBM.

trailing can: A principal module that is part of a shield machine such as an MTBM or tunnel boring machine (TBM); its use depends on the installation dimensions required and the presence of an articulated joint to facilitate steering.

trenching: See **open-cut**.

trenchless technology: A family of construction techniques for installing or rehabilitating underground infrastructure with minimal disruption to surface traffic, businesses, and residents. Also includes technologies for inspection, leak location, and leak detection with minimal disruption and minimal excavation from the ground surface.

tunnel horizon: The vertical band within the ground where the tunnel is excavated. Includes one tunnel diameter above the crown and one tunnel diameter below the invert.

two-pass: A procedure by which the carrier pipe is installed within a jacked casing.

unanticipated return: See **inadvertent return**.

upsizing: Process of creating a larger diameter MTBM than a machine with the original factory configuration. Also referred to as *skinning*. Also, any method that increases the cross-sectional area of an existing pipeline, replacing it with a larger cross-sectional area pipe.

utility monitoring point (UMP): An instrument designed to measure movement of an individual utility when exposed to construction activities.

utility tunneling: A construction method for excavating an opening beneath the ground without continuous disturbance of the ground surface. The excavation is of sufficient diameter to permit personnel access and to allow excavation, transport of spoils, and erection of a ground support system.

waler: Beam used within a braced excavation as support to control wall deflections.

water jetting: Process of using the internal cleansing mechanism of the cutter head, by which high-pressure water is sprayed from nozzles to help remove cohesive soils.

yaw: The angle of the MTBM to the left or right of a theoretical vertical plane passing through the MTBM's longitudinal axis.

zone of influence: Volume of ground that could possibly be affected by settlement or heave from tunnel mining, shaft excavation, pile driving, or other construction activities.

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CHAPTER 4 ACRONYMS, ABBREVIATIONS, AND INITIALISMS

ADR	alternative dispute resolution	gpm	gallons per minute
AIME	American Institute of Mining, Metallurgical, and Petroleum Engineers	GPS	global position satellite
ANSI	American National Standards Institute	hd	hydraulic head (pressure) as measured at the point of inflow into the core
API	American Petroleum Institute	hp	horsepower
ASCE	American Society of Civil Engineers	HQ	wire line tube size for rock coring
ASTM	American Society for Testing and Materials	ID	inside diameter (of pipe)
ATV	Association representing German specialists working in the fields of wastewater, waste, and water management	IJS	intermediate jacking stations
		ISRM	International Society for Rock Mechanics
AWS	American Welding Society	JF	jacking force
AWWA	American Water Works Association	JHA	job hazard analysis
BVR	boulder volume ratio	MOP	manual of practice
CFR	Code of Federal Regulations	MSDS	material safety data sheet
CMP	corrugated metal pipe or plate	MTBM	microtunnel boring machine
CPT	cone penetrometer testing	NASTT	North American Society for Trenchless Technology
CSI	Construction Specifications Institute	NOAA	National Oceanic and Atmospheric Administration
CSM	cutter soil mixing	NQ	wire line tube size for rock coring
CVR	cobble volume ratio	NSF	National Sanitation Foundation
dia	diameter	OD	outside diameter (of pipe)
DIN	Deutsches Institut für Normung (German Institute for Standardization)	OSHA	Occupational Safety and Health Administration
DIP	ductile iron pipe	PAC	polyanionic cellulose
DSC	differing site conditions	PCP	polymer concrete pipe
EBR	environmental baseline report	PPE	personal protective equipment
EN	Norme Européenne (European Standard)	PTM	pilot tube method
EPB	earth pressure balance	PVC	polyvinyl chloride
FEMA	Federal Emergency Management Agency	QA/QC	quality assurance/quality control
F_p	face pressure component	RCP	reinforced concrete pipe
F_R	frictional resistance	RQD	Rock Quality Designation (index)
FRMP	fiberglass-reinforced polymer mortar pipe	SBU	small boring unit
FS	factor of safety	SDS	safety data sheet
ft/hd	feet per hydraulic head	SPT	standard penetration test
g	gravity (the acceleration of mass on Earth)	TBM	tunnel boring machine
GAB	guided auger boring	tph	tons per hour
GBM	guided boring method	TTC	Trenchless Technology Center
GBR	geotechnical baseline report	UCS	unconfined compressive strength
GDM	geotechnical design memorandum	UMP	utility monitoring point
GDR	geotechnical data report	USACE	United States Army Corps of Engineers
GDSR	geotechnical design summary report	USCS	Unified Soil Classification System
GIR	geotechnical interpretive report	UTRC	Underground Technology Research Council
		VCP	vitrified clay pipe

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CHAPTER 5

SUMMARY OF PRACTICE

Microtunneling was developed in Japan in the mid-1970s and in Germany in the early 1980s. A common factor that allowed microtunneling's development in Germany and Japan was the use of equipment in areas of uniform soils, where it could easily be optimized.

The first use of microtunneling technology in North America was in 1984. It involved using a machine with an inside diameter (ID) of 1,800 mm (72 in.) to install 190 m (615 ft) of pipe under Interstate 95 in Miami, Florida.

A milestone in the acceptance of microtunneling occurred in Houston, Texas, in 1987 as part of the River Oaks Project. The project involved more than 6.1 km (3.8 mi) of sewer system installed with microtunneling. The pipe diameters installed were 250, 450, and 525 mm (10, 18, and 21 in.).

The growth of microtunneling has been steady since the 1990s, mainly on projects requiring installations below the groundwater table, with minimal surface disruptions, and through difficult ground conditions. Engineers are increasingly designing projects that require microtunneling methods. These standard guidelines were written and originally published in 2001 to aid in that process. This document is the first major update and revision of the guidelines.

Trenchless technologies continue to improve. Microtunneling machines, jacks, alignment control systems, separation plants, and ancillary equipment have become more reliable, more adaptable, and more sophisticated. Although microtunneling is not always more expensive than other pipeline installation methods,

it is typically perceived as such. Therefore, smaller diameter installations that once were completed with microtunneling are now more frequently constructed with guided auger boring (GAB), the pilot tube method (PTM), small boring units (SBUs), or pipe ramming, where conditions permit. Microtunneling is now typically associated with installations below the groundwater table, with diameters ranging from 600 mm to 2,400 mm (24 in. to 96 in.).

Microtunneling continues to expand its capabilities. Slurry microtunneling is now used in full face rock as well as with groundwater depths of 30 m (100 ft), or 3 bar of pressure, and with curved alignments. Auger microtunneling, which lost market share to slurry microtunneling, is still available and is beginning to resurface with the development of the pilot tube guidance system and the use of hollow-stem augers. Successful microtunneling projects continue to depend on the following elements:

- Proper characterization of subsurface conditions, with the use of appropriate geotechnical investigation methods;
- Detailed planning;
- Proper design;
- Development of fair contract procedures;
- Selection of appropriate equipment;
- Selection of appropriate pipe materials; and
- Use of skilled operators and crews.

Commentary. None.

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PART II

PLANNING ASPECTS

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CHAPTER 6

INITIAL CRITERIA

Microtunneling is a specialized construction method suitable for many conditions where open-cut construction methods are not cost effective, are too disruptive, or are not technically possible. Other trenchless methods for installing pipe include but are not limited to the following:

- Horizontal directional drilling,
- Auger boring,
- Pipe jacking,
- Use of a pilot tube,
- Utility tunneling,
- Pipe ramming, and
- Earth piercing (use of piercing tools).

There are significant differences between the above construction methods, in application and in results. The engineer should review each of these methods to determine whether it can meet project requirements and be used successfully under the specific conditions that will be encountered during the pipe installation process.

Microtunneling is appropriate for a wide range of conditions. Careful evaluation of the following parameters will determine the applicability and constructability of a potential microtunneling project:

- Geotechnical conditions and geology, including groundwater;
- Potential for encountering objects (both natural and remnants from past construction) along the proposed alignment;
- Environmental conditions, including potential for contaminated ground or groundwater;
- Available work space for shafts;
- Required drive lengths;
- Pipe diameters;
- Cost and construction schedule;
- Alignment (straight and curved);
- Site access;
- Depth;
- Grade;
- Tolerances;
- Effects on surface activities;
- Location of existing, abandoned, or proposed utilities;
- Rights-of-way; and
- Location of existing, abandoned, or proposed structures.

Commentary. The engineer of a pipeline project is encouraged to consider other trenchless methods before concluding that microtunneling is the best construction method available. Manuals and reports on engineering practice (known as MOPs) have been written by ASCE for different construction methods. The reader should consult the references below and others (see Chapter 2 and the References at the end of this document) for

additional information on available trenchless construction methods:

- MOP 89: *Pipeline Crossings* (out of print)
- MOP 106: *Horizontal Auger Boring Projects*
- MOP 108: *Pipeline Design for Installation by Horizontal Directional Drilling*
- MOP 112: *Pipe Bursting Projects*
- MOP 115: *Pipe Ramming Projects*
- MOP 120: *Trenchless Renewal of Culverts and Storm Sewers*
- MOP 125: *Pipelines for Water Conveyance and Drainage*

Most trenchless installations, including microtunneling, follow straight alignments. Pipelines designed with curved alignments require special planning, equipment, and pipe considerations. See Sections 9.2 and 13.7 for more details on the design and construction issues related to curved microtunnel drives.

6.1 PIPE DIAMETER AND GRADE

Pipe diameter must be of sufficient size to meet the project requirements.

For gravity sewers, the grade should be established to meet the project requirements and to accommodate the pipe diameter selected. Flow capacity (Q) and cleansing velocities must be considered for gravity sewers.

For pressure lines, the engineer has more flexibility with regard to grade and depth of the pipeline—an important consideration when ground conditions improve with depth. Increasing the depth of the pipeline may result in more favorable ground conditions for microtunneling and may provide increased clearance from existing utilities and other facilities, depending on the site-specific conditions.

Commentary. Design grades and tolerances are typically specified for gravity systems. Grades and tolerances are typically less critical for pressure pipelines and dry utilities and conduits. When it is possible to do so, contractors prefer to drive the MTBM upgrade to allow fluids to drain back by gravity to the jacking shaft.

When grades are relatively flat or critical, or when unfavorable ground conditions exist, a two-pass approach should be considered.

Pipe diameter can have a significant effect on the cost and risks of a microtunneling project. In addition, the pipe must be of sufficient diameter to accommodate the spoils removal system, the guidance system, power lines, and control lines to the MTBM. The smallest microtunneling systems in use have an inside diameter (ID) of 300 mm (12 in.).

The use of multiple pipe diameters increases the cost of a microtunneling project. Combining pipe diameters reduces the

need for multiple MTBMs or upsizing and may decrease project costs.

A larger diameter microtunnel may be more cost effective than a smaller diameter microtunnel and may lower the project's risks. Diameters also significantly affect the length of drive that will be possible. Larger diameter MTBMs generally have more power, thrust, and torque for processing rocks and buried objects in the tunnel horizon and can achieve longer drives. Heat buildup in smaller diameter MTBMs can cause laser refraction. In addition, intermediate jacking stations (IJSs) are typically not practical for IDs of less than 1 m (39 in.).

When a project needs multiple sizes of pipe installed, some contractors might choose to mobilize one MTBM and then upsize it to complete the other diameters. If an MTBM is upsized by adding an outside "skin" (increasing the machine's diameter by adding plate steel to the diameter), then the geotechnical conditions, including the potential for handling cobbles, boulders, and other objects, should be taken into consideration. A skinned MTBM, though larger in body diameter, still has the same machine motor, and so it may be insufficient for excavating through complex geology. In addition, the crushing chamber size does not increase when the MTBM is upsized.

The planning phase is the time to determine whether the pipe diameter can be increased to achieve more cost-effective construction. The pipe grade may need to be modified if the pipe diameter is changed.

The engineer should consider including at each manhole location an additional drop of 30 mm (0.1 ft) or more, plus the grade tolerance, to assist in maintaining the design grade during microtunneling construction. Over the length of a project, these drops can be used to help maintain the overall grade in the pipeline if one segment of the pipeline becomes misaligned during construction. See Section 12.2 for grade tolerances and Section 20.7 for steering issues and corrections during construction.

6.2 DEPTH

The depth of the microtunnel alignment should be determined on the basis of the following considerations:

- Gravity flow or pressure pipeline;
- Project performance (i.e., hydraulics; connections; operations and maintenance);
- Geotechnical conditions and geology;
- Existing buried infrastructure;
- Future construction and changes to the ground surface elevation;
- Potential and tolerance for settlement or heave; and
- Potential and tolerance for inadvertent returns of slurry and lubricant.

Commentary. Microtunneling is not as adversely affected by depth as open-trench construction is. The major costs of deeper microtunnel construction are shafts and/or pits. Therefore, microtunnel elevations can be adjusted to take advantage of better ground conditions, to avoid other infrastructure, or to meet future needs. Adjusting depth is not always possible for gravity systems with flat slopes.

Selection of minimum ground cover should take into consideration the ground conditions and use of the surface space above the pipe. Minimum ground cover is typically a function of gravity versus pressure flow, pipe diameter, and geology. As the ground cover decreases, the risk of surface settlement and heave increases, as does the chance of an inadvertent return of slurry or lubrication. Ground treatment before microtunneling may be

required to control surface settlement or heave with minimal ground cover.

Typically, a minimum cover of two pipe diameters or 2 m (6 ft), whichever is deeper, is needed to allow sufficient depth for the ground to arch over the new installation and for the slurry and lubrication to be confined. With special measures, such as ground improvement, microtunneling has been successfully used with ground cover as little as 1 m (3 ft). Evaluation and calculation of the potential settlement and heave risk should be performed by the engineer to assist in the selection of the appropriate ground cover.

Potential ground settlement or heave and the owner's tolerance to the attendant risk should be taken into consideration when establishing the depth and the amount of ground cover (see Section 13.5).

6.3 ROUTING

The routing of the pipeline can be affected by many factors, including the following:

- Required start, intermediate, and end points;
- Geotechnical conditions and geology;
- Surface topography;
- Existing utilities;
- Physical features, natural or human-made;
- Available rights-of-way;
- Maintenance of traffic during construction;
- Surface access; and
- Contaminated ground and groundwater.

Commentary. Refer to ASCE Manual of Practice 46, *Pipeline Route Selection for Rural and Cross-Country Pipelines* (out of print), for more information.

Open-cut trench construction typically needs to remain within the roadway right-of-way to maintain direct access to the pipeline after installation. By contrast, microtunneling can be used to cross under variable topographic features. The technology has been used to install pipelines under hills and embankments and between valleys, eliminating the need to follow the surface roadways. Microtunnels can be installed under environmentally sensitive areas instead of around them.

Areas of contaminated ground or groundwater should be avoided, if possible, especially at shaft locations. See Sections 6.5 and 7.6 for discussions on sustainability and contaminated ground and groundwater. Nevertheless, if the groundwater is contaminated but the contaminant constituent (such as fuel oil) is floating on top of the groundwater, then the microtunnel alignment may be able to pass under the floating contaminant, provided that the ground the microtunneling machine is passing through is not highly contaminated. All of these issues are even more complex and expensive to handle when open-cut trench construction is used.

In urban areas, shafts and working spaces may be located on side and/or cross streets and out of intersections, if possible, to keep the main working areas off the roadways with the largest amounts of traffic.

Underground utility density is also an important consideration. Refer to ASCE/CI 38, *Standard Guidelines for the Collection and Depiction of Existing Subsurface Utility Data*, for guidance on the collection of existing utility data. The preferred microtunnel route and alignment should consider minimizing effects on utilities in terms of shaft installation, emergency access, and movement of utilities caused by surface subsidence or heave.

Trenchless methods in general provide solutions and options not typically available when open-cut trench construction is used.

6.4 SHAFTS AND WORKING SPACE

Shafts are required to launch the microtunneling machine, jack the pipe, and retrieve the machine. During the planning stage, working space on the surface to accommodate support equipment adjacent to the jacking shafts should be identified. Space is required for

- Control room;
- Power source;
- Lubrication system;
- Lifting equipment;
- Pipe storage;
- Slurry separation system;
- Temporary muck storage; and
- Other equipment necessary to support the microtunneling operation.

Jacking shafts should be of sufficient size to accommodate the earth support system, the jacking system required for the pipe diameter being installed, slurry system valves and pump systems, shaft access devices for personnel, safety equipment, tools, and personnel work space.

Reception shafts need space for lifting equipment to retrieve the microtunneling machine.

To the extent possible, shaft locations should be selected that avoid overhead and subsurface utilities. For all shafts, maintenance of traffic during construction, including access for emergency vehicles, should be considered.

Shafts are typically located at changes in alignment (both horizontal and vertical), changes in pipe diameter, and changes in geotechnical conditions and geology. Shaft spacing and locations should also be based on practical drive lengths and project design constraints.

Commentary. Microtunneling proceeds from a jacking shaft to a receiving shaft. The receiving shaft can be smaller and is needed only when the microtunneling machine is being retrieved or when the permanent structure is being built. Shafts not in use can be covered for public safety or until needed if located in the roadway. Excavation of the jacking and receiving shafts before microtunneling is useful in confirming geotechnical subsurface conditions before jacking pipe. Early excavation of shafts should be considered when complex or difficult ground conditions are expected.

The shafts' construction generally affects the public longer than the actual microtunneling operation and can present more risks to the owner, such as settlement, break-in or breakout failures, shaft wall-to-floor failures, settlement of the machine outside the shaft, and utility disruptions.

When locating shafts, consideration of drive lengths may be weighed against challenges found at the shafts' locations. Some common challenges are utilities (overhead and underground), settlement of sensitive features, traffic disruptions, vibrations during shaft construction, and disruptions to residences and businesses.

A jacking shaft is in constant use and contains crews and hydraulic and electrical equipment that must be protected. The jacking shaft should be protected from flooding and cannot normally be covered other than with a removable security cover during nonworking hours. The dimensions of the jacking shaft need to take into account walers, bracing, thrust blocks, entry

and exit seals, ladders, ventilation ducts, sump pumps, jacking frame length, and the MTBM or pipe segment length (whichever is longer). Vertical space above and adjacent to the shaft should be free of overhead utilities.

The staging area adjacent to the jacking shaft is needed for cranes, loaders, bentonite mixers, pallets of bentonite, the spoil separation plant, the control room, the power plant, utility pipes, jacking pipe, slurry pumps, trucks, and wastewater treatment. Historically, minimum staging areas are 7,000 ft² (256 m²) for jacking shafts and 4,000 ft² (146 m²) for receiving shafts. This working space includes the shaft footprint. Working spaces can be reduced by storing jacking pipe in other areas and using gantry cranes.

In planning the geotechnical investigation, consider that shaft elements often extend some distance below the actual pipe elevation. Geotechnical borings should extend below the shaft and microtunneling depths.

Effects of work zones on traffic and other surface activities should be considered.

Groundwater and flowing ground conditions require close attention when designing entry and exit seal designs. In challenging ground conditions, consider using multiple sealing systems. When complex geology is present, consider locating shafts at the interface between different geological features, to allow for changes in cutter wheel design to better suit the changing geology.

All shafts need crane access to build the shaft and then position and remove the MTBM, pipe materials, and support equipment. Analyses of safety hazards should be completed for all critical lifts. Overhead wires should be identified before mobilization of equipment.

Engineers not familiar with microtunneling should consult with machine manufacturers and contractors, and should review published case histories to establish a practical drive length for the pipe diameter being considered. The maximum shaft spacing should be selected to be within 10 percent of the identified practical drive length. The practical drive length should be based on acceptable engineering calculations and should consider the project geology, pipe diameter, pipe materials, tracking equipment capabilities and accuracy, cutter tool wear and survivability, and known buried objects.

Shafts should be designed with sumps and sump pumps to handle any seepage of groundwater or construction fluids that can enter the shaft at any time.

Public safety should be considered when selecting shaft locations. This consideration may include covering the shafts during nonwork hours and the use of fencing, K-rails, and security services.

6.5 SUSTAINABILITY

Microtunneling is a construction method that minimizes environmental impacts (such as noise and dust), minimizes the volume of excavated materials, and reduces traffic impacts (such as traffic delays and traffic detours). Microtunneling provides greater sustainability benefits for owners and society than do construction methods such as open-trench construction. Therefore, microtunneling should be considered during the planning phase of any civil infrastructure project. See ASCE's Policy Statement on Sustainability for more information.

Commentary. Sustainability comprises a set of environmental, economic, and social conditions in which society as a whole has the capacity and the opportunity to maintain and improve the

quality of life indefinitely, without degrading the quantity, quality, or availability of natural resources and ecosystems.

It is ASCE's policy to promote the goal of sustainable civil infrastructure. As stewards of the built environment, civil engineers must lead the way in planning, designing, and constructing sustainable infrastructure.

Microtunneling offers a sustainable alternative to open-trench construction by avoiding disturbances of ground surfaces, which would require reworking of roads, landscapes, or buildings. Where possible, the spoils from the excavation and the separated slurry waste from the microtunneling operation should be used directly (or indirectly) as backfill.

During the planning phase of a project, equipment should be evaluated for the polluting effect of its emissions. Therefore, equipment age, efficiency, and filtering systems for exhaust are important. Also, running and idling time of all equipment should be limited, to the extent feasible. The carbon footprint of a microtunneling project is typically less than that of other, non-trenchless construction methods, and can be calculated using a tool on the NASTT website (www.nastt.org).

The volume of excavated material is far less with microtunneling than with open-trench construction, and this element also makes microtunneling a more sustainable alternative.

Use of a drilling fluid as part of the microtunneling process allows the slurry and drilling fluid to be recirculated within the tunneling process, thus promoting conservation.

Microtunneling does not promote the need for dewatering, and therefore does not remove or change the groundwater table or cause the migration of a contaminated plume.

6.6 REGULATIONS

The engineer should identify and meet all federal, state, and local regulations when planning and designing the project.

Commentary. Procedures to identify and meet all regulations should be incorporated into the planning process for a project. At a minimum, the engineer should identify the following:

- Need for environmental documentation and permits
- Local regulations for discharges and handling of materials
- Erosion control measures
- Wetland delineation and protection requirements
- Noise limitations

See Section 7.7 for additional special considerations to be addressed during the planning and design phases of a project. See Chapter 24 for regulations to be followed during construction.

CHAPTER 7

SITE CONDITIONS

The owner and the engineer should visit the proposed site during the initial planning phase and periodically throughout design to identify site conditions and constraints. A survey of existing site conditions, both surface and subsurface, should be performed in the early stages of planning.

Commentary. This information is useful in determining the feasibility, alignment, grade, and physical constraints of the proposed pipeline project. This work is required independent of the construction method selected. People who understand constructability issues should be consulted and included in the site visits.

7.1 LAND USE

To provide insight into anticipated subsurface conditions, potential future conflicts, and other potential problems, an investigation should be undertaken into the history of the area proposed for microtunneling operations, as well as current zoning and plans for future development.

Commentary. A review of historical drawings, construction records, land use maps, and aerial photographs can provide insight into past land use that might affect the feasibility of microtunneling and related risk. Review of aerial photographs, taken in different eras, can be particularly useful in assessing changes in land use and investigating evidence of previous construction. A survey of the project corridor must be done to identify existing infrastructure above and below ground level. Existing property access and traffic conflicts along the corridor need to be identified, reviewed, and understood. To the extent possible, potential future land uses that could restrict the location of structures or affect the alignment need to be identified. Additional site visits may be required in areas of rapid development.

7.2 EASEMENTS AND RIGHTS-OF-WAY

The availability of easements, rights-of-way, and, if necessary, property acquisitions should be determined during the planning process. Early determination of unacceptable corridors may require changes in the alignment. Alignment changes are typically less costly and less disruptive to the project schedule if done early in the planning process.

Commentary. For some pipeline projects, microtunneling or another trenchless approach may be the only acceptable method for construction within an easement.

The project owner and the engineer should evaluate different alignments. Consider keeping the alignment alternatives within the public rights-of-way rather than extending them onto private property, which would require acquisition or easement.

Typically, there is one alignment alternative that can remain within the public rights-of-way. The cost and schedule of acquisition of a private property or easement must be considered. It is possible that an alignment completely within the existing rights-of-way could be longer and more difficult, resulting in a total higher cost than a shorter alignment requiring private easements. Efforts to acquire the needed private rights-of-way or easements should start as early as feasible to avoid delays to the project. Easement and rights-of-way acquisitions are often critical to the construction schedule.

Permanent easements for shafts and surface structures should take into account future maintenance and repair needs. Temporary easements are often required for support equipment and storage of materials during construction. A drawing of the site (with all support materials, equipment layout, and shaft locations) should be developed to scale to aid in determining the construction footprint and property needs.

Depth of the pipeline and geotechnical conditions should be considered when evaluating future effects and property easement costs. When reviewing and acquiring easements, consider widths for future maintenance or repair needs and the most likely inspection and repair methods that will be used. Microtunneled pipes are typically inspected and repaired with a trenchless method. The diameter of the microtunnel often dictates the trenchless repair method, which in turn dictates future needs for access and permanent easement.

In some jurisdictions, a subsurface easement can be obtained rather than acquiring the property or obtaining a surface easement. Obtaining subsurface easements is sometimes easier and less costly than surface easements, but they come with restrictions. When a subsurface easement is obtained, a temporary construction easement may also be required for access to the tunnel horizon from the surface. Subsurface easements could also affect future maintenance and operations.

7.3 UNDERGROUND UTILITIES AND OTHER FACILITIES

Underground utilities and other facilities along or crossing the microtunnel alignment must be identified during the planning process. The utilities within the zone of influence must be identified in accordance with state utility-protection laws. Use of ASCE/CI 38, *Standard guidelines for the collection and depiction of existing subsurface utility data*, is recommended along with the *Common Ground Alliance Best Practices*.

Commentary. Utilities and other facilities within the proposed alignment may be affected and may therefore affect the success of the microtunneling drive. During the planning phase, it is important to identify those utilities that will have a significant cost, time, or “fatal flaw” impact on the tunneling component of

the project. Professional judgment is required to determine just how much utility investigation is necessary, but at minimum, identifying known and suspected utility owners within the tunneling easement and asking them for their records is an important first step. ASCE/CI 38 contains guidance on conducting utility investigations. At a minimum, a Quality Level D (QLD) (records research) utility map should be produced. If an accurate topographic plan has been generated at this stage of the project, this QLD utility map can be upgraded to Quality Level C (QLC) (records research correlated to an accurate site survey) at minimal cost and time. If high-risk utilities are identified through this process, further actions—such as upgrading specific utilities to Quality Level B (QLB) (surface geophysical methods) and Quality Level A (QLA) (exposure and test holes)—may be prudent.

Some states have a one-call planning or design ticket. In some cases, using these services may supplement or replace the use of ASCE/CI 38, which may not identify all utilities or provide the detail necessary to the exercise of prudent professional judgment. The engineer must also consider that other objects (such as abandoned utilities, concrete thrust blocks, concrete encasement, manholes, valves, and vaults) may affect the project alignment. Microtunneling through previous excavations may have an adverse effect on the project, even if there is no direct conflict with an existing utility. For example, backfill material from an existing trench could create a mixed-face condition in an otherwise homogeneous material.

The use of the QLD map, the QLC map, and/or the one-call information allows the engineer to make planning decisions on alignment, depth, and necessary utility relocations before tunneling, as well as other risk-management issues.

In some cases, a more complete subsurface investigation may be warranted. The combined use of geophysics and test holes may be prudent. ASCE/CI 38 contains some information on the current capabilities of advanced geophysics to image utilities and other features. However, this step is usually reserved for the design stage (see Section 11.5).

7.4 BURIED OBJECTS

The potential for encountering buried objects along or crossing the microtunnel alignment should be identified and evaluated during the planning process. The pipeline alignment should be adjusted, if possible, to avoid known buried objects.

Commentary. It is possible to microtunnel through some buried objects if provision is made for them before the drive. Buried objects should be defined, and methods for resolution should be addressed, in the planning process. Additional assessments of potential buried objects within the microtunnel horizon should be undertaken during the geotechnical investigation phases (see Section 11.3).

Buried objects can be remnants from past construction or natural in origin. Examples of remnants from past construction are building tiebacks; debris infill; abandoned utilities; wood, concrete, or steel piles; old foundations; utility trenches; and past excavations. Examples of naturally occurring objects are trees, tree trunks, root systems, cobbles, boulders, voids filled with material that differs materially and substantially from the surrounding ground conditions, and flood debris.

A buried object that is encountered and stops or impedes the forward progress of the MTBM along the design path, including tolerances is an obstruction. A rescue shaft or other means is required to remove the obstruction or recover the MTBM.

7.5 IN-LINE MICROTUNNELING

In-line microtunneling is used to replace an existing pipe in place. The new pipe must be larger than the existing pipe, with the existing pipe enveloped by the face of the MTBM. The existing pipe must be of a material that can be easily broken down and ingested into the face of the MTBM.

Commentary. In-line microtunneling may be considered when pipe bursting is impractical, when an existing pipe has insufficient required flow capacity, when there is insufficient horizontal space in the corridor, or when known objects (remnants from past construction or natural) impede the installation of new pipe at the same vertical alignment. In-line microtunneling may also be considered when the degree of upsizing required for hydraulic capacity exceeds the capability of pipe bursting or another trenchless method.

In-line microtunneling has been used to replace existing pipe materials consisting of clay, unreinforced concrete, and brick (as of 2014). A brick sewer's original method of construction can make in-line microtunneling difficult.

As-built records should be reviewed to determine if an underdrain or bedding system was used to install the original pipe. If this is the case, the system should be backfilled with a low-density, low-strength flowable fill, or plus the underdrain or bedding at the shafts before commencing in-line microtunneling operations, to prevent loss of drilling fluid and slurry into the existing underdrain pipe or bedding materials.

If a slurry microtunneling machine is used, the existing pipe should be backfilled with a low-density, low-strength flowable fill before commencing in-line microtunneling operations, to prevent loss of drilling fluid and slurry through the existing pipe. Depending on the size of the in-line microtunnel, the MTBM will encounter a mixed-face condition. See Section 11.3.4 for additional discussion.

All other in-line microtunneling operations are the same as standard microtunneling operations, including the use of jacking and receiving shafts.

Bypass pumping of existing flows may be required and should be investigated as part of the feasibility review. In some jurisdictions, there may be requirements to handle and dispose of pipe sediments separately; therefore, a determination of the chemical makeup of existing pipe material may be required to determine handling and disposal requirements.

The relative positions of the old existing pipe and the new in-line microtunneling pipe are important. See Fig. 7-1 for examples of positions where in-line microtunneling will work and positions that should be avoided.

Position A: The centerline of the existing pipe and the in-line microtunnel are at the same line and grade, and the new installation is larger. *Preferred and recommended.*

Position B: The centerline of the existing pipe and the in-line microtunnel are at the same line and grade but are offset, although the new installation is large enough to include excavation below the existing pipe. *Difficult and requires caution.*

Position C: The centerline of the existing pipe and the in-line microtunnel are along the same line but with different grades, although the new installation is large enough to include excavation of all portions of the existing pipe, from the jacking shaft to the reception shaft. *Difficult and requires caution.*

Position D: The centerline of the existing pipe and the in-line microtunnel are at the same line and grade, and the new installation is smaller than the existing pipe. This scenario must be avoided because it does not work. Because Position D is similar

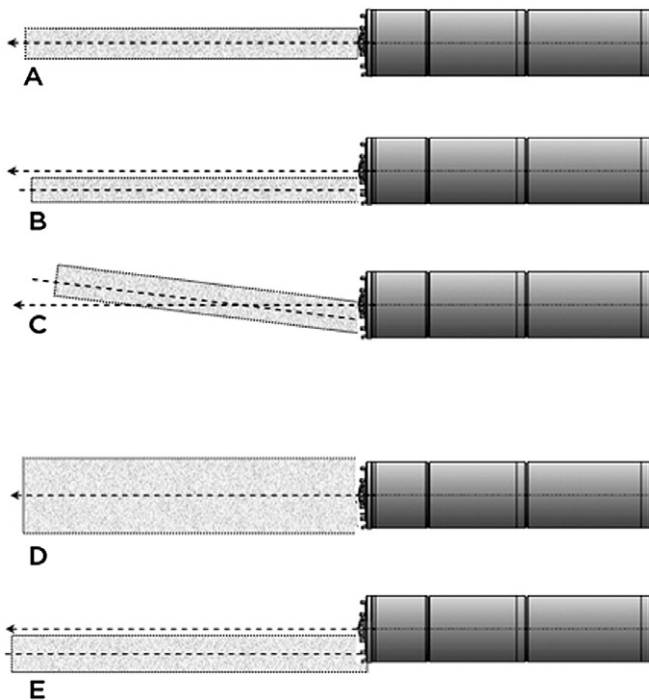


FIGURE 7-1. In-line microtunneling and the relative positioning of the old and new pipe. Position A is preferred. Positions B and C are difficult and require caution. Positions D and E should be avoided

to Position A, however, this scenario can work if the size of the in-line microtunnel is increased. *Avoid or upsize.*

Position E: The centerline of the existing pipe and the in-line microtunnel are at the same line and grade but are offset, and the new installation does not consume the entire existing pipe. This scenario must be avoided because it does not work. Because Position E is similar to Position B, however, this scenario can work if the size of the in-line microtunnel is increased. *Avoid or revise vertical alignment.*

Note that in-line microtunneling is not appropriate when the existing pipe is composed of asbestos cement. The reason is that the grinding of the asbestos cement pipe introduces asbestos contamination into the tunnel horizon.

7.6 CONTAMINATED GROUND AND GROUNDWATER

The type of contamination, location, degree of contamination, and extent of contaminated ground and groundwater should be determined to assess the effect of the contamination on the final alignment and construction materials, including pipe and joint materials. Contaminated ground and groundwater should be avoided, if possible.

Commentary. Disposal of contaminated ground and groundwater encountered during microtunneling operations increases the overall cost of the project. However, less contaminated material is excavated, handled, and disposed of through microtunneling; it therefore provides an economic advantage over open-cut methods. This advantage could result in an overall lower project cost.

Realignment of the project should be considered in an effort to avoid the contaminated ground and groundwater, and to avoid affecting the associated contaminated plume. See Sections 6.3 and 11.3.8 for additional discussions.

Processing of low-level contaminated materials through the separation plant quite often reduces the level of contamination. Explosive gases and volatile contaminants should be taken into consideration when determining the method of shaft construction, the type of ventilation, and the type of microtunneling machine to be used.

Consultation with specialty environmental and regulatory experts should be considered. Some areas of the country require the preparation of an Environmental Baseline Report (EBR) to document the location of contaminated materials, their potential effect on the environment, and mitigation methods (see also Section 11.3.8). The EBR should be considered a contract document.

7.7 SPECIAL CONSIDERATIONS

Special project considerations should be identified early in the planning process. Special considerations may include those listed here:

- Limited work hours;
- Restricted access;
- Environmental restrictions;
- Traffic disruptions;
- Haul routes and disposal sites;
- Service interruptions;
- Business disruptions;
- Noise limitations;
- Dust control;
- Light limitations;
- Visual effects;
- Odor control;
- Potential settlement of existing facilities;
- Utilities underground and overhead;
- Disposal of slurry, spoils, and groundwater; and
- Public events.

Commentary.

Limited work hours: Projects located in residential areas are typically restricted to 7:00 a.m. to 7:00 p.m. work hours, depending on the local ordinances. Work hours are typically restricted to nonpeak travel times when work is being done directly in the roadway. See Section 19.7 for additional discussion of work hours.

Restricted access: Microtunneling work is focused around the shaft locations. Shafts should be positioned in areas that offer access for trucks, employees, and supplies. The shaft sites should be fenced, with the general public restricted from entry. Anyone entering a construction site should be required to check in with the contractor and wear personal protective equipment (PPE) while on site.

Environmental restrictions: Microtunneling is often used to cross under environmentally sensitive areas. Shafts and associated worksites should be located in such a way as to avoid the defined limits of the environmental area. Some environmental areas may have nesting birds or other wildlife that should be considered and protected. If an environmental document has been prepared for the project, the document may contain mitigation requirements to be followed. Restrictions may apply to the time of year when the work can occur.

Traffic disruptions: Microtunneling and the associated shafts are typically located within public rights-of-way and in city streets. Since the microtunneling equipment is modular, it can be set up in a single traffic lane. Occupation of the traffic lane requires redirection of traffic past the shafts and the worksites.

Management of traffic can disrupt traffic flow. Traffic control plans should be developed for all microtunneling projects located in city streets. The engineer of the microtunneling project should look for shaft locations that minimize traffic disruptions or consider traffic detours where appropriate.

Haul routes and disposal sites: Spoils from the microtunneling process are generated and must be disposed of. Owners may have to identify potential disposal sites for the contractor. Once a disposal site is identified, haul routes need to be identified. Additional truck traffic will be generated on a daily basis as trucks enter and leave the jacking shaft sites.

Service interruptions: Microtunneling is used to install new pipelines. Once the main lines are installed, service connections are added. The process of adding service connections to new lines causes temporary interruption of services. If new lines are installed adjacent to existing lines, service interruptions can be kept to a minimum.

Business disruptions: Since microtunneling is typically done in city streets, there may be businesses along the alignment. The businesses at the shaft locations are most affected. Shafts should be located where they do not block driveways or sidewalks and prevent customers from reaching the adjacent businesses. The owner of the microtunneling project should initiate a public outreach program to keep businesses and residents along the project alignment informed of the project's activities and schedule.

Noise limitations: Noise will be generated from the construction equipment. This noise includes the hum from generators and other idling equipment as well as backup alarms on trucks. The slurry separation plant also generates noise and vibrations as the slurry is fed through the plant.

Dust control: Microtunneling involves excavation of the ground. Depending on the moisture content of the spoils, the excavated material can become dry and cause dust to be formed. If the spoils are too wet, mud and fluid spoils can be deposited on the ground surface, where they will eventually dry out and also form dust. Trucks entering and exiting the sites typically transport the dust to surrounding areas. The contractor should make efforts to keep the work sites clean. Dust control may include sweeping the road surfaces on a regular basis, using water trucks to dampen the dust, placing covers over the trucks as they exit, and washing the truck tires as they leave the site.

Light limitations: Microtunneling involves the forward advance of pipe into the ground. Sometimes a 24-hour operation

is needed to keep the pipe moving forward. In these circumstances, lights will be needed at the shaft locations to allow work to continue through the night. The setup and orientation of the lights should provide sufficient illumination for the workers without disturbing adjacent residents.

Visual effects: Equipment will be staged around the shaft locations and will be visible for the duration of construction.

Odor control: Odors may be noticeable to individuals passing by the worksites.

Potential settlement of existing facilities: Microtunneling involves excavation of the ground to allow forward advance of jacked pipe. This excavation process can result in over excavation and settlement of the ground. Existing utilities and facilities can be affected by any settlement that occurs.

Utilities underground and overhead: Microtunneling is used to install infrastructure pipelines, typically within the city's rights-of-way. Other utilities are in the same rights-of-way. Shafts installed to gain access to the microtunnel horizon will encounter these other utilities. Utilities in conflict with the microtunneling project will need to be relocated. Microtunneling involves the construction of shafts and the lowering of equipment into the shafts. As a result, cranes are needed during the microtunneling process. The location of overhead wires relative to the boom and swings of the each crane needs to be considered.

Disposal of slurry, spoils, and groundwater: The microtunneling process generates drilling fluid, slurry, and spoils during excavation. This material must be collected and disposed of. The water content of the material dictates whether the material can be hauled off site in sealed trucks. Construction water, groundwater, and surface runoff accumulate in the shafts and must be pumped to the surface. It is sometimes possible to obtain a permit to discharge water collected from the site into the local sanitary sewer system. The volume of material to be disposed of from microtunneling is significantly less than that generated by open-cut construction. The volume of spoils is limited to that produced by the shaft and pipe excavations.

Public events: Public events can occur at any time of the year. Most events are well planned and prepared in advance. Typically, a local government's parks and recreation department maintains a list of upcoming events. If a public event is planned for the period when the project is under way, the timing of the contract and the construction should be adjusted to minimize effects on the project.

CHAPTER 8

PRELIMINARY GEOTECHNICAL CONSIDERATIONS

In the planning phase of the project, a preliminary geotechnical investigation should be performed to identify the general subsurface conditions and any special subsurface conditions. The extent of the investigation required varies, depending on known local conditions.

Commentary. In areas of widespread uniform soil formations, it may be possible to use existing data to identify the soil types likely to be encountered. In areas of mixed, varying, or unknown existing conditions, a preliminary investigation should be undertaken to identify the general types of geotechnical conditions. A second function of the preliminary geotechnical investigation is to identify any environmental concerns.

Initial site investigation work, also known as a *desktop study* (since much of the data can be gathered at the engineer's desk through phone calls and Internet searches), is typically conducted during the planning stage of a project. It should include a site visit and a review of all readily available sources of

additional information concerning the site. The sources of additional information include geologic and topographic maps, aerial photographs, soil survey reports, construction case histories, professional papers about the area, historical documents concerning prior site usage, and discussions with local building officials and local contractors. In general, it is prudent to find out as much as possible about the site before planning and implementing subsurface investigations.

A qualified engineer familiar with the alignment corridor should determine the extent of the preliminary geotechnical investigation that will be required. Preliminary borings may be needed during the planning phase in order to understand alignment alternatives and to fill information gaps. The borings should be deep enough to capture conditions that, if present, would affect construction. (Recommendations for a complete geotechnical investigation, with borehole spacings and depths, are discussed in Chapter 11.)

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CHAPTER 9

PROJECT LAYOUT

9.1 LAYOUT FOR EFFICIENT MICROTUNNELING

Microtunneling projects should be laid out with attention to the benefits and constraints unique to this construction process. The following recommendations address some of these concerns:

- Minimize the number of pipe diameters. Changing pipe diameters may require a different MTBM and/or a completely new system. Minimizing the number of different pipe sizes, while still fulfilling the hydraulic and flow design requirements, reduces the overall equipment requirements of the microtunnel project.
- Construct microtunneling drives in both directions from a jacking shaft, when possible. To save money and time, microtunneling operations may be advanced both upstream and downstream from the same jacking shaft. Reception shafts are smaller and less costly. When practical, layouts of jacking shaft areas should be planned so that areas difficult to access contain only reception shafts.
- Optimize drive lengths consistent with pipe size, pipe material, type of MTBM system used, and geotechnical conditions. Longer drives are possible with larger pipe sizes.
- Evaluate different depths, pipe sizes, and gradients. Increasing the depth, pipe size, and gradient may not have the same effect on a microtunneling project as it would on projects using open-cut construction methods. Generally, the cost of constructing a deeper shaft is the only added cost associated with increased depth.
- Optimize alignment. Optimum alignment requires the smallest number of shafts without exceeding the practical drive length of the microtunneling system and the jacking capacity of the pipe being used, and it follows the most direct route.

Shaft locations are generally at manhole locations, at changes in direction, at high and low points of a pressure line, and at other permanent structures located along the pipeline. Local operation and maintenance (O&M) practices may require additional appurtenances.

Commentary. Where local agency or owner constraints limit spacing of manholes and appurtenances, it may be practical to reduce drive lengths to accommodate maintenance or other limitations. Another alternative to reducing drive lengths may be to construct intermediate manholes and appurtenances around the microtunneled pipe on a long drive.

Locations of shafts should be considered with respect to their effects on surface activities as well as with respect to changes in geology, as discussed in Sections 6.3 and 6.4. If possible, jacking shafts should be positioned to avoid having drives start in relatively soft ground conditions and then enter into hard ground

conditions. Mixed-face (soil and rock) conditions should be identified and avoided, if possible.

9.2 HORIZONTAL AND VERTICAL ALIGNMENT

Pipe layouts that use microtunneling traditionally have used alignments consisting of straight drives. Curved microtunneling increases a project's risk and should be approached with caution, using the appropriate equipment, pipe, and means and methods.

Slope (grade) should remain constant for the length of the drive. Differences in elevation between individual drives should be accommodated at the shafts. Changes in grade and alignment should typically be made only at shafts. Vertical curves are possible but have rarely been done.

Curved microtunnel drives may be considered for the purpose of avoiding conflicts with utilities and structures. Curved drives may allow the number of shafts to be reduced or the shafts to be moved to avoid conflicts with facilities or traffic. Curved microtunnel drives require careful attention to detail, however, and must be designed and constructed by qualified and experienced professionals.

Commentary. Choose vertical alignments once end-point requirements have been met. Take into consideration the raising or lowering of alignments to tunnel through favorable geology. When evaluating alignment alternatives, also consider the predicted settlement trough and settlement-sensitive types of structures that fall within the settlement trough.

The length of a microtunnel drive should be evaluated relative to the risk of completing the drive. Risks to consider include exceeding the jacking capacity of the pipe, hitting buried objects, and reducing the effective life of the cutting tools on the machine.

Adding curves, either vertical or horizontal, into the alignment increases the project's risk and complexity. The potential increased risks associated with a curved microtunneling alignment are the following:

- Increased jacking forces through the curve(s);
- Increased overcut and plowing of the MTBM and pipe segments through the curve, increasing the potential for surface settlement;
- Jacking through a curve, which may straighten the alignment and could make it difficult to install the pipeline within the specified line and grade tolerances;
- Uneven load transfer between pipe segments, increasing the potential for point loading of the pipe, which leads to damage or cracking of the pipe;
- The possibility (depending on the radius of the curve) of leaving pipe joints open upon completion, thus allowing for infiltration or exfiltration of the pipeline if a joint has not been properly designed;

- Reduction of the effective drive length because of inefficient transfer of load from one pipe segment to the next, thus increasing the need for more shafts or intermediate jacking stations (IJSs);
- Rotation (rifling or spinning) of the pipe segments as they are pushed through the curve, with potential damage to lubrication ports and slurry lines (rifling of the pipe segment containing any portion of the guidance system may require the resetting or surveying of the guidance system);
- The need (depending on the radius of the curve) for the MTBM to have self-cleaning articulation joints that do not become packed with cuttings when the steering joint is articulated to a more closed position; and
- The possibility that the jacked pipe may not follow the MTBM's path in the curved segment.

However, there are potential benefits to the use of a curved alignment, including the following:

- Avoidance of conflicts with utilities;
- Avoidance of known buried objects;
- Reduced disruption of traffic;
- Greater flexibility in shaft locations;
- Reduction in the number of shafts; and
- Reduced overall project cost.

Ideally, curved microtunneling is a collaborative effort among the engineer, contractor, equipment suppliers, and pipe manufacturers. However, the traditional design-bid-build or design-build contracting practices in the United States do not favor this type of arrangement because selection of the lowest price bid to perform the highly sophisticated work can be problematic. The increased level of risk and cost created by a curved alignment is not typically justified.

Examples of situations where a curved drive is a viable alternative in North America include the following:

- Surface easement and environmental protection constraints cannot be resolved to allow completion of pipeline alignments using straight segments; and
- Highly complex and congested underground utilities in public rights-of-way prohibit shaft construction in locations where straight drives would be feasible.

Curved microtunneling is also more expensive than microtunneling in a straight alignment. The increased costs result from lower production rates caused primarily by potentially higher number of IJSs, higher pipe costs, and additional surveying requirements. The advance rate may also be slower, simply to allow the pipe to follow in the excavated curve. Microtunneling alignments typically follow curved streets by using straight drives that cross from side to side of the right-of-way, under existing traffic and utilities.

Line and grade tolerances for curved alignments may need to be increased.

See Section 13.7 for additional discussion of curved alignments.

9.3 SERVICE CONNECTIONS

Service connections should be made at manholes and structures to minimize initial cost as well as the cost of future maintenance.

Commentary. Historically, pipelines have been laid in trenches. Service connections (drainage and sewer laterals) are made at any location along the pipeline because access is available

during construction. Pipelines with multiple service connections tend to allow the infiltration of groundwater over time, however, and can be difficult to inspect and rehabilitate.

Microtunneling allows the installation of a new pipeline from a jacking shaft to a receiving shaft. Such shafts are the only access points to the new pipeline and are typically converted into manholes. In an effort to reduce the number of service connections between the shafts, the manholes become the primary connection points. Service connections can be made to the manhole in any direction and from any desired depth. This radiating pattern of service connections to a manhole was first developed in Berlin, Germany, and is often referred to in the literature as the *Berlin construction method*. When this method is used, manholes become access points to the sewer laterals, allowing for simpler inspections and routine service. Inspection and rehabilitation of the main pipeline is also simpler, since the service connections are made only at the manholes.

Where connections directly to manholes are not allowed, or where such connections are not feasible, shallow collection lines constructed directly above the microtunneled line, or offset and parallel to the microtunneled line, can be used to convey local services to manholes. These shallow collection lines are referred to as *riders*. In some instances, other physical constraints, such as numerous utility conflicts, may require connections to be made directly to the new pipeline. If that is the case, excavations are required from the surface to the pipe at the point of connection.

9.4 EFFECTS ON ADJACENT STRUCTURES

Adjacent structures should be identified, and the effect of microtunneling in close proximity should be determined. Adjacent structures include aboveground buildings, substructures, vaults, and utilities. The evaluation should include the horizontal and vertical position relative to the microtunnel. The zone of influence of both the microtunnel alignment and working shafts should be identified. The age of adjacent structures and utilities should be considered along with the materials of their construction, their construction history, their sensitivity to ground movement, and their foundation types. The effects of nearby microtunneling can include ground movements, increases or decreases in ground and groundwater pressures, and reaction loads during jacking.

Commentary. Microtunnel construction normally has minimal or no effect on structures or utilities adjacent to the alignment. It may be possible to microtunnel close to existing structures or utilities. Objects in the backfill, or very hard and soft backfill adjacent to the structure or utilities, can cause problems with the microtunnel. For slurry systems, the possible loss of slurry should be taken into consideration. Ground movement in soft ground conditions must also be considered. See Section 13.5 for the evaluation of settlement risks. Monitoring of the actual volume of spoils recovered and comparison of recovered material with the theoretical volume may be helpful. The theoretical volume of excavated material should include a bulking factor.

A survey of adjacent structures and utilities should be undertaken before construction (see Section 23.1). The survey should include photographs and videos of existing structures, including any visible cracks. The installation and use of geotechnical instrumentation may be required for early detection of ground movement so that corrective action can be implemented (see Chapter 25).

With respect to existing utilities, their material construction and age should be considered. For example, older clay pipe may

collapse or fail if the ground surrounding the pipe moves. Any older pipe with unrestrained joints may move, and its joints may open. Welded steel, fused plastic pipe, electrical distribution cables, and communication cables can tolerate some movement.

The zone of influence is the volume of ground above the springline of the new pipe or bottom of a working shaft that would be affected by a potential movement of earth. The potential earth movement depends on the geology (including

groundwater) above these points. The zone of influence is generally accepted to be a 1:1 horizontal:vertical (H:V) slope for most geological conditions, but the ratio could be higher or lower. For example, loose granular soil or very soft clays may have a zone of influence of 2:1. For very dense or stiff soils, the zone of influence may be reduced to 0.5:1. The microtunneling project's design engineer should consult an experienced geotechnical engineer to determine the zone of influence on the basis of the project's geology.

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CHAPTER 10 COST CONSIDERATIONS

Cost is an important consideration when comparing microtunneling with other construction methods. Direct costs should not be the only consideration. Microtunneling has other benefits that should be considered when evaluating construction options.

Commentary. Microtunneling has proven to be an economical choice over traditional open-cut construction when all direct costs are included and when the depth of the installation is approximately 6 m (20 ft) or deeper. The lower direct costs that increase the competitiveness of microtunneling or reduce the depth at which microtunneling becomes competitive are related to dewatering, contaminated ground and groundwater, traffic control, project restoration, and more direct routes that can reduce a project's footage.

Other benefits that should be considered include reduced traffic effects and reduced environmental impact.

10.1 DIRECT COST OF MICROTUNNELING

The direct cost of microtunneling is influenced or controlled by several factors. These factors include the following:

- Geotechnical conditions and geology;
- Objects along the alignment;
- Groundwater;
- Volume of competing projects and number of qualified bidders;
- Labor conditions;
- Complexity of the project;
- Location;
- Depth and diameter of shafts;
- Length of microtunneling drives;
- Total footage;
- Mobilization, setup, and moving costs;
- Microtunneling penetration rates;
- Makeup time;
- Spoil and slurry handling and disposal;
- Curved alignments using straight drives;
- Curved drives;
- Type and diameter of jacking pipes;
- Effects of the microtunneling project on existing utilities and structures;
- Special requirements and/or restrictions;
- Project owner's reputation; and
- Availability of suitable microtunneling equipment.

Commentary. *Geotechnical conditions and geology:* Microtunneling is appropriate in a variety of ground conditions and is favored when the pipe is well below the groundwater table. The MTBM is designed to counterbalance earth and groundwater pressures. Costs increase as the clay content

increases because it takes longer to excavate and process spoils. Costs are higher for excavation in rock because advance rates are slower and rates of equipment wear are higher. Larger diameter machines fare better with rock than do smaller machines. Large-diameter MTBMs generate more spoils and require larger diameter pipe. Since larger MTBMs excavate more ground per foot of advance, they require larger slurry lines and separation plants to support faster penetration rates.

Objects along the alignment: Hitting one or more objects can slow or stop the forward progress of the machine and cause delays. Delays mean increased construction time, which in turn increases costs. When buried objects are encountered, additional costs are incurred for the construction of rescue shafts or for the use of alternative tunneling methods. Even objects that do not stop the MTBM can damage equipment or accelerate wear.

Groundwater: Generally, microtunneling is the preferred construction method when high groundwater head is encountered, because dewatering along the alignment is not needed. However, with increased depth, the groundwater head increases. This increase requires more attention to shaft construction, shaft and pipe joint watertightness, and sealing details. Costs of pumping and disposal of nuisance water increase with increasing shaft depth.

Volume of competing projects and number of qualified bidders: The number of projects under construction, recently awarded, and in the bidding phase affects the backlog of the contractors. Contractors who have no backlog or a minimal backlog may set their bid prices lower just to have work for their employees. As contractors' backlogs increase, they can be more selective about which projects to bid on and the prices they set for the work. The level of a project's risk, whether real or perceived, is critical to their interest in bidding. The number of qualified bidders also affects costs. Increased competition tends to reduce bid prices.

Labor conditions: The location of the project, and whether the area is subject to union rates or prevailing wage rates, influences the number of workers and their costs.

Complexity of the project: As the complexity of the project increases, so do the project's risks and pricing. Increased complexity may reduce the number of bidders and drive costs up.

Location: If a project is located close to any established microtunneling contractor, the mobilization costs may be lower. If a microtunneling contractor is on a local project, then mobilization costs are lower for other projects in the area if similar equipment can be used.

Depth and diameter of shafts: The cost of a shaft incrementally increases with its depth and diameter, presence of groundwater, issues related to rock versus soil conditions, and complexity of site access. Shaft shape may also increase cost. Going from a circular shape to a rectangular has the additional cost of the internal support system (walers, struts, and bracing) that is needed. In larger diameter shafts, the increased volume

of excavated ground for a circular shaft may offset the cost of the internal support systems for rectangular shaft. See Chapter 14 for more discussion on shafts.

Length of microtunneling drives: Longer drives are more expensive because of the equipment required, including intermediate jacking stations (IJSs), slurry pumps, ventilation, and guidance systems. Other non-equipment costs include production efficiency, the strength of the jacking pipe, and survey requirements. As the drive gets longer, more equipment is required to operate the slurry system, and the guidance system may need to be improved because the typical laser dot becomes too large or too weak to effectively guide the MTBM. As the drive gets longer, jacking forces also increase. The contractor may elect to use a stronger jacking pipe, up to some practical limit, or to install IJSs. IJSs can be expensive, and their cost needs to be factored into the pipe cost. Some contractors place one IJS directly behind the MTBM, to allow full recovery and reuse. When IJSs are used, there are additional labor costs for installing, operating, and then removing them upon completion of the drive. The jacking pipe material may affect the cost of the IJSs. There is a trade-off among pipe costs, the use of IJSs, the production loss associated with the use of IJSs, and construction of a new shaft.

Total footage: As the total footage of a project increases, the unit cost per foot decreases. The first drive requires mobilization of equipment, construction of two shafts, and setup of all equipment. The second drive may require only an additional receiving shaft and a turn of the jacking equipment in the jacking shaft. The second drive has a lower unit cost than the first drive, since the cost of the mobilization and jacking shaft are now spread across the two drives. Once the MTBM is on site, and when drive lengths and ground conditions remain relatively constant, the incremental unit cost for microtunneling tends to decrease because the initial costs are spread over more footage, and because the crew tends to become more productive through repetition of tasks.

Mobilization, setup, and moving costs: A major cost of a microtunneling project is the equipment mobilization, initial setup, and total number of equipment moves. Mobilization costs include preparation of the MTBM with new cutting tools, shop testing and refurbishing of ancillary equipment, and transport of the equipment to the jobsite. Depending on the diameter of the MTBM, power generation needs, and the type of slurry separation equipment required, 8 to 15 trailer loads will be needed. Setting up all of the microtunneling equipment and spoils separation plant can take two weeks or more, depending on the complexity of the project. To move all the equipment to a new shaft location takes time. If two drives are possible from the same jacking shaft, the process of preparing and turning around the jacking frame for the second drive can be done in less time. Consequently, it is most economical to design for long drives that are consistent with the equipment's capability, that minimize the number of required equipment setups, and that maximize the number of shafts from which multiple drives can be executed.

Microtunneling penetration rates: The MTBM's ability to progress through the ground affects the project costs. Generally, a soft-ground MTBM penetrates faster through sand than clay. Depending on the MTBM's size, gravel, cobbles, and boulders have lower penetration rates. Operator errors and failure to counterbalance face pressures in sands can lead to settlement, resulting in additional surface-restoration costs. Penetration rates for a rock MTBM decrease in clay.

Makeup time: One component of the advance rate is the makeup time, which is crew dependent. Makeup time is the time

it takes to retract the jacks, install a jacking pipe, reconnect the MTBM utilities, and restart the MTBM system. It also includes the welding time for steel casing. As more pipe is added, the crew generally becomes more efficient. When shorter pipe lengths are used, the number of pipe sections increases. Longer pipe sections require longer jacking shafts, and so makeup time must be evaluated against the increased length of the jacking shaft and the project's longer duration. Makeup time is independent of the advance rate.

Spoil and slurry handling and disposal: Slurry requires processing. Costs increase with increased size and complexity of the slurry separation plant. Factors that affect plant size include the following:

- Increased fines content;
- Efficiency of the separation plant in removing the fines content;
- Properties of the slurry to transport cuttings;
- Amount of the slurry that can be reused;
- Requirements for additional on-site storage; and
- Time required to separate the spoils.

All spoils generated at the site require disposal. Increasing the distance to the final disposal site increases cost. Contaminated ground or groundwater also increases the cost of disposal.

Curved alignments using straight drives: A curved alignment is commonly accommodated with a series of straight drives using angle points within the shafts. This procedure tends to increase costs because of the number of shafts required to complete the curve increases.

Curved drives: Curved drives can accommodate curved alignments with a minimal number of shafts. However, a curved drive increases costs for pipe, IJSs, and guidance systems. Costs are also increased because of reduced production rates, increased survey checking by specialized crew, monitoring and instrumentation of joint gaps, and special MTBM setup needs. The cost of a curve needs to be evaluated against the cost of adding shafts. The cost of a vertical curve needs to be evaluated against the cost associated with deeper shafts and straight drives.

Type and diameter of jacking pipes: The cost of jacking pipes can vary considerably. Pipe design elements that affect cost include pipe material (e.g., concrete, ductile iron, fiberglass, polymer concrete, steel, and vitrified clay), pipe diameter, jacking capacity, pipe length (other than standard), joint configuration, linings, and coatings. Because of the proprietary nature of several of the available kinds of pipe, these products may ship only from a single production facility, or from a particular domestic port. Transportation costs are affected by the pipe's weight, diameter, and travel distance.

Effects of the microtunneling project on existing utilities and structures: The condition of the existing utilities or structures needs to be assessed. Utilities may require relocation, replacement, or incorporation of protective measures. Structures may require protective measures.

Special requirements and/or restrictions: Any special requirements imposed on a project increase the project's costs. Owners and engineers should work with third parties to minimize such special requirements. Typical special requirements or restrictions may include the following:

- Limits on work spaces;
- Limits on work hours;
- Clearance of road surfaces during peak commute hours;
- Limits on truck trips, pipe delivery, or spoil removal;
- Limits on noise; and
- Dust control and frequency of road cleaning.

Project owner's reputation: An owner may have a reputation regarding contracting requirements and risk sharing. This reputation can increase or decrease a contractor's contingencies on a project, and its willingness to bid.

Availability of suitable microtunneling equipment: The diameter and capabilities of the MTBM equipment are relatively fixed by MTBM manufacturers. Therefore, the size of the casing or jacked pipe becomes critical to the question of which available MTBM fits the pipe and the project's requirements. If the MTBM requires such modifications as upsizing or a new or different cutter wheel, the project cost per linear foot increases as the amount of microtunneling footage being bid decreases. If the project is small enough, it is often cheaper to upsize the casing or jacked pipe instead of modifying the equipment. This approach may or may not require a new or upsized cutter wheel, which is affected by the ground conditions as well as by the size of the pipe. Finally, this cost effect is highly dependent on the bidding contractor's equipment inventories as well as on whether the existing available MTBMs are compatible with the range of anticipated subsurface conditions, or those of the MTBM manufacturers. See Section 12.2 for a discussion of two-pass microtunneling and upsizing of MTBMs.

10.2 INDIRECT SOCIAL EFFECT COSTS

The indirect social effect costs of a construction method should be factored into the selection process of the method. Indirect or social effects to be considered include the following:

- Vehicle traffic disruption;
- Pedestrian disruption;
- Disruption to business activities;
- Loss of tax revenue;
- Disruption to residences;
- Carbon footprint;
- Noise;
- Dust;
- Odors;
- Restoration costs for pavements;
- Public safety; and
- Effects caused by physical land required.

The cost of traffic disruptions is one indirect cost that can be easily considered. At a minimum, the engineer should use both the planned pipeline route and the proposed detour routes, if applicable, to determine the number of vehicles affected by construction in the road, the potential detour routes, and traffic volumes.

Commentary. The cost of indirect social effects, including traffic disruptions, has been documented in Boyce and Bried (1994a, 1998); Vickridge, Ling, and Read (1992); and Sterling (1994). When alignments are through areas with heavy traffic, the indirect cost to the public may be the overriding consideration in choosing microtunneling. Microtunneling operations affect an area from the shaft construction, through microtunneling and site restoration. Properly conveying the duration and impacts before construction can reduce the costs of mitigating indirect social impacts during construction or prevent social impacts from becoming direct costs.

Vehicle traffic disruption: When large numbers of vehicles are slowed by pipeline construction in a roadway, the time lost by the drivers and passengers is considered a social effect. Although no specific dollar amount can be determined for this disruption, there is a social expense in terms of people's wasted time, which increases the anger of motorists and the potential for accidents.

During a construction project, an owner or municipality is likely to receive and will need to deal with more complaints from the public. Microtunneling minimizes the number of disturbed areas in the roadway, thereby minimizing vehicle disruption. When microtunneling shafts can be moved onto side streets, the amount of vehicle disruption can be further reduced. Microtunneling work areas can also be set up with linear footprints, closing only one lane of traffic at a time. Microtunneling is more effective at reducing vehicle disruption than are other construction methods (such as open-trench construction).

Pedestrian disruption: When large numbers of pedestrians are affected by pipeline construction, the time lost is considered a social effect. Though no specific dollar amount can be determined for this disruption, there is a social expense in terms of people's wasted time, increasing the anger of the pedestrians, and increasing the potential for accidents. During a construction project, an owner or municipality is likely to receive and will need to deal with more complaints from the public. Microtunneling minimizes the number of disturbed areas along sidewalks, thereby minimizing pedestrian disruption. Microtunneling work areas should be set up with linear footprints, minimizing the closure of sidewalks. The work space should be bounded by fencing to provide pedestrian safety. If a sidewalk closure is needed, provisions should be made for pedestrians to use crosswalks on both sides of the work space, with plenty of signage to direct them to the designated safe walkways.

Disruption to business activities: Depending on the setup of the work spaces, access to businesses may be temporarily disrupted. Microtunneling projects are typically positioned within the roadway. Whereas parking spaces in front of stores may be needed during construction, access to the businesses and store fronts should remain open. Signage may be needed to let pedestrians and drivers know that businesses have remained open. Shafts should be positioned so as not to block the driveways of parking lots used or needed by the businesses.

Loss of tax revenue: Loss of tax revenue, though considered by some to be an indirect social cost, actually is not and should not be considered. Consumers buy products when and where they want or need. If access is restricted to a particular store, then the consumer will simply purchase the product at a different location. The amount of tax revenue generated and collected for a government entity is independent of where a product is purchased.

Disruption to residences: With any construction, there are disruptions to the public. Trenchless methods, including microtunneling, are less disruptive than open-cut trenching. Any effect related to social costs has to do with the loss of time to pedestrians and vehicles, as discussed above. Complaints concerning the amount of disruption to a specific neighborhood are typically a function of the income of the residents and reflect the fact that they know which people to call in their local government.

Carbon footprint: The amount of carbon dioxide generated by a project is now a consideration. As a result, programs exist to determine a project's carbon footprint. Microtunneling and trenchless methods use fewer surface vehicles, and historically they have generated a lower volume of carbon emissions than open-cut trenching.

Noise: With any construction, noise is generated. The issue here is any noise above normal background noise. The noise generated by microtunneling typically comes from a single source and can be mitigated with the use of sound walls, selection of equipment, and/or operational changes to the equipment. Local ordinances may affect whether work can be completed during off-hours. The social effect cost is an additional cost to

the contractor, which is paid by the owner for the mitigation measures.

Dust: Dust is generated because of the excavation of spoils and the movement of equipment. Microtunneling tends to reduce the generation of dust when slurry systems are used. Dust is mitigated by the use of water trucks, covered dump trucks, and tracking mats. Any fugitive dust can be cleaned with street sweepers. The social effect cost is an additional cost to the contractor, which is paid by the owner for the mitigation measures.

Odors: Odors can be generated from equipment (fumes) and chemicals used in various processes. In general, these odors are limited. In some cases, excavations for shafts in contaminated ground may generate odors that can be offensive to the public. In areas where submerged marine clays may be encountered, disturbance of the marine clay may release odors from decayed marine life. These odors may be difficult to mitigate during construction but can be avoided with proper planning of the alignment.

Restoration costs for pavements: Municipalities as well as regional and state agencies regularly renew pavements. These restoration projects generally consist of full-depth reconstruction, at a significant cost to these agencies. To maintain the structural integrity of the pavement as a whole, these agencies may require full sectional reconstruction of the pavement and not just patches to portions of the pavement. There is a direct cost to the contractor and, ultimately, to the project sponsor when full sectional reconstruction is required. There is also lost cost to the agency that expended monies to replace the pavement in the first place. Coordination, when possible, with the agency responsible for pavement maintenance should be conducted to adjust schedules and cost. In other words, microtunneling work should be done before any planned pavement replacement. Where older pavement is concerned, incisions caused by shaft excavations and by relocation of associated utilities may reduce the structural integrity and life expectancy of the pavement. This disruption may require expenditures of monies sooner than expected by the agency responsible for pavement maintenance.

Public safety: Microtunneling is one of the least invasive trenchless installation methods because the footprint of the operation can be relatively small. However, surface support equipment for the microtunneling operation has a significant footprint at the jacking shafts. The footprint may affect pedestrian use of sidewalks, reduce travel lane widths, or block access to hydrants or buildings in ways that affect public safety. Coordination with local emergency services should be conducted during the design and construction of a microtunneling project. There is an associated cost for these additional services, which is generally borne by the local taxpayers.

Effects caused by land requirements: The most cost-effective alignment for microtunneling is a straight horizontal alignment with a vertical alignment on a constant slope. Based on the geology, the vertical alignment should be consistent in multiple reaches, unlike the horizontal alignment, which must be maintained within a given easement, right-of-way, or utility corridor. It is almost always necessary to provide a receiving shaft at every horizontal point of intersection. Receiving shafts require a smaller footprint, since they are used primarily to allow the MTBM to be recovered and moved to the next location. Presumably, a receiving shaft is needed to connect two intersecting reaches of piping. The jacking shaft, which is often strategically located at a major horizontal point of intersection, requires sufficient area to stage multiple pieces of equipment in direct support of the microtunneling process as well as multiple sec-

tions of jacking pipe staged for installation. The engineer should consult with an experienced contractor or MTBM equipment vendor to ensure that there is sufficient space and land to support the microtunneling operation and its support activities.

10.3 ENVIRONMENTAL COSTS

The effect of a construction method on wetlands and environmentally sensitive areas should be considered in the evaluation and selection of construction methods.

Commentary. Microtunneling can significantly reduce the construction effect on wetlands and other environmentally sensitive areas. The effect of construction through contaminated ground and groundwater should be taken into account where potentially contaminated ground and/or groundwater have been identified during the investigation process. A contingency plan identifying methods for handling contamination may be required as part of the contract documents when construction is to be through potentially contaminated areas. Where possible, shafts should be located outside the contaminated area or constructed using methods designed to minimize the migration of contaminated ground and groundwater. Microtunneling below the level of the contaminated material, especially through impervious layers, can significantly reduce or eliminate the problems associated with microtunneling operations through these areas. Jacking and receiving shafts should be located as far as possible from critical environmental features, stream banks, and wetland buffers. It is important to perform hydrogeological studies to inform the owner, the engineer, and the contractor about potential groundwater issues that could affect water flow to wells, springs, seeps, or other groundwater expressions.

10.4 RISK ANALYSIS

A risk register should be started for the project during the planning phase. The document is a living document and should be updated throughout the project's life. The engineer, with input from the owner, should identify and analyze the risks associated with viable construction methods. It may be helpful in some cases to conduct an alternative analysis of microtunneling and other methods that accounts for both initial and long-term direct costs, indirect costs, and the effect of construction on third parties.

Commentary. The process for developing a risk register includes the following activities:

- Hold a brainstorming session, with key members of the design team and the owner, to identify project hazard and risk scenarios for each construction method proposed.
- Group the hazard and risk scenarios into related categories (e.g., shafts, microtunneling, public outreach, permitting, property acquisition, safety, and contractual issues).
- Identify the causes of each identified hazard or risk scenario.
- Determine the probability or likelihood that a specific hazard or risk scenario will occur.
- Determine the potential effect of the occurrence in terms of relative cost and/or scheduling delay to the project.
- Rate the hazard and risk scenarios on the basis of their likelihood and severity.
- Develop mitigation measures to reduce the probability and/or severity of hazard or risk scenarios.

Table 10-1. Risk Table with Severity of Effect and Probability of Occurrence

Very high (5)	5	10	15	20	25
High (4)	4	8	12	16	20
Moderate (3)	3	6	9	12	15
Low (2)	2	4	6	8	10
Very low (1)	1	2	3	4	5
Impact Probability	Rare (1)	Unlikely (2)	Possible (3)	Likely (4)	Most likely (5)

- Rerate the hazard or risk scenarios with mitigation measures incorporated.
- Update the risk register with newly identified hazard or risk scenarios as the project advances in design and into construction.

As an example, the severity of the effect and the probability of a risk are both ranked from 1 to 5. Each risk ends up with a ranking between 1 and 25 (Table 10-1).

Risks ranked in the red (black) zone should be mitigated during design into the yellow (gray) or green (white) zone. Risks ranked in the yellow (gray) zone should be mitigated during design into the green (white) zone, to the extent possible.

The risk register becomes a project management tool for making sure that mitigation measures, once identified, have been incorporated into the design.

Risks may exist for one or more drives on a microtunnel project that could prevent the drive from being completed successfully. The effect of failing to complete a drive should be considered. A contingency plan should be prepared, identifying at least one alternative construction method.

10.5 CONTINGENCY COSTS

There are unknown or unforeseen conditions associated with any underground project. Contingency costs should be identified and included in any construction cost budgets prepared by the owner.

Commentary. Despite completion of a risk register, as outlined in Section 10.4, some risks cannot be eliminated or reduced by design or construction mitigations alone. To allocate these risks, the owner should consider adding contingencies to the construction cost budget. These contingency costs can be held outside the construction contract or included as allowances within the bid schedule. The allowances are fixed dollar amounts, set by the owner, to cover additional items for the project that could not be quantified at the time of the bid. As an example, allowances can be included for disposal of contaminated ground.

The cost of a rescue shaft is also an appropriate allowance item. The design team for the project may know of or anticipate potential objects within the microtunneling alignment that could slow or stop the forward progress of the MTBM. However, the exact location, surface accessibility, frequency of use, size, and depth of the rescue shaft are not known. To ensure that all bidders include this item, an amount allowance should be included in the bid schedule. If and when an MTBM becomes stuck, the owner and the contractor can develop a rescue plan. The cost for the rescue shaft can then be established on a time and materials basis, with funding provided from the allowance item.

When owners show their willingness to identify and share project risks, the bidding contractors are more likely to include fewer contingencies in their bids. See Section 15.8 for more discussion on dispute resolution as a method available to allocate and share risks on a project.

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PART III

DESIGN ASPECTS

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CHAPTER 11

DETAILED SITE INVESTIGATIONS

11.1 GENERAL

This section addresses site investigation issues that are important for detailed design and construction of microtunneling projects. These site investigations include the following:

- Geotechnical investigation;
- Utility surveys (both aerial and subsurface);
- Investigation of traffic flow and access for vehicles and pedestrians;
- Environmental investigation;
- Investigation of flood zones; and
- Seismic investigation.

It is not the intent of this section to provide detailed guidance on conducting all site investigations for a specific microtunneling project. All projects are different and require site-specific investigations.

Commentary. None.

11.2 GEOTECHNICAL INVESTIGATION APPROACH AND METHODS

Detailed geotechnical investigations should be used to obtain factual information about the distribution and engineering characteristics of soil, rock, and groundwater at a site, to the extent required to produce a sound, safe, and economical design and allow reliable construction planning for the work.

The approach and methods may include the following:

- Review of existing information;
- Aerial photography;
- Historical maps;
- Test pits;
- Trenches;
- Borings;
- Large-diameter drilled holes;
- Cone penetrometer testing (CPT);
- Geophysical surveys; and
- Laboratory testing.

Commentary. Microtunneling equipment can be selected, set up, and operated to provide satisfactory results for a wide variety of ground conditions. However, a given combination of equipment, setup, and operating practice cannot be expected to perform effectively in all ground conditions. It is important that ground conditions and their range of variability be reliably established so that the project can be designed, the contract documents can be prepared, and the appropriate equipment can be selected, set up, and operated properly. Problematic ground conditions can reduce the feasibility of microtunneling or eliminate it as an

option for construction. Early identification of problematic ground conditions enables the designer to confirm the selection of microtunneling and prevents loss of time in pursuing an option that will not be successful. The potential range and likelihood of the ground conditions that can adversely affect the performance of microtunneling equipment must be identified as early as possible during design.

The site investigation requirements for microtunneling are no different in principle from those for other tunneling projects. In most circumstances, the required subsurface information can be obtained, interpreted, and applied to microtunneling with the proper application of existing technology. As with all underground projects, experience and good judgment are critical ingredients throughout the site investigation's development, implementation, and interpretation process.

Initial site investigation work, which is typically conducted during the planning stage of a project, should include a site visit and a review of all readily available sources of information concerning the site. See Chapter 8 for additional information to be collected during the desktop study phase.

An important decision is to determine how many test borings to drill, and to what depth. At a minimum, test borings should be taken at all jacking and receiving shaft locations. Additional borings are often appropriate for microtunneling drives longer than 90 m (300 ft). If surface access is limited, other exploration methods should be used to supplement the boring information.

Test borings should be drilled below the microtunnel and shaft inverts to identify changes in the strata below the microtunnel and shaft that could affect the construction. The additional boring depth is needed to collect the geotechnical information for the shaft foundation. A distance of at least 3 m (10 ft) below the tip elevation of any proposed shaft ground support system is recommended. If soft, compressible soils are present at or near the invert, some but not necessarily all of the borings should be extended into firm ground.

On some projects, the microtunneling horizon is changed or lowered during the design phase. For these cases, new borings should be drilled to extend at least one pipe diameter or 1.5 m (5 ft), whichever is deeper, below the new microtunneling horizon.

The final decision on the number and depth of borings should be made by a geotechnical professional on the basis of the importance and complexity of the project and the nature and complexity of the ground conditions. The more difficult and costly the proposed project, and the more complex the ground conditions, the more extensive the subsurface investigation program should be.

Important decisions that must be made relative to the test borings involve the sampling interval and the need for laboratory and field testing. The frequency and type of samples that must be obtained are a function of the project type and ground condi-

tions. Samples should typically be taken at intervals no greater than 1.5 m (5 ft) and at changes in strata. To identify thin strata that could have significant effects on microtunneling, continuous sampling from one diameter above the pipe to one diameter below the pipe is prudent. Classification of soil and rock can be accomplished by using disturbed sampling procedures. For laboratory strength and compressibility tests, more costly undisturbed samples are required. Field tests that might be used consist of in situ vane testing, hand penetrometer testing, point load testing on rock, pressuremeter testing, and in situ pump testing.

At some sites, it may be prudent to supplement the test boring program with other forms of subsurface investigation, such as test pits or large-diameter drill holes. Test pits and large-diameter drill holes can be indispensable methods of subsurface investigation, especially for gravels, cobbles, and boulders, or where subsurface conditions are extremely variable or where areas of undocumented fill may exist. Test pits and larger diameter drill holes provide the following:

- Large-volume samples;
- Exposure of undisturbed ground for observation;
- Direct evidence of the type and distribution of objects that exist in the ground; and
- Important facts about ground behavior and groundwater inflows.

Geophysical techniques and other emerging methods may be effective in helping to interpolate subsurface conditions between the test borings. Although a complete listing of all exploration methods is beyond the scope of this section, some examples are listed here:

- Seismic refraction and reflection surveys;
- Ground-penetrating radar;
- Electrical resistivity studies;
- Pilot tube insertion;
- Horizontal directional drilling; and
- Pipe ramming.

These methods have been shown to be effective in some types of subsurface profiles. In general, such applications should be used if they provide useful information at a nominal cost relative to the proposed design and construction requirements.

Other issues of importance to the success of a subsurface investigation program include good-quality exploration and sampling services and continuous on-site monitoring of the work by a trained geotechnical professional representing the engineer. Thorough observation of field activities must be provided so that the results of the fieldwork can be accurately and correctly interpreted.

A common and accepted practice is to conduct a detailed site investigation in phases, whereby each phase of the work provides the information needed to plan the next phase of study for site-specific problems. The first phase is to conduct a desktop study, as discussed in Chapter 8. The next phase may consist of a few test borings, supplemented with one or more observation wells. Samples should be obtained for laboratory testing. Upon completion of this work, it is possible to establish a working model for the site and to go back to the field with a considerable amount of additional information about ground properties, stratigraphy, and groundwater conditions. Use of the observation wells allows the collection of groundwater information to establish the fluctuation of groundwater levels. Groundwater samples can also be analyzed for water quality and corrosivity. The next phase of investigation becomes not only an opportunity to obtain new information but also a test of the validity of the model.

Depending on the outcome of this test, the geotechnical professional is able to question his or her preconceptions about the site and to conduct various “what if” scenarios regarding important geological assumptions. Depending on the outcome of this work and the importance of specific conclusions relative to subsurface conditions, it could be prudent to go back for an additional phase of even more detailed investigations to resolve remaining ground condition uncertainties.

11.3 IMPORTANT GEOTECHNICAL CHARACTERISTICS

11.3.1 General The following geotechnical characteristics should be determined for microtunneling projects:

- Soft ground conditions;
- Rock conditions;
- Mixed-face conditions;
- Gravels, cobbles, and boulders;
- Groundwater conditions;
- Potential buried objects;
- Potential for naturally occurring gas, such as methane and hydrogen sulfide;
- Contaminated ground or groundwater; and
- Human-made and environmentally sensitive features.

Commentary. The potential effect that these characteristics exert on construction and the effects that construction may have on site features are summarized. Methods for obtaining the required data are also described. The relative importance of individual characteristics varies from project to project, and no particular significance should be placed on the order in which they are listed and discussed.

11.3.2 Soft Ground Conditions The term *soft ground*, as used in the underground construction industry, typically refers to soil. For cohesionless soils (predominantly sands and gravels), the important characteristics that should be identified include the following:

- Grain size distribution;
- Coarse fraction, i.e., gravel, cobbles, and boulders, if present;
- Unit weight;
- Hydraulic conductivity; and
- Density (typically in terms of standard penetration test N-values).

For cohesive soils (clays and silts), the important characteristics that should be identified include the following:

- Moisture content;
- Plasticity index;
- Unit weight;
- Shear strength;
- Compressibility;
- Consistency (typically in terms of standard penetration test N-values);
- Grain size distribution and hydrometer analysis; and
- Coarse fraction, i.e., gravel, cobbles, and boulders, if present.

Laboratory testing should be done in accordance with ASTM standards outlined in Section 2.2.1.

Commentary. It is important to understand the soil conditions in terms of these characteristics because they affect most, if not all, aspects of the microtunneling process. Some ground can

Table 11-1. Tunnelman's Ground Classification for Soils

Classification	Behavior	Typical Soil Types
Firm	Heading can be advanced without initial support, and final lining can be constructed before ground starts to move.	Loess above the water table; hard clay, marl, cemented sand, and gravel when not highly overstressed.
Raveling Slow Raveling Fast Raveling	Chunks or flakes of material begin to drop out of the arch or walls some time after the ground has been exposed. This is caused by loosening or overstress and "brittle" fracture (ground separates or breaks along distinct surfaces, as opposed to squeezing ground). In fast-raveling ground, the process starts within a few minutes; otherwise, the ground is slow raveling.	Residual soils or sand with small amounts of binder may be fast raveling below the water table and slow raveling above. Stiff fissured clays may be slow or fast raveling, depending upon the degree of overstress.
Squeezing	Ground squeezes or extrudes plastically into tunnel, without visible fracturing or loss of continuity, and without perceptible increase in water content. Ductile, plastic yield and flow are caused by overstress.	Ground with low frictional strength. Rate of squeeze depends on degree of overstress. Squeezing occurs at shallow to medium depth in clay of very soft to medium consistency. Stiff to hard clay under high cover may move in combination with raveling at excavation surface and squeezing at depth behind surface.
Running Cohesive Running Running	Granular materials without cohesion are unstable at a slope greater than their angle of repose (from ± 30 to 35 degrees). When exposed at steeper slopes, they run like granulated sugar or dune sand until the slope flattens to the angle of repose.	Clean, dry granular materials. Apparent cohesion in moist sand or weak cementation in any granular soil may allow the material to stand for a brief period of raveling before it breaks down and runs. Such behavior is cohesive running.
Flowing	A mixture of soil and water flows into the tunnel like a viscous fluid. The material can enter the tunnel from the invert as well as from the face, crown, and walls and can flow for great distances, completely filling the tunnel, in some cases.	Below the water table in silt, sand, or gravel without enough clay content to give significant cohesion and plasticity. May also occur in highly sensitive clay when such material is disturbed.
Swelling	Ground absorbs water, increases in volume, and expands slowly into the tunnel.	Highly preconsolidated clay with a plasticity index in excess of about 30, generally containing significant percentages of montmorillonite clay.

Source: Adapted from Heuer (1974).

include coarse fraction that significantly affects the ground behavior and microtunneling performance.

The Tunnelman's Ground Classification system, first described by Terzaghi (1950) and slightly refined by Heuer (1974), is designed to describe ground behaviors and their effect on larger, conventionally constructed soft ground tunnels. The system is also a useful tool for evaluation of soft ground conditions as they relate to microtunneling. Descriptions of various ground behaviors, in terms of the Tunnelman's Ground Classification system, and their effects on microtunneling are presented elsewhere (see Bennett et al. 1995). General ground behaviors include the following:

- Firm;
- Raveling;
- Squeezing;
- Running;
- Flowing; and
- Swelling.

Table 11-1 provides a general description of different ground behaviors.

11.3.3 Rock Conditions Based on appropriate descriptive criteria (i.e., ASTM, ISRM), the field subsurface and surface site investigation strategy for projects in which rock is anticipated should focus on determining the following:

- Depth and extent of bedrock;
- Rock type (e.g., lithology, classification);
- Rock quality designation (RQD);
- Weathering and alteration index;
- Discontinuity set frequency and spacing;
- Discontinuity surface characteristics (e.g., aperture, infill material, roughness, shape, joint roughness coefficient);
- Presence of fracture-controlled groundwater;
- Discontinuity orientation (e.g., strike and dip direction); and
- Point load strength.

Conventional rock coring accompanied by downhole acoustic and/or optical televiewer (A/O TV) logging can be a cost-effective way to arrive at the field data outlined above, and it also allows the user to collect data on discontinuity orientation.

After the field investigation phase, laboratory testing should be completed on select samples obtained in the field, in order to arrive at reasonable estimates of the following:

- Hardness (e.g., Mohs Scale);
- Unconfined compressive strength (UCS);
- Intact rock or joint direct shear strength;
- Tensile strength (e.g., Brazilian);
- Punch penetration;
- Abrasiveness (e.g., Cerchar, Taber);
- Mineralogical (thin sections);
- Boreability; and
- For clay shale, slake durability and swelling tendencies.

Depending on the depth of burial, the geologic region of the proposed work, and the project risk, additional field testing may be necessary, including determination of in situ rock mass modulus of deformation or horizontal stresses through pressure-meter testing. In addition, packer testing (e.g., using the Houlsby method) can be useful in determining fracture-controlled hydraulic conductivity and behavior of discontinuities under controlled head conditions.

Commentary. Advances are being made in rock cutting, but microtunneling in rock currently presents significant challenges. The challenges include the need for high-thrust-capacity cutters with bearing and mounting assemblies that are small enough to fit to a small-diameter cutter wheel and robust enough to resist excessive wear over long distances. Proper cutter spacing and configuration to achieve efficient rock spalling is also a challenge, especially for MTBMs smaller than 1,500 mm (60 in.). Unlike the cutters on larger hard-rock tunnel boring machines, cutters on smaller MTBMs cannot be changed out during a drive,

because there is no access to the cutter wheel. For certain MTBMs with a diameter of 1,500 mm (60 in.) or larger, it may be possible to access the back of the cutter wheel to change tooling and cutters. However, depending on ground conditions, it may be necessary to stabilize the ground ahead of the MTBM or use compressed air to allow manned intervention to the back of the cutter wheel.

Because of legal restrictions, use of an air lock may also require the use of an MTBM approximately 2,400 mm (96 in.) in diameter. A variance from the federal OSHA 1926.800S standards is required because of the size of microtunneling. Obtaining this variance takes substantial planning and approval time prior to the start of microtunneling. It is possible that a variance may not be granted. It is important to verify the federal and state rules governing compressed air operations; the rules vary according to the air pressure required. Use of an air lock and interventions should be avoided, if possible.

Reliable exploration and test methods are available for determining the rock mass characteristics listed above. Drilling and sampling procedures that both maximize core recovery and minimize core damage are essential to the development of a reliable rock mass characterization program. Triple-tube (e.g., HQ, NQ) coring generally provides the least sample disturbance. In collecting rock cores, it is wise to take photographs with a scale and color guide immediately upon opening the sample. Once the sample is opened or moved, it begins to deviate from its in situ condition because of exposure to the atmosphere and vibration.

Unconfined compressive and tensile strength, together with abrasiveness, are key parameters for estimating boreability. Point load tests and Schmidt hammer tests can be used to supplement laboratory-determined unconfined compressive strength data and to rapidly obtain preliminary strength data in the field. The boundary between soil and rock unconfined compressive strength is commonly described as 140 psi (20,000 psf).

Weathering index, fracture frequency, joint characteristics, and overall rock mass quality are also important for estimating boreability. This information, coupled with field permeability testing, is necessary for estimating potential drilling fluid losses that may occur through open fractures and joints. Estimation of block sizes can be an important factor in evaluating the potential for blocks to become wedged in the cutter wheel or to become separated or dislodged above the pipe and to cause gouging of the pipe or couplings. RQD is widely used as an indicator of overall rock quality. Rock quality determinations can be refined, if necessary, with various empirical rock mass classification schemes (e.g., RMR and Q) that were developed for construction of larger tunnels (Barton et al. 1974; Bieniawski 1974).

Bedrock mineralogy affects the efficiency of MTBM operations at the advancing face. Preparation of thin sections and evaluation by a qualified geologist yield important details on sample mineralogical composition at the intact rock scale. For example, a high percentage of siliceous mineral content (e.g., quartz) typically results in increased abrasivity of the rock, which may result in accelerated rates of bit or cutter wear. In addition, the same potentially problematic mineralogy at the advancing face may also pose problems with respect to increased base friction (e.g., increased jacking or thrust forces) parallel to the drive length, which may affect the need for and quantity of lubricating fluids.

Boreability is the capacity of the rock in front of the MTBM to be ripped, chipped, and broken into smaller pieces to allow the advance of the MTBM. This parameter is controlled by the frequency of discontinuities of the rock, best determined on the

basis of RQD (ASTM D6032), as well as by the unconfined compressive strength (ASTM D7012), tensile strength (ASTM D3967), hardness (ASTM D5873), and abrasiveness of the intact rock (ASTM D7625). The four properties of the intact rock are determined by conducting a series of laboratory tests (see the specific ASTM reference cited above for appropriate testing procedures). The boreability of the rock also affects the tooling efficiency and the tooling survivability. Boreability typically falls into four categories: no effect on tooling, minor effect, difficult, and rock dominance. Some rock may be too strong, unfractured, and abrasive to be cuttable by typical tooling. Drive lengths and MTBM diameters may require modification. In some cases, other construction methods should be considered.

11.3.4 Mixed-Face Conditions Mixed-face conditions, which are defined as distinct variations in ground conditions within the cross-sectional area of the face, such as soil over rock, should be identified during the design phase, to the extent possible. Mixed-face conditions are to be avoided, if possible.

Commentary. Mixed-face conditions can result in uneven forces on the MTBM, thereby reducing steering control and presenting significant challenges to alignment, grade control, and stability of the face in addition to presenting the risk of over excavation and stoppage of the machine's advance. Mixed-face conditions should be identified during the design phase, and modifications should be made to the vertical and/or horizontal alignments so as to avoid these conditions, to the extent possible. This identification may require the use of extremely close-spaced borings or probes to locate the mixed face. In addition, geophysical surveys complemented by borings can provide indications of the top of the rock and hard layers.

If mixed-face conditions cannot be avoided, they should be documented in the contract documents, and the contract documents should require bidders and/or contractors to assess the risk and develop strategies to address the mixed face. These potential problems can sometimes be partially or completely mitigated with the design of the cutter wheel, use of ground stabilization, and careful operation of the MTBM. For example, when advancing from a full face of soft ground into a zone with a mixed face of hard and soft ground, there is a tendency for the hard ground to deflect the MTBM. Accordingly, where this condition is present, better line-and-grade control can be achieved in some cases by advancing the MTBM from hard ground into the softer ground. The MTBM operator may need to reduce the advance rate and steer into the hard ground of the mixed face to maintain line and grade. Allowance and adjustments to the construction schedule may be required.

If a mixed-face condition is known to exist along the alignment, contingency plans should be developed by the engineer during the design phase and by the contractor before construction. These plans might include placing a shaft at the mixed-face location so that one drive can be in the soft materials and the other drive in the hard materials. There could also be a bid item for rescue shafts in case they are needed. Consideration should be given to grouting the soft ground side of the mixed face and making the entire face of approximately the same strength. See Section 21.5 for additional discussion of contingency plans.

11.3.5 Gravels, Cobbles, and Boulders The extent, frequency, size distribution, and physical properties of gravels, cobbles, and boulders should be determined. Microtunneling drives should be located to minimize the amount of gravels, cobbles, and boulders to be encountered, if possible. If these conditions cannot be avoided, the MTBM should be operated and selected with fea-

tures designed to reduce risk when excavating through cobbly and bouldery ground. If anticipated cobble and boulder size and number are expected to obstruct the MTBM, allowance for obstruction removal or MTBM rescue should be provided in the bid documents.

Commentary. Intensely packed gravels or the presence of cobbles and/or boulders encountered by an MTBM can bring a microtunneling project to an abrupt halt. If rock pieces are encountered within the sampler during the geotechnical investigations, steps should be taken to determine whether the rock pieces are naturally occurring gravel or were generated by the sampler encountering and fracturing cobbles and/or boulders. Where there is a potential for gravels, cobbles, and boulders, large-diameter boreholes (up to 1 m [3 ft]), roto sonic borings, test pits, or trenches can be useful in obtaining representative samples for grain size analysis and determining the size and distribution of the gravels, cobbles, and boulders. Samples of the matrix material should also be obtained and tested.

The presence of gravels, cobbles, and boulders is a function of the geological setting. Settings include glacial deposits, moraines, tills, lag zones, alluvial deposits, colluvium, and weathered rock (core stones). The gravels, cobbles, and boulders can be isolated or nested, depending on their position within the deposits.

The unknown orientation of the odd-shaped cobble or boulder is important. The best method of sizing cobbles and boulders is based on determining the largest dimension of the rock in any direction. The physical properties of the cobbles and boulders should be determined in accordance with the rock testing outlined in Section 11.3.3. Rock type and mineralogy of the gravels, cobbles, and boulders should also be determined, to help in selecting the best equipment for the project.

One way to document the frequency of the cobbles or boulders is to determine the cobble volume ratio (CVR) and the boulder volume ratio (BVR). The CVR or BVR is the volume of cobbles or boulders in relation to the excavation volume, reported in terms of percentage. CVR and BVR are best determined by digging test pits or trenches with a calculable volume. For individual isolated cobbles or boulders in a test pit, CVR or BVR can be as small as 0.01 percent. A zone of nested cobbles or boulders in a test pit can have a CVR and BVR as high as 35 percent. The CVR and BVR values can be checked and confirmed when the jacking and receiving shafts are excavated. The use of CVR and BVR could be extended to each individual drive length.

The properties of the matrix material surrounding the cobble and boulders are important to understand. Stiff and densely packed material around the cobbles and boulders can keep the rock pieces in place and allow the cutter wheel to chip and break them up. The smaller chips and rock fragments can then enter the openings in the cutter wheel for further processing within the machine's crusher. A soft or weak matrix allows cobbles and boulders to move freely in the face. In this case, the cobbles and boulders can become ball bearings in front of the machine, and the tooling will be unable to cut them. Over time, the matrix material will be excavated, leaving behind cobbles and boulders nested in front of the MTBM at or near the invert.

If the matrix material is soft or weak, an open cutter wheel should be considered, to allow the cobbles and boulders to fall into the machine's crusher for processing. Ground treatment of the matrix material is also a consideration. If the matrix material is treated, it can be changed from soft and weak to hard and stiff in an effort to keep the cobbles and boulders in place for the cutter wheel and tooling to work.

The type and style of cutter wheel and tooling and the size of the aperture are important considerations in developing a strategy for excavating through gravels, cobbles, and boulders.

Special consideration of the drilling fluid type and additives may be required in soils with gravels, cobbles, and boulders and low fines content. See Section 19.4 for more discussion on drilling fluid design.

In situations where numerous cobbles and/or boulders are expected, the microtunnel diameter should be increased to allow easier excavation of the cobbles and/or boulders. Ideally, manned intervention can be used to remove nested cobbles and boulders collecting in the face and impeding the forward progress of the MTBM. Typically, this would require an MTBM with a diameter of 1,500 mm (60 in.) or larger. The use of a skinned MTBM should be restricted because the MTBM capabilities are based upon the manufactured MTBM and could be less. A larger MTBM allows better handling of cobbles and boulders because of the following factors:

- Higher horsepower and thrust are available.
- Larger aperture and openings in the face can be realized (which allows crushing and passing of larger particle size).
- Personnel access to the face is possible if the MTBM was manufactured to allow access.

If densely packed gravels are encountered, they can bind the cutter wheel or plug the slurry lines. Gravels are sometimes encountered as the bedding materials for utility trenches, or as backfill to structures and vaults. Understanding the MTBM's position relative to all utilities and buried facilities is important.

11.3.6 Groundwater Conditions The groundwater level and piezometric levels in confined aquifers should be determined by installing observation wells and/or piezometers at shaft locations. Consideration should be given to adding piezometers at intermediate points for longer drives. The hydraulic conductivity of water-bearing strata should also be determined.

Commentary. Groundwater conditions have a significant influence on ground behavior, and therefore on the design and construction of the jacking and receiving shafts, the selection of the MTBM, and the operation of the MTBM. Piezometers installed at shaft locations during design can later be used during shaft excavation to confirm shaft watertightness.

Hydraulic conductivity of water-bearing strata can be estimated with the use of grain size correlations and borehole permeability tests. For larger projects where highly permeable soils are anticipated with significant groundwater control issues during construction, pumping tests may be warranted. If the microtunnel is above the groundwater table and excavation is being carried out through dry granular soils, consideration should be given to the composition of the drilling fluid, to ensure support and transport of the excavated soil. Dewatering in the immediate vicinity of active microtunneling operations should be avoided, since it can result in the following:

- Loss of drilling fluid and slurry in the face of the MTBM because the slurry is being drawn into the dewatering wells, a situation that can have a detrimental effect on the operation of the MTBM; and
- Increased friction on the pipe, and higher jacking forces because of increases in effective stress and loss of lubrication.

These dewatering effects should be considered in the selection of shaft excavation support, since some systems require dewater-

ing in permeable soils. In addition, since MTBMs are capable of controlling groundwater pressure at the face of the excavation, dewatering along the pipe alignment is not necessary and can be detrimental for the reasons given above.

The use of watertight shoring systems that do not require dewatering during tunneling operations normally reduces the risk that dewatering will affect the microtunneling operation.

11.3.7 Potential Buried Objects The likelihood of buried objects, their nature, and their relative sizes should be established by the site investigation.

Commentary. The ability to identify buried objects goes back to doing site history investigations, as outlined in Chapter 8. Not all buried objects may be found before construction. Buried objects may or may not cause problems for microtunneling.

Obstructions are defined as any buried objects that lie completely or partially within the cross-sectional areas of the planned pipeline excavation and that impede continued forward progress of the MTBM along the design path and within allowable tolerances. Occasionally a stuck MTBM can be freed by judicious manipulation of the auger or slurry system, or by counterrotation of the head in quick succession, or by adjustment of jacking pressure. However, when obstructions are encountered, the face typically must be exposed to allow removal. Access to the face may be gained by sinking a rescue shaft at the face, by sinking a shaft off-line and hand mining to the face, or by hand mining from the receiving shaft. Shafts and hand mining are expensive, as is MTBM downtime, and so encountering obstructions is an expensive problem that often delays project completion. Consideration should be given to increasing the size of the MTBM and pipe string to a size that allows the larger diameter to consume the buried object(s). Typically, increasing the MTBM's diameter to 1,500 mm (60 in.) or larger allows personnel interventions to the cutter wheel.

Establishing the likelihood of buried objects requires an evaluation of information from a variety of sources, including regional and site geology reports, site history, geophysical surveys, borings, and test pits. Field investigations, including test borings, test pits, and, under some circumstances, geophysical surveys, can serve to verify preliminary conclusions relative to the potential for encountering obstructions.

Borings and test pits provide physical evidence of the nature, location, frequency, and size of buried objects that may be encountered. Geophysical surveys can sometimes provide insight into the relative sizes of buried objects, the frequency of their occurrence, and locations of anomalies, depending on the depth of interest and resolution. Indications of anomalies, based on geophysical surveys and other information, should be verified with borings and test pits.

Although there are many approaches to locating potential obstructions, it must be recognized that locating all of them along a given alignment is often not possible within practical budget and time constraints for site investigations. This situation results in the need to develop plans for dealing with obstructions, technically and contractually, before construction begins. See the commentary in Sections 12.3, 20.6, and 27.6 for additional information regarding buried objects and obstructions.

Wood typically becomes an obstruction because of its fibrous nature. Wood fibers tangle in the rotating part of the MTBM. Wood fibers can block the openings on the slurry return screens. Backflushing of the slurry lines is required to help move the wood fibers and clear the blockages. Wood can also be difficult to cut; as a result, it is pushed during the excavation process and disturbs the surrounding ground.

Wood within the ground can be naturally occurring or remnants from past construction. Naturally occurring wood includes buried trees, roots, logs, or buried marshes. Wood from past construction includes timber piles, timber roads, and buried fill.

Wood typically exists in three different age states: young (new), middle-aged, and old. Young and middle-aged wood tends to splinter and is more prone to binding the MTBM and blocking the screens. Old wood tends to be decomposed and is less disruptive to the excavation process.

11.3.8 Contaminated Ground or Groundwater Determination of the potential for encountering contaminants should be completed during the planning phase. If contaminants are found, then determination of the nature and extent of the contaminants must be undertaken during the site investigation.

Even if contaminants are not identified during the planning phase, the site investigation should still look for and test for contaminants.

Commentary. As with any type of construction, unanticipated ground and/or groundwater contamination has cost and scheduling effects on projects. Careful planning and execution may minimize these effects. Even if ground or groundwater contamination is anticipated, its presence can dramatically increase the cost of a project, depending on the types of contaminants and their extent.

However, microtunneling can be advantageous when contaminated materials are encountered because the quantities of contaminated ground and groundwater that must be handled are significantly less than would be handled by open-cut methods.

Sources of information regarding the potential for contamination include health department records; fire, planning, and building departments; and local or regional agencies responsible for pollution control or water quality. In areas where "paper studies" indicate the potential for contamination, chemical analyses of ground and groundwater samples should be undertaken to identify their nature and concentration. This information is necessary to determine the appropriate procedures for disposing of ground and groundwater during construction.

Personnel training for the handling of contaminated ground and groundwater should be implemented and followed per local, state, or federal requirements during all phases, including site investigation and construction.

11.3.9 Human-Made and Environmentally Sensitive Features Existing site features that could affect shaft construction and microtunneling operations should be identified during the site investigation.

Commentary. Existing utilities may be located by consulting as-built records or by contacting locating services, although locations of abandoned utilities are often not known. Potholing should be used to confirm locations of utilities that are near to or intersected by the planned pipeline. Historic buildings and environmentally sensitive areas usually require evaluation on a case-by-case basis to ensure their protection.

One of the biggest problems is buried footings and foundations left behind in the ground. Demolition of human-made structures typically involves removing all aboveground portions (the visible parts) and then cutting the top few feet of the foundations below the ground surface. The remaining structures are left behind as potential objects for the MTBM to encounter. It is important to investigate the history of the proposed alignment and to look for structures that were once in place. Review of historical photographs over set periods of time is one way to identify potential human-made and buried objects.

Microtunneling can avoid the surface effects to environmentally sensitive areas by tunneling under the features. Environmentally sensitive areas include the following:

- Sources of water supply;
- Islands and island corridors;
- Beaches;
- Dunes;
- Wetlands and wetland transition areas;
- Breeding areas;
- Migratory stopover areas;
- Wintering areas;
- Prime fishing areas;
- Migratory pathways;
- Water areas supporting submerged vegetation;
- Shellfish-harvesting waters;
- Forest areas;
- Habitat for federal and state endangered or threatened plant and animal species;
- Federal and state wilderness areas; and
- Areas designated as wild, scenic, or recreational, or as developed recreational rivers.

11.4 GEOTECHNICAL REPORTS

All subsurface data collected in connection with the geotechnical investigation, professional interpretations thereof, and design and construction considerations should be summarized in project reports. As the next step, the geotechnical data should be evaluated and baseline conditions should be established. All of the data and baseline conditions should be included in the contract documents.

Commentary. The geotechnical data report (GDR) contains all of the factual geotechnical information for the project, including explorations, laboratory and field testing results, geophysics information, and historical geotechnical data. This report typically does not include interpretation or recommendations and should be included in the contract documents as a stand-alone volume or provided on CDs, depending on the size of the project and the amount of information.

Often, it is necessary to produce a series of geotechnical design memoranda (GDMs) or a geotechnical interpretive report (GIR). These memoranda or reports typically present summaries of the geotechnical data, interpretations of the data, information about earth pressures to be used for design, discussions of the expected behavior of the ground, and other geotechnical information as well as design recommendations, such as for appropriate tunneling and shaft types and systems. Since the reports or memoranda are typically prepared before design, there is much said in these reports with respect to the project that may not be applicable at the time of bid. In addition, there are typically many broad statements about what may or may not occur during construction. Though such reports are useful to the designer, they can be misleading or even erroneous at the time of bid. It is strongly recommended that these reports not be made a part of the contract documents. It is standard practice to disclaim these reports and supersede them with more appropriate language and baseline statements in the contract.

Increasingly, another stand-alone report is prepared, known as a Geotechnical Baseline Report (GBR). The GBR is typically prepared with the owner's input as the design is being completed, and it serves to define the anticipated ground conditions and as the definitive geotechnical baseline for use in the resolution of disagreements, disputes, or claims related to differing

subsurface conditions. See *Geotechnical Baseline Reports for Construction: Suggested Guidelines* (2007), a joint publication of the American Society of Civil Engineers (ASCE) and the American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME). The GBR is a contract document and is used in connection with dispute resolution during construction. See Section 15.8 for more discussion.

11.5 UTILITY SURVEYS

All utility information (aerial and subsurface) needs to be collected in connection with the project alignment. The information should be collected in accordance with ASCE/CI 38, *Standard guidelines for the collection and depiction of existing subsurface utility data*.

Commentary. Existing utilities should be dealt with in an iterative manner similar to that discussed in Section 7.3. During the design phase, it may be possible to make small changes in the microtunneling alignment and shaft locations, to avoid conflict and the necessity for a potentially costly and time-consuming utility relocation. At this stage of the project, knowing the existence and exact location of any utility within the construction zone is necessary for success. Using the procedures set forth in ASCE/CI 38 to obtain Quality Level B (QLB) and Quality Level A (QLA) utility data is a prudent way to accomplish this, especially if the utility expectancy is complex and/or risky. Note that using a "design" ticket from a state's one-call service can only result in Quality Level C (QLC) or Quality Level D (QLD) information.

Utilities should be shown on the contract drawings. Information such as that found in ASCE 38 (e.g., achieved utility quality level, diameter, depth, material type, owner, trenched cross section, and backfill material used) should be included, if available.

If utilities conflict with the new pipeline or shafts, relocation plans should be prepared. Relocation of the utilities can be done before the project or as part of the project.

Utilities that remain in place and are close to the project may require protection and should be monitored during construction. Evaluation of settlement and heave risks may be appropriate, depending on clearances and ground conditions. See Section 13.5 and Chapter 24 for discussions of potential ground settlements and geotechnical instrumentation plans.

11.6 TRAFFIC FLOW AND ACCESS FOR VEHICLES AND PEDESTRIANS

All traffic information needs to be collected in connection with the project alignment. The information should be used to identify shaft locations and staging areas and to develop means of minimizing disruption to vehicles and pedestrians.

Commentary. Since traffic control and construction costs can be calculated for each alternative, these costs can be compared with the relative benefits for each method. The benefits of each method's traffic control strategy may include such elements as reduced delay and fewer miles to travel.

Costs of traffic control and construction need to be reviewed as part of the evaluation. These costs should be readily available as part of the design effort. These costs should include an appropriate allowance for traffic control enforcement, where appropriate.

Alternative traffic management strategies through work zones may have substantial consequences for project costs. For

example, if a given method involves time restrictions on construction activities, such as construction during off-peak hours only, the effective workday may be reduced by 25 percent or more. Thus the overall duration of project construction increases, substantially affecting project costs.

Similarly, other methods may influence the available work area during certain phases of construction, such as work space limitations that may be caused by parallel or adjacent construction activities, with the potential consequence of increased project costs.

The cost to vehicles for traffic delays must be accounted for. This is the cost of time, in dollars per vehicle, for delay of traffic through and around construction activities.

The extent and nature of the data requirements for the evaluation process are largely dependent on the nature, complexity, and setting of the proposed construction.

In this context, roadway occupancy refers to the degree to which the roadway is "occupied" by construction activities and therefore unavailable for normal traffic flow. Because different types of work zone strategies occupy varying amounts of the roadway, they also have varying effects on capacity of the facility.

Strategic roadway occupancy locations are an important part of the design and construction phase of any rehabilitation or replacement project.

The most undesirable location would be at the intersection of two roadways. This location would interfere with vehicles turning into, out of, and through the intersection and would cause delays in traffic. The second most undesirable location would be one needed for activities that would completely occupy the width of the roadway, thus closing all vehicle traffic from the roadway. Roadway occupancy within 15 m (50 ft) of an intersection is also undesirable. This location, depending on the direction of traffic flow as well as which side of the roadway is occupied, would interfere with vehicles turning into, out of, and through the nearby intersection, thus causing delays in traffic.

The most desirable location for roadway occupancy would be approximately midblock, between two intersections, and to one side of the roadway. This location would allow vehicle traffic to occupy most of the roadway, with little or no disruption.

The second most desirable location for roadway occupancy would be an area at least 30 m (100 ft) from an intersection and occupying the middle of the roadway. This location would not, in general, interfere with turning movements from and through the nearby intersection. In addition, vehicle through-traffic on the roadway could squeeze by the construction area with minimal interference with traffic flow.

In addition to strategic location of roadway occupancy areas to minimize disruption to vehicle traffic, disruption to commercial and business activities must also be considered and evaluated. Locations of abutters' loading docks, as well as locations of entrances to parking areas and of points of public access to abutters' buildings, must be known when the selection of a roadway occupancy area is being considered.

An abutter's loading dock is the lifeblood of its business operations activities. To disrupt the ability of an abutter to conduct business would cause economic loss to that business. Disruption of access to an abutter's parking area is also undesirable, although not as important as disruption to the abutter's loading docks. Nevertheless, it is an inconvenience that costs the abutter in terms of delay to its business activities and employees. Employees are required to seek alternate parking areas, and so their personal lives and business activities are affected.

11.7 ENVIRONMENTAL CONDITIONS

Environmental conditions include but are not limited to contaminated ground and groundwater, wetlands, endangered species' habitats, and naturally occurring hydrocarbons and asbestos. Microtunneling can be used to reduce or eliminate the effect of construction on these environmental conditions.

Commentary. When sensitive or contaminated conditions are found along a project alignment, microtunneling can reduce or eliminate the project's effects.

Effects to the surface or near the surface environment can be reduced by locating jacking and receiving shafts outside the sensitive area and tunneling under the sensitive area. In this case, the effect is avoided by simply passing under the sensitive area. If the distance is too great for a single tunnel drive, consider placing a receiving shaft at the midpoint and tunneling to the receiving shaft. This method reduces the effect because a receiving shaft is smaller than a jacking shaft, and it requires less equipment in support of its operation. The volume of spoils is increased by the volume of the shaft. Alternately, consideration should be given to increasing the tunnel diameter to achieve a longer drive distance. In this case, the increase in the excavated volume is based on the increase in the tunnel diameter and not on the presence of an additional shaft.

For contaminated ground, the volume of spoils can be reduced by locating the shafts outside the contaminated area and then tunneling through the contaminated area. In this case, the volume of contaminated ground is reduced to the excavated tunnel volume. If the distance is too great for a single drive, consideration should be given to placing a receiving shaft at the midpoint and tunneling to the receiving shaft. Adding a receiving shaft increases the volume of spoils by the shaft volume, which is typically less than in open-trench construction. Constructing a receiving shaft creates less volume than constructing a jacking shaft. Alternately, consideration should be given to increasing the tunnel diameter to achieve a longer drive distance. In this case, the increase in the excavated volume is based on the increase in the tunnel diameter, which may be less than the combined volume of the shafts and the tunnel.

When excavating affected ground, consideration should always be given to the total excavated volume, shafts, and tunnels in arriving at a final design. Also consideration should be given to requiring a slurry separation plant to reduce the water content of the spoils. This typically makes the spoils easier to handle and less costly to dispose of.

For contaminated groundwater or slurry water that becomes contaminated, the volume of slurry water that requires treatment and/or disposal is limited to that within the slurry system. The volume of slurry water can be further reduced by using a spoil separation plant that removes the spoils from the slurry stream and recycles (reuses) the drilling fluid.

Unintentional hydraulic gradients can introduce cross-contamination by causing flow of contaminated groundwater into areas of noncontaminated ground or groundwater. These gradients are typically created by dewatering of the shaft or possibly loss of (i.e., insufficient) microtunneling pressures. This issue is a third-party risk that could have significant monetary compensation requirements. The effects of dewatering should be considered by the engineer during the design process.

11.8 FLOOD ZONES

Floods, storm surges, and flash flooding conditions should be considered when identifying and selecting shaft locations and

microtunnel alignments during design. Designing for potential high water conditions can minimize flooding of shafts, minimize damage to microtunneling equipment, prevent pollution from petroleum products, and protect personnel.

Commentary. Assessment of flooding risk should include review of information from FEMA, NOAA, and USACE as well as local knowledge, to determine whether storm effects are probable along with their expected seasonality and the frequency of their occurrence. In some locations, flooding may last for weeks and affect the duration of the project. The ground conditions may become saturated and can affect the shafts and the microtunneling operation.

If flooding conditions are a significant risk, the design may need to be revised in light of the following considerations:

- Design shafts that address the full hydrostatic pressure condition (i.e., increase pressure to the top of the shafts, or plan to flood the shafts to eliminate unbalanced hydrostatic pressures).
- Design for fully saturated ground conditions.
- Design for increased uplift that may occur during flooding, or determine the maximum groundwater level and the method of monitoring that is allowed before flooding occurs.
- Increase pipe joint pressure ratings to accommodate installation and operational high-water pressures caused by flooding. Design may need to consider 10-, 50-, 100-, or even 500-year flood frequency events.
- Develop plans for evacuation, and note that access roads may flood before the project site floods.
- Protect or remove sensitive equipment, oils, and fuels.
- Develop plans to protect the project site from flooding when not occupied, including weekends and holidays.
- Develop plans to monitor water levels before and during large rainstorm events.
- Design shafts, pipes, and operations that prevent destabilization of flood-control levees and embankments. Recognize that localized levee slope failures can occur during floods.
- Plan construction operations to avoid periods with a high potential for flooding (i.e., hurricane season, rainy season).
- Develop start-up plans after a flood event. Plans should include mobilization of pumps, disposal of water from the shaft, and cleanup of the surface, shaft walls, and equipment within the shaft.

In areas prone to flooding, the owner should consider including allowances or contingencies in the contract to handle such special events, the associated cleanup, and the loss of time to the

construction schedule. A major flood event is typically considered a Type 2 differing site condition (see Chapter 27).

11.9 SEISMIC CONDITIONS

Seismic conditions, including fault offset and related ground responses, should be considered in shaft location selection, microtunneled alignments, design of the jacking and receiving shafts, and the installed utility service. The seismic considerations include protection of personnel and minimization of damage from fault offset, ground shaking, landsliding, liquefaction, and tsunamis.

Commentary. Assessment of seismic risk should include review of information from the U.S. Geological Survey, applicable state geological surveys, and local knowledge to determine the location, style, extent, recurrence interval, and probability of seismic conditions relative to the project. Seismic events are of relatively short duration, but their effects can be long-lasting and can severely alter ground conditions at shaft locations and along microtunnel alignments.

If seismic conditions are a significant risk, the design should consider the following:

- Flooding, for areas prone to tsunamis (see Section 11.8);
- Seismic pressures on shafts, in addition to earth and hydrostatic pressures;
- Automatic seismic shutoff valves on piping;
- Damage evaluation and start-up plans after a seismic event;
- Ground improvements, including those required by local codes, for critical lifeline or hazardous utilities in areas prone to liquefaction or seismically induced landsliding;
- Local code requirements for setbacks from, and/or crossing requirements of, active faults;
- At fault crossings, the anticipated direction, style, recurrence interval, and displacement of a single characteristic seismic event;
- Fault crossing direction with respect to anticipated style and sense of fault displacement (the tensile versus compressive strength of the same material varies);
- Oversized casing around product utilities to allow fault offset or creep deformation to be accommodated by the casing and void space (air or crushable material) between casing and product utility; and
- Tie-in valving, to be outside the limits of possible fault movement to allow for quick temporary overland bypassing of fault-offset damaged utilities.

In areas prone to seismic events, the engineer should consider the need for ground improvement and the possibility that the project could be affected and delayed if an earthquake occurs during construction.

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CHAPTER 12

MICROTUNNELING EQUIPMENT

12.1 GENERAL

A discussion of microtunneling equipment in the context of the design aspects of a microtunneling project is applicable only to the extent that the engineer must set forth in the contract documents the minimum capabilities of the equipment and the performance of the equipment. The actual selection of the specific equipment should be the responsibility of the contractor, subject to the established minimum requirements and review during the submittal process. The following sections present discussions of some of the basic elements of typical microtunneling equipment as well as considerations for establishing minimum equipment capabilities and performance requirements.

Commentary. None.

12.2 MTBM

The microtunneling boring machine (MTBM) should meet the following minimum capabilities and performance requirements:

- It must be capable of providing positive face support (both earth and water pressure).
- It must be articulated in such a way as to enable controlled steering in both the vertical and the horizontal direction, to a tolerance of plus or minus 3 percent of the MTBM diameter on grade or 25 mm (1 in.), whichever is greater, and 6 percent of the MTBM diameter on line or 50 mm (2 in.), whichever is greater, and from the design grade and line.
- All MTBM functions must be controlled remotely, typically from a surface control unit.
- The MTBM's rotation must be capable of being controlled. The cutter wheel must be capable of bidirectional drive. Antiroll fins or grippers may also be installed, but not as the primary method of rotation control.
- It must be possible to inject lubricant around the exterior of the jacked pipe at the back of the MTBM.
- With proper operation of the MTBM, it must be possible to control heave and settlement to acceptable tolerances, as indicated in the contract documents.

Commentary. There are two basic types of MTBMs, commonly referred to as *slurry-type* and *auger-type* (Fig. 12-1), which are distinguished by how a positive supporting pressure is maintained at the excavation face and how excavated material is conveyed from the MTBM to the jacking shaft.

In a slurry-type MTBM, the machine mixes the spoils with drilling fluid in a chamber located behind the cutter wheel to form a slurry. This pressurized slurry serves three purposes. It provides equalized earth and groundwater pressure, helps limit settlement, and conveys the cuttings back to the ground surface.

The slurry system can be thought of as a closed-loop system. Clean (processed by the separation plant) drilling fluid is piped from the ground surface to the MTBM, and slurry with cuttings is piped back to the ground surface. The cuttings are removed, and the slurry is recirculated back to the MTBM, without the need for routine personnel entry into the pipe being jacked.

In an auger-type MTBM, the spoils are conveyed from the MTBM to the jacking shaft by a continuous-flight auger enclosed in a separate casing inside the pipe being jacked. Positive pressure is maintained at the excavated face by regulating the volume of spoils removed relative to the advance rate of the MTBM. Auger-type machines are still used by contractors who purchased them years ago. It is possible for these types of machines to be proposed and used on a project. For the most part, the pilot tube method has replaced auger-type MTBMs. Projects involving pipeline installations of 1,200 mm (48 in.) and smaller at or near the groundwater table typically use the pilot tube method or auger boring with a guide bore or pilot tube.

For all the methods noted here, spoils are removed without the need for routine personnel entry into the pipe being jacked.

The applicability of a specific type of MTBM (slurry or auger) is dependent on ground conditions, including such factors as groundwater levels, ground permeability, whether the soil to be excavated is cohesive or granular, and such other factors as drive length and pipe diameter. Fig. 12-1 shows the typical thought process used to determine whether an auger-pilot tube or slurry machine is to be used. A short drive is defined as covering any distance less than 166 m (500 ft). This distance is based on general industry practice and may vary according to the project conditions, the pipe diameter, the equipment, and the contractor's ability.

The information provided in Table 12-1 represents the capabilities of various manufacturers' current equipment for the various ground conditions.

For some projects, only one type of MTBM may be specified; for many projects, however, it may be appropriate to allow either type of MTBM. Where either type is allowed, information about anticipated ground behavior should be provided to the contractors in the contract documents so they can make the equipment selection, with the key performance requirement being that the MTBM, regardless of its type, must be capable of providing positive face support.

Under mixed-face or other adverse ground conditions, any line or grade tolerances may not be achievable. On some projects these tolerances may not be needed, or they can be relaxed, at least in the horizontal direction. However, relaxing the tolerance too much may result in poor steering, which can lead to broken pipe and increased jacking loads. The engineer should use good judgment, knowledge of the ground conditions, and project requirements when establishing these tolerances.

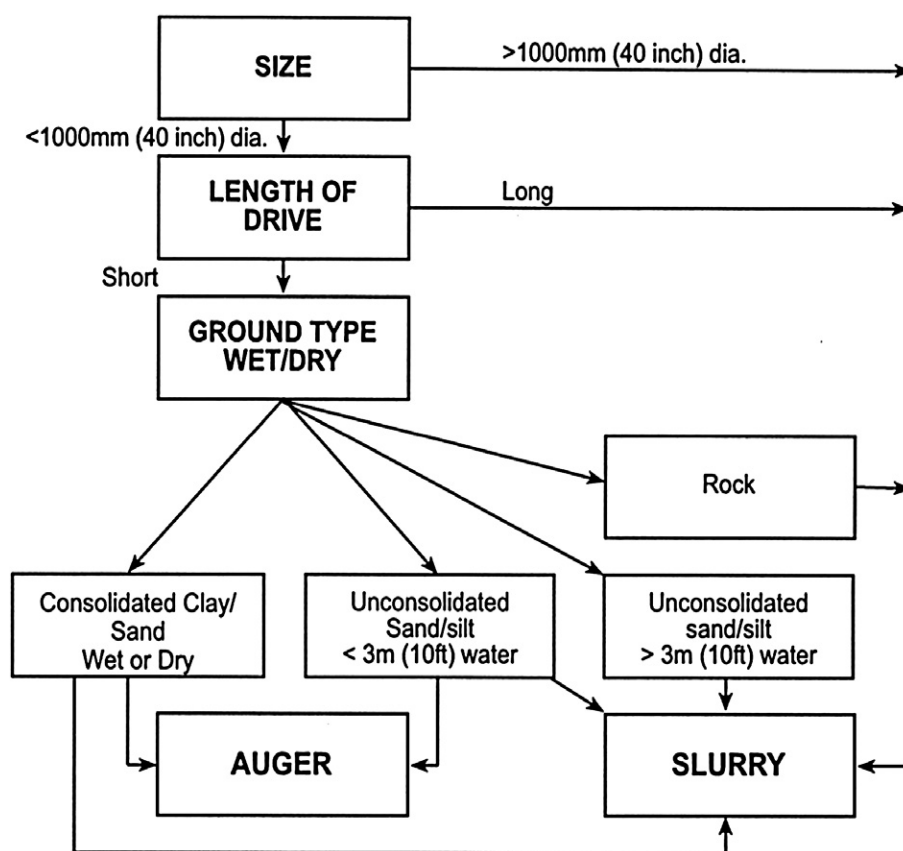


FIGURE 12-1. Slurry-type and auger-type MTBMs

Table 12-1. Microtunneling Capabilities for Different Ground Conditions

Ground Conditions	Microtunneling	
	Slurry	Auger
Soft to very soft clays, silts, and organic deposits	Y	M
Medium to very stiff clays and silts	Y	Y
Hard clays and highly weathered shales	Y	Y
Very loose to loose sands below water table	Y	N
Very loose to loose sands above water table	Y	Y
Medium to dense sands below the water table	Y	N
Medium to dense sands above the water table	Y	Y
Gravels and cobbles less than 150 mm (6-in.) in diameter	Y	Y
Soils with significant cobbles, boulders, and objects larger than 150 mm (6-in.) in diameter	M	M
Weathered rocks, marls, chalks, and firmly cemented soils	Y	Y
Slightly weathered to unweathered rocks	M	N

Y = Yes. Generally suitable for experienced contractor using appropriate equipment.

M = Marginal. Difficulties may occur, and some modifications of equipment or procedures may be required.

N = No. Substantial problems; generally unsuitable or not intended for these conditions.

The engineer should consider using a two-pass approach when the design grade and line tolerances must be less than 3 percent of the MTBM diameter or 25 mm (1 in.) on grade, or 6 percent of the MTBM diameter or 50 mm (2 in.) on line. The two-pass approach allows for the installation of a larger diameter casing. The larger diameter casing is installed with more relaxed tolerances, consistent with ground conditions and MTBM size. Once the larger casing is installed, the final pipeline can be inserted and adjusted to meet the tighter line and grade tolerances required for the project.

Drive lengths affect achievable accuracy. Drive lengths greater than 183 m (600 ft) will require one or more of the following measures: (1) careful use of the laser, (2) additional physical survey checks, (3) use of advance guidance systems, and (4) relaxation of the specified line and grade tolerances. See Section 19.5 for a discussion on the use of advanced guidance systems.

Some specifications have required certification from MTBM manufacturers that the equipment is capable of satisfactory machine performance on the project. This requirement can be impossible to obtain for the following reasons:

- Contractors have purchased MTBMs from one or multiple manufacturers. Once the MTBM is in the contractor's possession, the machine manufacturers have no record of the use, modifications made, or maintenance done to the MTBM. It is impossible for the machine manufacturers to certify the MTBM that is no longer in their control.
- The second major issue is the history of the microtunneling industry and machine manufacturers. Many of the MTBMs

are manufactured overseas and to obtain certification requires reshipment back to the home country resulting in delays to the project. Many of the MTBMs were manufactured by a company no longer in business. There is no manufacturer to certify the MTBM. Asking for MTBM certification is impossible and unobtainable.

Adding the requirement of MTBM certification has been abused by some in an attempt to shed risk on a project. Manufacturers are generally unwilling to certify machine performance. Therefore, this requirement is impractical and should not be included in the specifications. Manufacturers also do not certify repairs and refurbishment of an MTBM. These types of requirements should not be included in the specifications.

Belly pans are used to increase a portion the MTBM's outside diameter (OD). Belly pans provide a nonuniform increase in the OD. A belly pan normally increases the bottom 120 to 180 degrees of the OD. The remaining portion has a different OD. The belly pan is shorter than the MTBM steering head or the MTBM body to which it is attached. This type of upsizing provides a protrusion and does not place the entire MTBM body close to the excavated ground. Upsizing with a nonuniform design may lead to difficult steering, steering failure, or, in some cases, stoppage of the MTBM's advance.

The machine manufacturer should provide guidance on the upsizing capability of the individual MTBM, and the upsizing should be based on the project requirements. For the smallest MTBMs, upsizing is typically limited to the next manufactured outside diameter. The larger MTBMs may be capable of being upsized by two or three manufactured outside diameters. Upsizing does not change the overall capability of the original MTBM. Upsizing does not increase the volume of the internal crusher or its ability to ingest and process larger material like cobbles and boulders. The body and the upsizing should be designed to be constant, uniform, and free of protrusions.

To affect steering, the MTBM's body moves opposite to the desired steering direction (steering head) and allows the steering head to penetrate the face in the new direction. The MTBM's body side wall presses against the ground as the equal and opposite force to the steering action. The area without a belly pan is permitted to move further, increasing the amount of steering angle required to create an equal steering action. The space between the MTBM and the excavated ground is subject to infilling by the ground, slurry, and lubricant. The steering joint should be designed as self-cleaning, allowing material that accumulates in the steering joint to be pushed out of the pinch point when steering is activated. If the steering joint is not self-cleaning, the MTBM steering joint may become packed with material until the MTBM loses all steering ability.

The term *plowing* refers to when the MTBM is articulated to steer in one direction but the MTBM continues in an undesired direction. Plowing can occur during vertical and horizontal steering. Plowing may be caused by excessive gauge overcut. Plowing can also occur in very soft or very loose soils when the MTBM is advancing at a maximum speed or penetration rate. The inability of the MTBM to react to steering is related to insufficient soil resistance causing the MTBM to go in an undesired direction, thereby rendering the steering mechanism ineffective. Adding a longer articulated trailing can to the MTBM may provide a more effective steering system.

12.3 CONTROL SYSTEM

The main control system is used to remotely control the micro-tunneling operation and should provide the following minimum

required information to the operator for the successful operation of the MTBM:

- Deviation of the MTBM from the required line and grade of the pipeline, normally by reference to a laser beam;
- Pitch, roll, yaw, and drift of the MTBM (see Fig. 12-2);
- Jacking forces from main jacks and intermediate jacking stations (IJSs);
- Torque and revolutions per minute (rpm) of the cutter wheel;
- Jacking rate and total distance jacked;
- Measurement of steering cylinder position and pressure;
- Volume of slurry flow in both the supply and the return sides of the slurry loop;
- Indication of slurry bypass valve position; and
- Measurement of the slurry pressure in the slurry chamber.

Commentary. The main control system from which the micro-tunneling operations are conducted is normally containerized and located close to the jacking shaft. This equipment provides the remote control for the MTBM, spoil removal, and jacking system and may also include automated data collection.

Other parameters may be provided to help with the operation of the equipment, but the above should be required. Depending on advance rate and ground conditions, the above parameters should be logged in real time and over short distances, either manually or by a computer system, to provide a complete record of each drive conducted.

Deviation: The deviation is the distance from design line and grade, as measured from the reference. If the reference is incorrectly set, then the resulting installation is likely to be out of tolerance or not as designed.

Pitch: Pitch is the measurement, in degrees, of the inclination or declination of the MTBM as measured at the point of instrumentation to a reference. The pitch is not an absolute pitch as measured to horizontal but as set to a reference, which is manually controlled. Most if not all MTBMs measure pitch within the MTBM body and not the steering head, since the body is a better indicator of where the pipeline is to be installed. The pitch must be read in conjunction with roll.

Roll: Roll is the rotation of the MTBM in a clockwise or counterclockwise manner, which can be measured in degrees. Roll is an absolute measurement, because it is measured against "no roll." Common controls for preventing roll are the use of a bidirectional cutter wheel, computer software that limits the amount of roll, antiroll fins in soil, and grippers or jacks in rock. One measure for reversing roll is to reverse the direction of the cutter wheel. Common sources of roll include normal operation and high-torque events like encountering a buried object.

Yaw: Yaw is left or right rotation within the horizontal plane (i.e., turning left or right). Yaw is the angle of steering left and right and can be measured from the MTBM body or the guidance reference (i.e., the laser).

Drift: Drift is the difference between the absolute calculation and the actual change in position. Plowing would be steering in one direction with the MTBM advancing in an undesired direction, where the drift equals the loss in effective steering.

Jacking forces from main jacks and IJSs: The main jacks are installed in the jacking shaft and provide the thrust to advance the MTBM. With small MTBMs (i.e., those with an inside diameter (ID) of less than 760 mm (30 in.), where personnel entry is either impossible or very undesirable), IJSs are not added to extend the jacking distance. The IJS requires personnel entry to remove the operational components of the IJS and allow the pipe string to be closed. The jacking distance is limited by the main

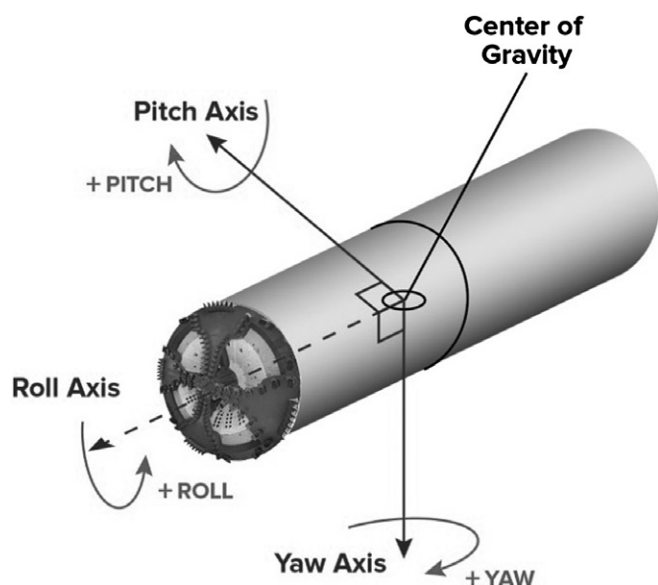


FIGURE 12-2. Pitch, roll, and yaw of MTBM measured from machine body or guidance reference

jacks' jacking capacity and by the ability of the jacking pipe to accept those forces, less some factors of safety. IJSs are added into the pipe string to increase the jacking distance. The IJSs are added to the pipe string as the frictional load increases, thereby allowing for long jacking distances. See Section 17.5 for more details.

Torque and rpm of the cutter wheel: Torque on an MTBM is typically measured with an indirect measurement. Torque spikes along with cutter wheel rotation can help locate the position of a mixed face. MTBMs with an electric drive are normally measured in amperes or percentage of amperes. MTBMs with hydraulic drives are normally measured in hydraulic pressure.

There are two types of MTBMs, those with fixed rpm and those with variable rpm. Torque is a function of rpm, and therefore both torque and rpm need to be measured and recorded. A freewheel measurement of torque needs to be recorded at the start of each drive, and if the MTBM can be run at variable rpm, then it needs to be recorded at several different rpms.

Jacking rate and total distance jacked: Jacking rate, along with torque spikes and additional information, can help demonstrate the encounter with a buried object. Total jacked distance helps keep the operator informed as to the actual location of the MTBM and the location of said encounter.

Measurement of steering cylinder position and pressure: The measurement of the steering cylinder's extension or retraction helps in understanding the amount of drift and the amount of steering action required to create steering reaction. The steering cylinder pressure helps in estimating the face pressure and the location of face pressure in a mixed face.

Volume of slurry flow in both the supply and the return sides of the slurry loop: Measuring the slurry volume in both the feed and return sides allows for estimating solids removal volume and balanced slurry flow.

Indication of slurry bypass valve position: This indication ensures that the bypass valve is closed, preventing flooding of the MTBM during pipe changeovers. This control can also be used during special operating conditions, to prevent over excavation.

Measurement of the slurry pressure in the slurry chamber: The measurement of slurry pressure during shutdowns allows for the direct measurement of the existing hydrostatic pressure, thereby allowing for an operation that is balanced with slurry pressure. The location of the shutoff valve as it relates to the slurry line pressure is important for balanced operation. If the pressure sensor is located high in relation to the slurry feed line, then the elevation differential needs to be calculated and used to create a balanced operation.

12.4 HANDLING OF SPOILS

The type of spoils handling should be limited in the contract documents only if job conditions so dictate.

Commentary. The two types of MTBMs (slurry and auger) have vastly different requirements relative to issues of spoils handling. For auger-type systems, handling spoils is relatively straightforward. Spoils are conveyed to the jacking shaft through the continuous-flight augers, where the spoils are typically deposited into skips, which are then lifted out of the shaft by crane. The spoils are then deposited directly into trucks for disposal, or they are temporarily stockpiled at the shaft site.

Slurry-type MTBMs can require a complex system of spoils handling, depending on the size of the MTBM and the type of ground conditions to be encountered. The system consists of feed and return pumps and a means of separating solids from the slurry. The slurry separation system can vary from a simple sedimentation tank or pond, where the solids are allowed to separate from the slurry by gravity and are removed by backhoe, to a complex system of mechanical separation methods, such as those that use shaker screens, sand cones, hydrocyclones, centrifuges, and vertical clarifiers.

The appropriate method of slurry separation for a specific project is a function of many factors, including but not necessarily limited to the following:

- Diameter of the pipe being jacked;
- Grain size and plasticity of the spoils;
- Depth and length of the drives; and
- Space constraints at shaft locations.

Selection of the specific means and methods of slurry separation should be left to the contractor, with the contract documents providing the information necessary for the contractor to select them. This information should include the anticipated range of the grain size and plasticity for the soils to be excavated as well as any special constraints, such as local regulatory requirements relative to the disposal of excavated material and spent or used slurry.

See the commentary in Section 20.5 for additional information regarding the slurry separation plant.

12.5 JACKING SYSTEM

The specific type and capacity of the jacking system should be selected by the contractor, with the contract documents providing constraints (if any) on jacking shaft size and pipe segment length.

The jacking system must have the capability to push the MTBM and pipe through the ground in a controlled manner. The system must apply a uniform force around the pipe with the use of a push ring specifically designed for the pipe being used. The jacking capacity of the system, including IJSs, must be greater than the anticipated jacking forces, with a reasonable factor of

safety. The use of IJSs can reduce the jacking force requirements of the jacks in the jacking shaft and/or allow longer pipe drives.

Commentary. Intermediate jacking stations are not normally installed on pipe smaller than approximately 900 mm (36 in.) in diameter, primarily because of space limitations for entry of personnel who are required to remove the jacks.

It is usually desirable to require the jacking system to have an automatic shutoff, to prevent overstressing of the pipe being jacked or to address situations where the jacking system exceeds the design capacity of the thrust wall. The vast majority of jacking systems use hydraulic jacks, and there are many variations relative to the specific layout of the jacking system. The layout is primarily a function of the length of the pipe segments being jacked and the size of the jacking shaft.

12.6 PIPE LUBRICATION SYSTEM

The ability to apply lubrication to the pipe on a routine basis to reduce jacking forces should be a requirement for the microtunneling system.

Commentary. Pipe lubrication systems typically consist of a mixing tank, a holding tank, and pump(s) to convey lubricant from the holding tank to application points at the rear of the MTBM. For pipes with an ID greater than about 900 mm (36 in.), one or more lubrication ports should be provided at a minimum as frequently as every third pipe, as well as at the back of the MTBM.

Ideally, the whole length of the pipeline should get adequate lubrication. Operation of the valves at the lubrication ports, to ensure that bentonite starvation does not occur, can be done

manually, where doing so is safe and where practical access can be gained, or it can be done by an automatic system operated remotely from the operator's control cabin. On long microtunneling drives, the preference is normally to use an automatic system.

The lubricant itself typically consists of a mixture of bentonite and water, but polymer lubricants are also often used. See Section 19.6 for more discussion on the lubrication design. Lubricant selection should be at the option of the contractor and should be selected on the basis of an evaluation of the ground conditions and groundwater (e.g., the presence of freshwater versus seawater). Any additives used should be NSF International Standard 60 approved. Though the option to lubricate is the contractor's decision, there are clear advantages to using a lubricant.

12.7 POWER SUPPLY

Available power sources and noise constraints should be identified in the contract documents.

Commentary. The power supply systems for microtunneling equipment typically consist of generator sets and hydraulic power packs (driven by electric motors or internal combustion engines), depending on the type of equipment used. It is also possible to draw electric power from local sources, if available. From a design standpoint, it is important to identify noise limits on power generation equipment and constraints on drawing power from local sources in the contract documents. The worst offender for noise is usually the diesel engine of the crane. Electric gantry cranes should be considered in noise-sensitive areas.

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CHAPTER 13

MICROTUNNELING DESIGN ELEMENTS

The design elements and the process for a microtunneling project are generally the same as for other underground projects. Design elements include the following:

- General conditions;
- Special or supplementary conditions;
- Technical specifications;
- Plans;
- Geotechnical investigation and reports;
- Design memoranda and technical memoranda;
- Design calculations and analyses;
- Permit applications; and
- Utility notifications and utility investigations.

The design is often completed in stages or phases that may include the following:

- Feasibility evaluation;
- Preliminary design;
- Final design;
- Bid services; and
- Design support during construction.

Each of these elements is well understood in the design community.

Commentary. Contract documents consist of the general conditions, special or supplementary conditions, technical specifications, plans, and geotechnical investigations and reports. See Chapter 15 for a discussion of these contract document elements. The following sections discuss the specific elements to be considered during preliminary and final design.

13.1 DESIGN MEMORANDA AND TECHNICAL MEMORANDA

Design memoranda (DMs) and technical memoranda (TMs) may be prepared during preliminary design to explain the following:

- Alternative alignments and construction methods under consideration;
- Hydraulic requirements;
- Potential environmental and permitting issues that must be addressed;
- Rationale for the design;
- Identified risks, including those related to settlement and jacking forces;
- Capture modeling and calculation results;
- Evaluation of pipe material options;
- Investigations of corrosion protection requirements; and
- Other issues specific to the project.

Design memoranda and technical memoranda help guide the final design effort. These memoranda are used to develop the

design drawings and specifications provided to all bidders on the project. DMs and TMs are not typically distributed to the bidders.

Commentary. None.

13.2 PIPE MATERIAL SELECTION

Numerous pipe materials are available for casing and carrier jacking pipe. See Chapters 16, 17, and 18 and Section 21.4 for discussion of pipe material aspects. The design criteria should be summarized for each project. These criteria include flow requirements, minimum and maximum diameters, joint characteristics, and dimensional tolerances. The decision process and the results of the pipe material selection can be summarized in a stand-alone or combined design memorandum or technical memorandum. The information should be appended to a Preliminary Design Report.

Commentary. Normally on pipeline projects (particularly in open-cut construction), the engineer specifies the allowed pipe materials, including internal diameter, class(es), joint type(s), and lining, coating, or other additional features deemed appropriate, to ensure adequate performance for the installation, the in-use loads (see Section 17.1), and the service environment. Typically, the engineer can identify all loads and conditions influencing the pipes' performance. However, for microtunneling installations, jacking loads significantly affect pipe design. The following are some of the pipe design factors that affect jacking loads:

- Straightness of pipe;
- Squareness of pipe ends;
- Allowable angular deflection;
- Smoothness of outside surface;
- Outside diameter of pipe;
- Configuration of pipe joints;
- Circumferential uniformity;
- Roundness; and
- Compression rings.

Recommended dimensional criteria are provided in Section 17.3. The dimensional criteria affect, to a significant degree, the microtunnel contractor's choice of pipe material, equipment, overcut, lubrication, and drive distance, some of which, to varying degrees, are outside the engineer's normal control. It may also be more economical for contractors to minimize the number of machine sizes used on a contract. The engineer must be aware of these constructability issues when selecting pipe materials, pipeline routing, drive distances, pipeline grade, and dimensions. These issues should be considered concurrently with hydraulic design issues during the planning and design stages, to ensure a final design that is both cost-effective and

hydraulically efficient. Decisions by the contractor may still call for changes in pipe design. However, if consideration of constructability comes early in the project development, it may reduce the number of such changes needed and thus ensure a more efficient and cost-effective project.

Jacking pipes are not in all cases manufactured to standard pipe diameters. The engineer should provide the minimum pipe performance requirements and allowable size ranges. With this information, the contractor can select from a list of acceptable materials and diameters to match available equipment and provide an economical solution. Design criteria included in the contract documents should include the following:

- Flow requirements;
- Factors of safety;
- Minimal diameter; and
- Maximum diameter.

For a casing used in a two-pass installation, the casing diameter should be sufficient to ensure that the carrier pipe can be installed to design line and grade. Casing pipe diameter and installation tolerances should be compatible with the objective of installation of the carrier pipe to design line and grade.

13.3 EARTH LOADS ON PIPE

Selection of the appropriate pipe class or stiffness requires a reasonably conservative estimate of earth loads acting on the pipe. Computation of earth load is significantly different for pipe tunneled through undisturbed ground and for pipe within an excavation. Pipe installed by microtunneling experiences lower earth loads than pipe installed in open trenches, because of the ground's natural arching ability.

Commentary. Earth loads on pipe installed by microtunneling are proportional to the diameter of the pipe and effective unit weight of the ground. Earth loads are inversely proportional to ground stiffness. For pipe installed in medium stiff to hard clay, or in very dense sand, the earth loads acting on the pipe are typically less than the diameter of the pipe multiplied by the effective unit weight of the ground. For pipe installed in soft clay or in loose to medium sand, the earth loads acting on the pipe may approach two to three times the diameter of the pipe multiplied by the effective unit weight of the ground. Earth loads on microtunneled pipe in stable rock and hard clay are negligible. Total overburden stresses should not be used to calculate earth loads on pipes except for relatively shallow installations in soft, squeezing clays and in loose to very loose sands. A conservative approach to determining earth load on a tunneled conduit is the conventional Marston formula as modified to include soil cohesion.

Consult the following references for guidance on determining appropriate earth loads on pipe during the design process:

- ASCE MOP 60/WEF FD-5 (2nd Edition);
- Bennett (1998); and
- Bennett and Cording (2000).

Earth loads acting on pipe that protrudes into or across locations of planned or potential future cut-and-cover excavations are larger because of reduced arching action in cut-and-cover excavations. Such locations include jacking and receiving shafts, manholes, and service connections. The conventional and more commonly used Marston approach may be used to estimate earth loads acting on pipe in such locations. The axial jacking forces required for installation are typically the critical load case that controls pipe class or stiffness. The pipe manu-

facturer should be consulted with respect to estimates of earth loads acting on the pipe.

13.4 EVALUATION OF JACKING FORCES

Anticipated jacking forces must be evaluated in terms of the baseline and anticipated ground conditions (assuming good workmanship and good operational practices by the contractor) to ensure that the pipe and the thrust reaction wall are designed to safely resist the jacking forces and that the jacking system has adequate capacity. An industry-accepted approach should be used to evaluate jacking forces during design. The specifications must require contractor submittals confirming that the jacking system has adequate capacity and that the pipe and the thrust reaction wall have been designed to safely resist the anticipated jacking forces.

Commentary. The contractor must provide microtunneling equipment, pipes, gaskets, and connections that are capable of providing and resisting the calculated jacking forces to a degree that is consistent with the contractor's selected means and methods and with the anticipated ground conditions.

The criteria for determining jacking forces should take into account the baseline ground conditions provided in the Geotechnical Baseline Report (GBR) and encountered in the geotechnical investigation.

Jacking forces consist of two primary components: the face pressure component and the frictional resistance component. The jacking force can be expressed as

$$JF = F_p + \sum F_R$$

where

JF = total jacking force;

F_p = face pressure component; and

F_R = frictional resistance (acting in the direction of pipe axis).

The frictional resistance (F_R) component can be expressed as

$$F_R = \sigma'_n \mu' A_c L$$

where

σ'_n = effective normal stress acting on pipe;

μ' = effective friction factor acting on pipe;

A_c = circumferential area of pipe OD; and

L = jacking distance.

The range of the anticipated face pressure component may be estimated as the product of the cross-sectional area of the MTBM head and the sum of the groundwater pressure and effective earth pressures. The actual face pressure is typically slightly greater than the combination of groundwater and active earth pressure acting on the head. If the actual face pressure falls significantly below the combination of groundwater and active earth pressure in unstable soils, ground losses at the face may occur, resulting in large settlements. If the actual face pressure significantly exceeds the combination of groundwater and passive earth pressure, heave may occur. Though heave is uncommon, it has been observed as a result of excessive face pressure on shallow, large-diameter microtunneling drives. Face pressure is typically a small component of the overall jacking force. However, face pressure may be the dominant component on large-diameter, short drives.

The jacking forces should be determined with the use of industry-recognized rational methods. Jacking force calculations require the selection of appropriate geotechnical properties on

the basis of judgment and the results of the geotechnical investigation.

The effective normal stress acting on the pipe may be evaluated with arching theory and is proportional to the pipe's outer diameter and ground conditions. Calculations of effective normal stress using total overburden significantly overestimate the normal stresses acting on the MTBM and the pipe except in relatively shallow installations (i.e., <2 diameters of earth cover).

The effective friction factor may be estimated as the tangent of the residual soil friction angle, reduced to account for the beneficial effects of lubrication, if used. Clay soils, under short-term undrained conditions, are often assumed to behave as purely cohesive materials (i.e., with a friction angle of zero). However, some drainage occurs along the interface as the drive progresses, resulting in frictional behavior.

For drives in very stiff to hard clay and stable rock, effective normal stresses acting on the pipe may be very small. The frictional component in these cases may be more accurately estimated as the buoyant weight of the pipe multiplied by the effective friction factor between the pipe and the ground.

A variety of factors affect jacking forces, including

- Soil type (clay, silt, sand, gravel);
- Soil consistency, which may be inferred from Standard Penetration Test (N) values;
- Effective normal stress acting on the pipe and the MTBM;
- Effective friction factor along the jacked pipeline;
- Lubrication;
- Overcut;
- Face pressures caused by earth and hydrostatic pressures;
- Steering of the MTBM and the pipe string;
- Pipe surface's roughness, dimensional consistency, surface porosity, and absorption; and
- Deflections at pipe joints.

Some of these factors are not explicitly taken into account when calculating jacking forces using available methods. The key to reasonable accuracy is judgment in selection of appropriate ground properties on the basis of a thorough geotechnical investigation and a fundamental understanding of the geotechnical characteristics that influence or control behavior.

Operational factors can exert significant influence on the estimation of jacking forces but are difficult to quantify. For example, if excessive drilling fluid pressure is used, fines may be forced around the cutter wheel and into the overcut, which significantly increase jacking forces. When microtunneling takes place in rock, rock flour can be forced into the overcut, significantly increasing jacking forces (and torque) if the cutter wheel muck scoops do not efficiently clean the invert. If lubrication is not used or is used ineffectively, jacking forces can significantly increase. These and other operational factors are typically not within the engineer's control. Therefore, the engineer making the jacking force calculations typically assumes proper equipment setup and good workmanship. All assumptions should be clearly stated when these and all other design calculations are made.

Methods of calculating jacking forces are provided in the following references:

- Bennett (1998);
- Bennett and Cording (2000);
- Pipe Jacking Association (1995b);
- Thomson (1993);
- Stein (2005); and
- Najafi (2013).

During microtunneling under high groundwater heads, the hydrostatic pressure on the face of the MTBM may push the

MTBM and the jacked pipe back into the shaft when the jacks are released during the initial stages of the drive. A pipe brake or clamp may be needed to resist the face pressures on the MTBM and prevent inward movement of the pipe string. This scenario should be checked when the jacking forces are calculated.

Any time high jacking forces are encountered, pipe failure may occur. Any misalignment of the jacking pipe may reduce the ultimate load where pipe failure occurs because of point loads.

The introduction of a curve into the microtunnel alignment increases the frictional resistance component of the jacking force. As the pipe enters and advances through the curve, the surface contact pressure of the pipe to the ground increases. The pipe also begins to plow and to compact the surrounding formation, which increases frictional forces.

13.5 EVALUATION OF SETTLEMENT RISKS

Settlement calculations should be undertaken and evaluated by the engineer. Existing utilities, roadways, railroads, foundations, structures, and other facilities must be protected against damage from ground movements resulting from microtunneling and shaft construction. The steps include the following:

- Identify facilities at risk for settlement.
- Establish recommended maximum allowable settlements for critical features through consultation with the facility owner.
- Evaluate potential settlements at each feature.
- Compare estimated settlements against recommended maximum allowable settlements for each feature, and develop mitigation measures to reduce effects on significant structures, if necessary.
- Develop a settlement instrumentation and monitoring plan for the project, and ensure that the plan is implemented during construction.
- Coordinate with permitting agencies and owners of critical facilities, a critical step in successful implementation of a proactive program to avoid damages that may occur as a result of microtunneling and shaft construction.

Commentary. The risk of ground movements (both vertical and horizontal) and the potential effect on the critical features should be evaluated. An acceptable approach is to identify sensitive and critical utilities, roadways, railroads, foundations, and other facilities that may be at risk from the microtunneling and shaft construction activities. It is a good idea to establish recommended maximum allowable settlements for critical facilities in coordination with facility owners. Suggested guideline settlement values for various facilities have been published (Wallin et al. 2008), but these values should be used as general guidance only and refined through coordination with facility owners. Potential settlements should be evaluated at those critical features. The calculated settlements should be compared with the recommended maximum allowable values, and the risks associated with the predicted ground movements should be evaluated. Mitigation measures should be developed, if necessary. An instrumentation and monitoring plan to protect critical facilities should be developed. The plan should be followed during construction. If the evaluation results indicate high risks of damage, it may be necessary to relocate certain utilities or protect them in place.

Ground movement associated with microtunneling can be divided into two types: large ground movements at isolated locations and systematic settlements along the pipeline align-

ment. Large ground movements are typically the result of over excavation at the face, which can occur because of improper setup of the MTBM and also because of rapid, unexpected changes in ground conditions (such as encounters with loose sand or hard objects), mixed-face conditions, artesian groundwater pressures, operator error, or some combination of these factors. Large ground movements must be avoided through proper setup of the MTBM, a thorough geotechnical investigation, and the efforts of a diligent, experienced, skilled operator. The importance of a highly skilled operator and a thorough geotechnical investigation cannot be overemphasized.

Systemic settlements are those settlements associated with closure of the overcut and the shield cut as the ground moves into the annulus. Additional systemic settlements associated with minor face losses may also occur. The risks of systemic ground movement associated with microtunneling should be evaluated with the approach first proposed by Peck (1969) and Schmidt (1969) for open shield tunneling, refined by Cording and Hansmire (1975), and adapted to microtunneling by Bennett (1998) and Bennett and Cording (2000). This approach has been described and extended to other trenchless construction methods by Bennett et al. (2008). The approach approximates the settlement trough above a microtunnel as an inverted normal probability distribution curve. Representative soil properties must be selected using judgment combined with the results of the geotechnical investigation. Systemic settlements are proportional to annular space and pipe outer diameter, and they are inversely proportional to soil stiffness, depth of earth cover, and offset distance from centerline of the bore. Calculated settlements increase as pipe diameter, shield cut, and overcut increase, and they decrease as soil stiffness, depth, and offset distance increase.

The first step in evaluating anticipated settlements is to calculate the annular space as the difference between the volume of the excavated tunnel and the pipe volume. The next step is to select the proportional amount of the annular space that is likely to contribute to the settlement trough volume. For hard clay and stable rock, the proportion of annular space that contributes to the settlement trough volume may be zero; for squeezing clay, very loose sand, or nonengineered fill debris, it may be 100 percent or even higher. For many soils encountered on typical microtunneling projects, such as medium to stiff clay and medium to dense sand, the settlement trough volume may be 25 to 75 percent of the annular space.

After the evaluation of anticipated settlements and the comparison to recommended maximum allowable settlements have been completed, unacceptable risks may exist for certain features, such as overlying utilities with insufficient clearance. Mitigation measures may be appropriate at these locations. Mitigation measures may include deepening the alignment, if possible, to ensure sufficient clearance between the crown of the microtunneling bore and the feature of interest. If sufficient clearance cannot be provided, other measures must be taken. Protective measures may include exposing and protecting the feature in place, grouting the annulus, reducing the shield cut, ground improvement by jet grouting or other means, or relocation of the feature or utility.

Ground movement risk associated with shaft construction can be evaluated with the approaches indicated in Clough and O'Rourke (1990).

The results of the ground movement calculations, the criteria for allowable limits, and the instrumentation and monitoring plan can be summarized in a technical memorandum during preliminary design.

During construction, an appropriate instrumentation and monitoring plan should be used to monitor critical utilities and fea-

tures. A formal instrumentation and monitoring plan may be appropriate for higher risk projects. The plan should specify the critical features, the locations and types of instruments, the monitoring frequencies and durations, and the maximum allowable and threshold values for settlements. The maximum allowable values should be less than the values that could result in damage to the feature or utility. Recent developments in technology have reduced the need for repeated personnel access to monitoring points, which may require lane closures or other disruptions. Examples include photogrammetric settlement surveys and laser profiling.

A preconstruction condition survey must be conducted before construction activities begin, to establish conditions before construction and protect against frivolous damage claims. Surveys should be continued during construction at established intervals, and for some time after construction is completed, until no additional movements are recorded between at least two successive surveys.

The contract documents should require that all settlement survey results be submitted to the engineer within a reasonably short time, such as 24 hours after the survey is conducted, so that the results can be reviewed and any required actions can be implemented in a timely manner.

13.6 DRILLING FLUID DESIGN

The drilling fluid should be designed to accomplish the following objectives:

- Counterbalancing of the earth and groundwater pressures at the face of the MTBM;
- Mixing of the drilling fluid with excavated material (slurry), and transporting the mixture to the surface for separation;
- Transport of the slurry to the separation plant without excessive pressures or velocity;
- Limits on over excavation; and
- Allowance for efficient excavation.

Commentary. Drilling fluid is water that may contain bentonite, polymers, or other additives to enhance its desirable engineering properties and aid in the separation of solids.

Bentonite is made from montmorillonite, which is a naturally occurring clay mineral. Bentonite improves microtunneling through noncohesive ground conditions. Montmorillonite is a sheetlike crystalline mineral with a large surface area and a small edge area. Sodium (Na^+) montmorillonite is characterized by its high capacity for water adsorption and swelling and by its ability to produce high viscosity and gelling when in suspension. When water is added to a mass of montmorillonite, the individual clay crystals move further and further apart as they adsorb the water molecules onto their surfaces. If sufficient water is added, the clay crystals may completely disassociate from each other. This disassociation is what occurs when sodium bentonite is added to water to form a suspension, which becomes the drilling fluid for the microtunneling operation.

Before the bentonite is added, soda ash is typically added to the water, to raise the pH to approximately 9 to 9.5. This step increases the yield (hydration) of the bentonite. The order of addition of the additives is water, soda ash, bentonite, polymers, and other additives, such as surfactants. The suppliers should be consulted when designing a drilling fluid.

The drilling fluid can be characterized by the following:

- Viscosity;
- Sand content;
- Density;

- Gel strength;
- Yield point; and
- Filtrate loss.

Viscosity is defined as resistance to flow. Field measurement of viscosity can be done quickly and simply with a Marsh funnel test (ASTM D6910). The Marsh funnel viscosity of water is 26 seconds per quart (s/qt) at sea level and 73°F. A drilling fluid typically starts with a Marsh funnel viscosity of 45 to 55 s/qt. Higher drilling fluid viscosities may be appropriate for poorly graded sands or gravelly soils. A high viscosity value may also result from incomplete separation of solids, which can lead to increased abrasion and wear of equipment.

Sand content is the amount of material retained on a No. 200 mesh sieve (2.5-in. diameter). It is expressed as a percentage. Sand content can be measured in the drilling fluid and slurry. The sand content of the drilling fluid should be kept below 2.5 percent. Sand content has a direct effect on the density of the slurry. Increases in sand content are typically caused by worn, improperly installed, or incorrectly sized screens. Lack of or worn desanding cones could also cause an increase in sand content. Sand content should be measured per ASTM D4381.

Density is the weight of the drilling fluid or slurry. Pump pressure and production rates decrease as the drilling fluid and slurry densities increase. Once the drilling fluid reaches 10 pounds per gallon (1.2 specific gravity), it should be diluted or removed from the circulation system. Water weighs 8.3 pounds per gallon (1.0 specific gravity). Density of the drilling fluid and slurry is tested per ASTM D4380.

Gel strength and yield point are infrequently determined for microtunneling. These properties are determined with a rotational viscometer, or V-G meter. This test instrument allows the viscosity of the fluid to be measured at a variety of shear rates. The most typical shear rates used are 600 and 300 rpm. Gel strength and yield point are used to determine the crystal-to-crystal interaction in static and dynamic conditions, respectively. Gel strength is a measure of the gelling or thixotropic property of the bentonite in a water-based suspension (drilling fluid or slurry) under static conditions. The gel strength of freshwater slurry increases with time after agitation has ceased. The mechanisms of gelation include edge-to-face, edge-to-edge, and face-to-face association between crystals. Gel strengths are measured at two separate intervals: 10 seconds and 10 minutes. Gels that increase dramatically over time are termed *progressive*. Gel strengths are determined by the following procedure:

1. Place a sample in the container of the V-G meter and immerse the rotor sleeve exactly to the scribed line.
2. Stir the sample on high speed for 10 seconds.
3. Allow the sample to sit undisturbed for 10 seconds. Start the V-G meter at the 3-rpm speed. The maximum reading is the initial gel strength (10-second gel) in pounds per 100 square feet.
4. Stir the same sample again for 10 seconds on high speed.
5. Allow the sample to sit undisturbed for 10 minutes. Start the V-G meter at the 3-rpm speed. The maximum reading is the 10-minute gel strength in pounds per 100 square feet.

Yield point is a measurement of the electrochemical forces in a drilling fluid or slurry under flow conditions. Yield point is dependent upon surface properties of the slurry solids, volume concentration of the solids, and the electrical environment of these solids. The yield point is calculated by first subtracting the 300-rpm reading from the 600-rpm reading to provide a result known as the *plastic viscosity*. The plastic viscosity value is then

subtracted from the 300-rpm reading to obtain the yield point value.

Filtrate loss is the amount of water that is lost from a drilling fluid to the surrounding formation. When water is lost from a bentonite-based fluid, the bentonite solids build up on the formation into which the water is moving. The layered bentonite solid that is formed is called a *filter cake*. For a filter cake to form, it is essential that the drilling fluid contain some particles of a size only slightly smaller than that of the pore openings of the formation. These particles (bridging particles) are trapped in the surface pores, and the finer particles are, at first, carried deeper into the formation. The bridge zone in the surface pores begins to trap successively smaller particles quickly until, usually in a few seconds, only water invades the formation. The water that enters the formation is known as *filtrate*. An American Petroleum Institute (API) fluid loss test consists of applying 100 psi of air pressure to a fluid sample in a standardized test receptacle. A disc of filter paper is first placed at the bottom of the test receptacle. This paper is used to allow the fluid to form a filter cake. In the standard field test, the air pressure is applied to the test fluid in the test receptacle for 30 minutes. The amount of water lost from the receptacle during that period is collected and measured to the nearest milliliter or cubic centimeter. Upon completion of the test, the filter paper and the filter cake are removed from the receptacle, measured for thickness, usually in terms of 32nds of an inch, and palpated for consistency. Subjective terms such as *thin*, *thick*, *soft*, *mushy*, *firm*, and *tough* are used to physically describe the filter cake. A desirable filter cake is described as thin (1 to 1.5 of a 32nd of an inch), tough, and slick.

The drilling fluid used in the microtunneling system should be engineered to the specific ground conditions encountered during the mining process. Testing should be undertaken to monitor the drilling fluid and slurry system. Typically, testing includes testing the drilling fluid being sent to the cutting head and testing the slurry returned to the separation plant. Testing of the drilling fluid and the slurry is outlined in Section 19.4.

In all ground conditions, a specific drilling fluid plan should be developed before mining. This plan includes benchmarks for density, viscosity, sand content, and pH as well as plans for resolving issues related to any concerns discovered by testing. In microtunneling through cohesionless soils with fines less than 15 to 20 percent, the use of an engineered drilling fluid can improve stand-up of the ground and should be considered.

In most projects, all of the materials added to the drilling fluid should be ANSI/NSF International 60 Clean Water approved. Soda ash is not ANSI/NSF International 60 Clean Water approved but may be required to control calcium and increase pH. Salt-water additives do not require ANSI/NSF International 60 Clean Water approval because saltwater is not potable water. Additives containing hydrocarbons should be avoided.

13.7 SPECIAL CONSIDERATIONS FOR CURVED DRIVES

Curved microtunneling may increase project risk and should be carefully considered and managed. The potential benefits should be carefully evaluated against the risks. Previous experience with the design and construction of curved drives is recommended.

Commentary. If an engineer needs to incorporate a curve into the alignment, simply showing a curve on the plan and profile sheets is not enough; see discussion in Section 9.2. The engineer

should incorporate the following requirements and design criteria into the contract documents:

- An initial tangent must be maintained for as long as possible before starting a curve (this initial tangent reduces the number of pipe segments that must go through the curve).
- The number of IJSs should be increased so that they can be used to push the pipe forward—into, through, and beyond the curve.
- Pipe diameters have to be large enough to accommodate personnel entry for survey checks.
- An appropriate guidance system is required. Traditional laser guidance systems are not appropriate. Skilled personnel are required for the guidance system's operation.
- The jacking pipe that is designed to progress through the curve(s) should be designed in its entirety. Such a design should include the allowable joint deflection, individual pipe segment length, pipe joint packing material, specials for the IJSs, and IJSs.
- The shield cut and overcut are larger with curve alignments and may increase settlement risk.
- Pipe joints, lengths, types, and reinforcements must be designed to accommodate the curve, transfer the jacking force, and maintain joint integrity.
- The MTBM's penetration rate is reduced as it mines through the curve. This lower penetration rate is required to ensure that the curve is built as it was designed.
- The MTBM and jacking pipe must be able to provide the designed radius.

- The guidance and steering system must be able to operate at the designed radius.
- Curved drives in hard and abrasive rock may not be economically feasible.

If curved microtunneling is to be incorporated, the specifications should require that one straight drive be completed first with the guidance and steering systems operating. The contractor must show that the end results are within tolerances before starting the curved drive. The contractor's plan, with design calculations, must be submitted to demonstrate how the MTBM and the entire system have been modified to complete the curve. The specifications should also require extensive monitoring of the excavation of the curve, including stop requirements, to determine whether the line, grade, and curve tolerances have been met, not met, or exceeded. The deflection of each joint must be monitored. This monitoring requires more time and effort on the part of the owner and the contractor.

If there is limited experience with curved alignments, prequalification based on prior experience may not be appropriate because it limits competition on the project. The challenge of pipe jacking through curve(s) is highly dependent on many factors such as the number, type, radius, length, and location of curve(s) in a microtunnel drive and the type of pipe and pipe joint used. As the difficulty and risk increase, an MTBM operator with curved alignment experience should be required. The operator may be in-house or furnished by a third party.

With curved alignments, additional geotechnical investigations and testing of samples are warranted.

CHAPTER 14

SHAFT CONSIDERATIONS

14.1 LOCATION

Jacking and receiving shaft locations should be selected to minimize the number of shafts and disruptions while not exceeding the practical limits of the drive lengths achievable for a given set of site conditions and constraints. Maximum drive length is a function of the microtunneling equipment used, the diameter of the pipeline, the jacking strength of the selected pipe, the ground conditions, lubrication, and maintenance requirements. Project risk increases when longer drives are used and shaft locations must be considered.

Commentary. In general, shafts are typically at manhole locations and at locations where changes in vertical and/or horizontal alignments are required. One strategy for minimizing the number of jacking shafts is to drive in two or more directions from one shaft. Curved alignments may reduce the number of shafts and their locations as well as their social, environmental, and economic effects.

Shaft locations should be selected to minimize the effect on traffic and businesses and to avoid site features that could result in inefficiencies during construction, such as low overhead utilities and/or underground utilities that cannot be relocated. Limitations on staging and laydown areas at shaft locations, work hour restrictions, and size limitations on the shafts must be clearly identified in the contract documents.

Receiving shafts are typically smaller, require less working space, and are occupied for shorter durations than jacking shafts. Where possible, receiving shafts should be located in areas with limited space, or where surface disruption needs to be kept to a minimum.

Some flexibility in shaft locations may be needed because a contractor may request alternate locations. The contractor may need some additional freedom with shaft locations to accommodate the ground support system and equipment while minimizing the effect on utilities and other adjacent structures. Alternate shaft locations require approval by the project owner.

14.2 SHAFT DESIGN CONSIDERATIONS

Design criteria and performance requirements for shafts should be provided in the contract documents. At a minimum, design criteria and performance requirements should include the following:

- Applicable codes and standards;
- Groundwater control and disposal constraints;
- Allowable settlement of the ground surface and structures adjacent to the shafts;
- Acceptable types of excavation support wall systems;
- Limitations on the use of external lateral support methods, such as tiebacks;

- Constraints and requirements relative to relocation of existing utilities within the footprint of the shafts; and
- Methods for exit and entry of the MTBM.

The shaft must be designed by or under the direct supervision of a registered professional engineer licensed in the state where the project is located. The design must be signed and stamped by the professional engineer.

Commentary. In addition to the minimum design criteria and performance requirements listed, additional information should be provided by the engineer to be used by the contractor's designer. This information includes the following:

- Soil, rock, groundwater, and minimum surcharge loading;
- Minimum factors of safety relative to bottom stability;
- Unacceptable types of excavation support wall systems; and
- Parameters for thrust wall design.

The use or nonuse of a dewatering system controls which ground support methods can be effectively used. In some cases, contaminated groundwater precludes the use of dewatering wells. Restricting groundwater disposal quantities, limiting discharge locations, and requiring special testing and water treatment may also preclude the use of dewatering systems. More often than not, the engineer needs to require the contractor to use a shaft support system that is watertight to avoid discharge-related permitting issues. Common shaft systems are identified in Table 14-1. Some of the shaft support methods are limited by depth, ground conditions, and groundwater.

Detailed design of jacking and receiving shafts is commonly the responsibility of the contractor, since these shafts are typically temporary structures. A notable exception to this generalization occurs when caissons or other support systems are used for permanent manholes, drop structures, or wet wells from which microtunneling operations are conducted. For these projects, the caisson or shaft is typically designed by the engineer for the long-term loads, and the contractor is required to check the design against temporary construction loads and provide additional reinforcing, as necessary.

In some ground conditions, removal of the shaft support system may displace the adjacent ground, resulting in settlement of manholes constructed inside the shaft and/or adjacent utilities. The contract should specify either that the shaft support system be removed carefully or that it be left in place. In some regions, only the upper few feet of the support system require removal.

14.3 GROUNDWATER CONTROL

Design of groundwater control systems for jacking and receiving shafts should be the responsibility of the contractor, subject to performance requirements established by the engineer.

Table 14-1. Common Shaft Systems

Type	Size and Shape*	Typical Depth*	Ground Type	Watertight	Remarks
Trench boxes and speed slide rails	Any size (width limited only by internal bracing)	20 ft	Most soils	No	Used above groundwater; short installation time
Soldier piles and wood lagging (or steel plates)	Any size (width limited only by internal bracing)	60 ft	Any	No	Used above groundwater; limited cantilever depth; sequential excavation and lagging installation
Soil nailing	Generally noncircular	50–60 ft	Most soils	No	Typically used above groundwater; requires specialized equipment; sequential excavation, used for portals
Liner plates	Any size (up to 30-ft dia)	100 ft	Soil with stand-up time	No	Flexible; adaptive to various sizes; can be expensive
Conventional excavation with rock dowels and shotcrete	Any size; up to 25-ft dia (deep); up to 40-ft dia (shallow)	1,000+ ft (deep), 200 ft (shallow)	Rock	No	Used in constructing mine shafts
Interlocking sheet piles	Any size (width limited only by internal bracing)	50 ft	Most soils (but trouble in cobbles and boulders)	Yes	Can be reused; inexpensive; used below groundwater; limited by crossing utilities; predrilling required in rock or bouldery ground
Secant piles	Circular (up to 35-ft dia)	110 ft	Most soils and weak rock	Yes	High cost; requires specialized equipment; limited by crossing utilities
Drilled shafts	Circular (up to 30-ft dia)	Up to 200 ft	Most soils and weak rock	Yes	High cost; requires specialized equipment; limited by crossing utilities
Cutter soil mixing	Any size (up to 35-ft dia)	90 ft	All soils	Yes	Similar to slurry walls; specialized equipment; generates heat with cement
Slurry walls	Any size (up to 135-ft dia)	200 ft	All soils	Yes	Specialized equipment; expensive; limited by crossing utilities
Ground freezing	Any size (up to 35-ft dia)	200 ft	Most soils	Yes	Difficult to freeze moving water or saline-rich water; high energy costs; allows flexible layouts and shapes
Caissons	Any size (up to 30-ft dia)	130 ft	Most soils	Yes	Typically part of the permanent works; can be used below the water table; limited by crossing utilities

*Maximum diameter and depth depend on ground conditions, groundwater, and equipment capabilities. There may be a trade-off between the maximum diameter and the depth that can be achieved for all methods.

Commentary. Microtunneling generally involves working below the groundwater table. One of the performance requirements is that the shafts be free of flowing or standing water while in use. There are two general approaches to groundwater control in connection with shaft construction and operation. One approach is to lower the groundwater table by pumping from dewatering wells or sumps. The other approach is to use an excavation support system that is relatively impervious so that infiltration into the shaft is kept to a minimum.

One of the key issues relative to groundwater control is the determination of whether groundwater lowering at shaft locations is acceptable, and to what extent. During design, an assessment of the effects of lowering the groundwater at shaft locations should be made. If it is determined that lowering of the groundwater in the vicinity of the shafts will have unacceptable effects (i.e., settlement of compressible ground or migration of contaminated plumes, among others), then groundwater lowering should be precluded as an option during shaft construction and operation.

The use of dewatering wells associated with microtunneling has other effects. The dewatering wells are designed to pull groundwater from the ground. The same wells can also pull the slurry and lubricants from the MTBM and pipe, preventing these fluids from doing their intended tasks. The use of dewatering wells should be limited when they are adjacent to a shaft location. Dewatering is not necessary and should be prohibited along the microtunneling alignment.

Other issues that should be addressed in the contract documents relative to groundwater control are constraints on disposal

of groundwater, the need to ensure that the groundwater control system does not result in the loss of fines from the ground surrounding the shaft, and the need to ensure that surface flooding and/or water that seeps through breaks or leaks in the pipe does not cause flooding of the shaft(s) and the microtunnel operation. Consideration needs to be given to minimizing the risk of water traveling along the annular space back to the shaft. This is true especially if nonwatertight shafts are used without seals.

Groundwater control when successfully launching and receiving an MTBM is a major concern. See Section 19.2 for more discussion on MTBM launching.

14.4 DESIGN OF ENTRY AND EXIT OF SHAFTS

Entry and exit seals should be provided at shaft walls to prevent inflows of groundwater, ground, slurry, and lubricants. Microtunneling designs and work plans must include provisions for portal stabilization at shaft penetrations. Contractors' work plans should address not only methods to prevent ground and groundwater inflows into the shafts during MTBM launching from the jacking shaft, during active pipe jacking, and during MTBM retrieval at the receiving shaft but also methods to prevent ground, groundwater, slurry, and lubricant inflows as portals are opened in the shaft support system.

Commentary. One of the more critical points in a microtunneling drive is when the MTBM begins penetrating the ground immediately outside the excavation support wall. If the ground at this location is not stable (i.e., if it has no stand-up time or is

under high groundwater pressures), significant loss of ground can occur. To minimize the risk of such occurrences, an assessment should be made of the potential for unstable ground to exist outside the shaft and of any special requirements that may be warranted. Special requirements, such as grouting of the ground outside the shaft or use of more sophisticated launching methods, should be indicated in the contract documents. The same assessment needs to be performed at the receiving shaft site. See Section 19.2 for more discussion on launching and receiving of the MTBM.

It is best practice to mount an entrance seal on the wall of the shaft before launching the MTBM. This is the case even where non-watertight shafts are used. Without a seal, it is impossible to pressurize and control the face of the MTBM. In addition, the absence of a seal could allow for shaft flooding if a water-bearing feature were to be encountered.

The seals are typically bolted or welded to the shaft wall (or a cast addition to the wall) and are fitted with one or more rubber gaskets that form a seal between the shaft wall and the MTBM and subsequently between the shaft wall and the pipe being jacked (see Fig. 14-1). In terms of performance requirements, the seal must prevent groundwater, if present, from flowing along the pipe into the shaft and must prevent the pipe lubricant and slurry used for face control from escaping into the shaft. Receiving shaft seals may be required, depending on the method of ground stabilization at the shaft and the presence of groundwater.

Microtunneling methods are typically used in soft, loose, and otherwise unstable ground conditions, often below the water table. Microtunneling equipment is specifically designed to operate in these ground conditions and to provide adequate ground support to prevent significant ground movements. Similarly, shaft systems are available to provide support of unstable ground conditions for microtunneling projects. However, a critical aspect of microtunneling projects has to do with situations in which penetrations must be made in the shaft support systems to allow exit or entry of the MTBM. Portal stabilization is required to minimize risks for the personnel and equipment in the shafts as well as for existing structures and facilities surrounding the shafts.

Portal stabilization can be achieved by a variety of means. As a starting point, the contract documents should establish

performance criteria for the stabilization approach—e.g., “No more than 5 gpm of groundwater and no more than 3 ft³ of ground shall enter the shaft during portal opening and MTBM launch or retrieval.” The specific ground conditions dictate the design approach needed to achieve the performance criteria. Potentially flowing ground conditions combined with high groundwater head may require double or triple seals, bladder seals, and/or a predesigned ground improvement approach (e.g., jet grouting, permeation grouting, or deep soil mixing). Lower risk conditions, such as slow-raveling ground may allow for less sophisticated and less costly approaches, such as guillotine walls or compaction grouting. Finally, certain shaft support systems offer the benefit of integral portal stabilization, thanks to shaft walls that can be directly excavated by the MTBM. These shaft systems include secant pile shafts (and cutter soil mixing, or CSM, shafts), auger-drilled shafts, slurry wall shafts, and sunken concrete caissons (combined with soft eyes).

Where conditions allow, the contract documents should allow the contractor to choose from a list of feasible portal stabilization schemes that can meet the performance criteria. The documents should also allow the contractor to propose additional approaches, subject to the owner’s or engineer’s approval. This approach brings the contractor’s expertise and preferences to bear on the issue of providing the required protection.

Finally, it is important to verify, in progressive steps, the effectiveness of the chosen stabilization methods before the shaft portal is opened to full size. One proven approach involves opening a hole approximately 3 in. (76 mm) in diameter in the shoring and then checking for ground and groundwater inflows. If inflows are less than the maximum allowable quantities, then the test location can be opened to the next suggested size of approximately 12 in. (305 mm). If inflows through the 12-in. (305-mm) test hole remain within allowable limits, then the portal can be opened to full size before the final check for inflow is made. If any of the test holes produce more ground or groundwater inflows than allowed, the test hole(s) should be resealed and additional stabilization measures should be taken before the test opening process is repeated. Fig. 14-2 shows the typical shaft wall seal, before and after MTBM placement.

14.5 THRUST BLOCKS

Thrust blocks are designed to distribute loads into the ground in a uniform manner so that any deflection of the thrust block is uniform and does not impart excessive loads to the shaft itself or cause the jacking frame to become misaligned. If the jacking frame becomes misaligned, the pipe can be point loaded, leading to pipe failure in some cases. If the jacking shaft is expected to be located in ground unable to support the maximum anticipated jacking force with an appropriate factor of safety, then ground improvement or other measures should be considered during the design phase and should be included in the contract documents.

Commentary. Thrust blocks are constructed between the jacking frame and the wall of the jacking shaft to provide a mechanism for transferring jacking loads into the ground. Detailed design of thrust walls, including establishment of allowable thrust load, is typically the responsibility of the contractor, subject to performance requirements established by the engineer.

Guidance for the design of thrust blocks can be found in the following references:

- Stein et al. (1989);
- Thomson (1993);

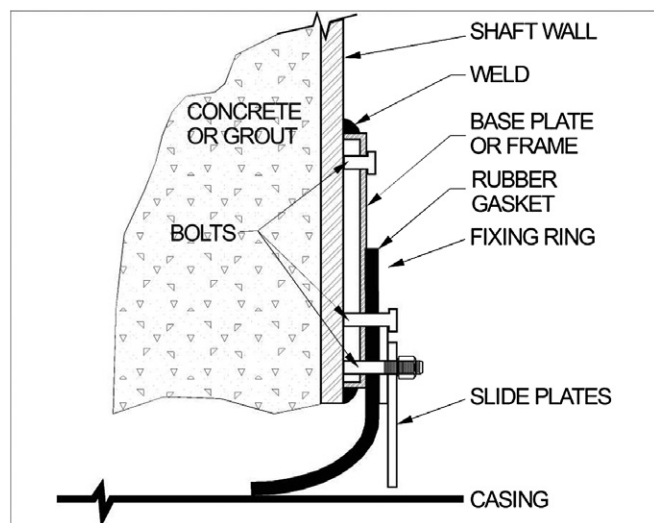


FIGURE 14-1. Cross section of typical shaft seal with rubber seal engaged

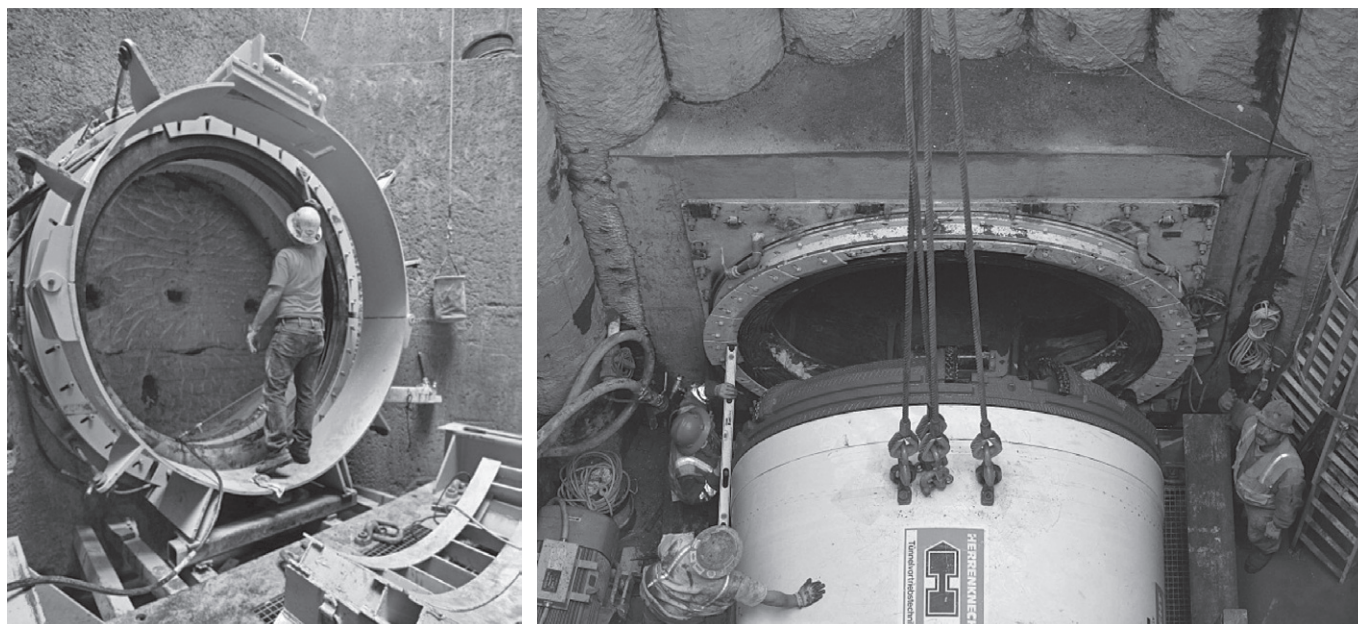


FIGURE 14-2. Shaft wall seal before (left) and after (right) MTBM placement

Source: Courtesy of Sue Bednarz, Jacobs Associates; reproduced with permission.

- Stein (2005);
- Najafi (2005); and
- Najafi (2013)

Depending on the magnitude of the jacking forces and the nature of the ground around the shaft, it may be necessary to “stiffen” the ground adjacent to the thrust block, using methods such as grouting, to keep deflections to acceptable levels. Jet grouted columns or other forms of ground improvement may be used to stiffen the ground around the shaft. To minimize the risk of unacceptable thrust block deflection, a preliminary assessment of jacking forces and thrust block deflections should be conducted. See Section 13.4 for more discussion on jacking forces. Special requirements that may be warranted, such as grouting of the ground outside the shaft (behind the thrust wall) or keying of the thrust block into the ground below the shaft, should be indicated in the contract documents along with payment provisions.

14.6 COMMON SHAFTS

Shafts used by multiple contractors should be identified and properly designed so that joint-use issues are properly identified and resolved.

Commentary. Common shafts are defined as shafts that may be used by two different contractors simultaneously or sequentially. They are typically used between adjacent construction contracts. The use of common shafts can result in disputes among contractors, which can ultimately have a detrimental effect on the project. Therefore, common shafts should be avoided, if possible. If common shafts cannot be avoided, the contract documents must contain a mechanism to minimize the potential for disputes between contractors and a means of resolving such disputes if they occur.

CHAPTER 15

CONTRACT DOCUMENTS

15.1 GENERAL CONSIDERATIONS

As with any civil works project, the bid and contract documents prepared by the owner and the engineer should provide the contractor with the necessary information to prepare bids, to plan the work, and ultimately to construct the project. These documents should include the following:

- General Conditions;
- Special and Supplementary Conditions;
- Technical Specifications;
- Plans; and
- Geotechnical information.

Commentary. Regardless of whether a pipeline is to be installed by microtunneling or other methods, it is important that the contract documents be complete, clear, and concise and that they be prepared by individuals with experience in the specified methods and technology. Although contract documents for microtunneling and for any other pipeline project have much in common, microtunneling is a specialty construction technique; as such, it entails some unique requirements that should be included in the contract documents.

The general conditions should be reviewed for project-specific requirements and for requirements that are compatible with the construction method. The general conditions of the contract specifications should include the following:

- Precedence of documents;
- List of geotechnical information provided;
- Issue resolution processes;
- Schedule constraints;
- Local requirements; and
- A section permitting value engineering proposals for opportunities to achieve significant economies and a better project (for example, see the commentary in Section 13.2 concerning the possible economy achievable by limiting the range of pipe sizes used on a contract, thus reducing the number of machines required).

The following sections discuss issues either unique to microtunneling or important enough to warrant discussion in these microtunneling guidelines.

15.2 GENERAL CONDITIONS

The general conditions describe the contract conditions governing construction projects for the owner agency or organization. The general conditions describe the following:

- Work to be performed;
- List of bid items;
- Legal framework for enforcing the contract, including measurement and payment;

- Acceptance tests;
- Period of performance;
- Insurance requirements;
- Change order procedures;
- Dispute resolution; and
- Liquidated damages for failure to complete the project in a timely manner.

Commentary. Some of the items listed above may be found in the Special and Supplementary Conditions, see Section 15.3.

15.3 SPECIAL AND SUPPLEMENTARY CONDITIONS

Special and supplementary conditions include supplementary or special conditions that may be necessary for the particular project. On some contracts, supplementary conditions may include the Geotechnical Baseline Report (GBR) and a differing site conditions clause.

Commentary. None.

15.4 TECHNICAL SPECIFICATIONS

The technical specifications include all the specifications that, taken together, establish the minimum requirements for the work to be performed, the materials to be used and installed, and the execution of the work. Technical specifications may be performance specifications or prescriptive specifications, sometimes referred to as means and methods specifications.

- Performance specifications establish the requirements for the product to be constructed, leaving means and methods to the contractor.
- Prescriptive specifications establish the requirements for the product to be constructed as well as the means and methods that the contractor is required to use in constructing the specified product. Prescriptive specifications create an implied warranty by the owner that if the contractor uses the means and methods required, with generally accepted industry good practices, the work product will meet the requirements of the contract.

Project technical specification packages are sometimes a mixture of performance and prescriptive specifications, even within an individual technical specification, if the engineer and the owner believe that it is necessary to limit the range of the contractor's options. Strictly defined performance and prescriptive specifications are less commonly used.

The technical specification package for a microtunneling project typically includes separate sections for each major item of work, including microtunneling, jacking pipe, shafts, portal stabilization, ground improvement, geotechnical instrumentation

and monitoring, contact and/or backfill grouting, groundwater control and disposal, and installation of carrier pipe in casing. The technical specifications should clearly describe each major item, relevant standards, design criteria, quality assurance and quality control (QA/QC), submittal requirements, work products, and execution.

The technical specifications for microtunneling projects should focus on performance rather than on means and methods. See the definition of performance specification in Chapter 3. As a general guideline, the technical specifications should convey the following information, at a minimum:

- Minimum amount of experience required for contractor's key personnel and other requirements for microtunneling contractors;
- Jacking pipe materials, coatings, joints, and treatments acceptable to the owner;
- Capabilities and performance requirements of microtunneling equipment;
- Constraints on construction sequence, noise, and work hours;
- Technical and contractual means of dealing with buried objects (see Sections 11.3.5 and 11.3.7);
- Horizontal and vertical alignment tolerances;
- Shaft requirements;
- Contaminated ground and groundwater disposal requirements;
- Drilling fluid, slurry, and spoil disposal requirements;
- Settlement and heave criteria; and
- Required submittals.

Commentary. The following comments expand on the subjects listed above:

- The specifications for minimum experience of the contractor's personnel should identify years of experience and number of projects with similar ground conditions. See Section 15.7 for more discussion on qualifications.
- Qualifications of the project manager, the general superintendent, and the machine operators are typically requested.
- The qualifications of the contractor's surveyor should include experience working in underground spaces.
- Regarding constraints on construction sequence, noise, and work hours, constraints on the contractor's sequence of operations should generally be kept to a minimum. However, sound social or technical reasons may sometimes dictate restrictions.
- Construction tolerances should be established on the basis of project needs. See Sections 12.2 and 17.3 for construction and material tolerances.
- Regarding contaminated soil and groundwater disposal requirements, many public agencies are now requiring a separate specification section that establishes the requirements for characterization, stockpiling and storage, transport, and disposal of excavated material and groundwater.
- Regarding settlement and heave criteria, microtunneling is often selected to cross under a road, building, railroad tracks, or other surface feature(s). In these instances, settlement and heave tolerances of the structures should be estimated, and criteria should be negotiated with the respective owners or agencies. Such criteria should be reasonable and achievable and should be clearly stated in the specifications, as should the consequences of exceeding the criteria. Where such criteria may not be reasonably met, the designer should specify a program of ground modification or structure support or should take other measures during

design to ensure that specified settlement and heave criteria are achievable.

- For required submittals, see Chapter 21.

Specifications are typically prepared in a standard format. The Construction Specifications Institute (CSI) format is the most common. Each owner has an individual preference, so check with the owner or control agency of the project. Typical technical specification sections include the following:

- Microtunneling;
- Shaft excavation and support systems;
- Pipe materials;
- Geotechnical instrumentation;
- Utility relocations;
- Ground improvement methods;
- Contact grouting;
- Installation of carrier pipe;
- Backfill grouting;
- Manholes; and
- Shaft backfill materials.

In a design-bid-build contract, it is the engineer's responsibility to prepare a set of specifications to define the "WORK" to be completed. The WORK is documented as a series of specification sections. Each specification section defines the qualitative and administrative requirements for products, materials, quality, and workmanship for a specific element of the WORK. Each specification section is formatted into three parts. PART 1 is the general portion of the specification section, containing administrative and procedural requirements that pertain only to that section. PART 2 is the products portion, containing requirements for the materials and equipment for that particular section. PART 3 is the execution portion, containing requirements for the actual work in that section.

In the interest of developing consistency across the project specifications, the sections should be written in the same format. The specifications should be written and reviewed for consistency between sections. Requirements for specific WORK should be established in one location. Common requirements for a separate piece of the WORK could be included in a separate section, provided that there is cross-referencing in both sections. Cross-referencing is important for tying together the requirements in the various specification sections.

For example, the typical concrete specification, Section 03300, addresses the general requirements for the concrete to be used on a project. Other elements of the WORK needing concrete (e.g., secant piles for the shaft construction), would then cross-reference the concrete specification. These elements must also reference the concrete specification section by number (e.g., "see Section 03300 for concrete specifications"). This method allows the concrete to be specified in one location and eliminates duplication of requirements. The bidder and contractor is thus directed, by number, to the appropriate section, where the concrete specification and any additional requirements are found.

15.5 PLANS

The design plans must clearly and concisely show the work to be done. As a general guideline, the plan and profile drawings should convey the following information, at a minimum:

- Number of microtunnel drives; their lengths, diameters, and slopes; and inlet and outlet elevations at manholes;
- All known above- and below-ground utilities and structures, both active and abandoned;
- Subsurface exploration locations and conditions;

- Shaft locations and work areas;
- Geotechnical instrumentation locations;
- Locations of known soil and rock horizons, or of groundwater contamination requiring special treatment or disposal requirements; and
- Groundwater depth and location.

Commentary. The plan and profile sheets should show the geotechnical boring logs. When these are included, the geotechnical information and the actual plan and profile drawings are to the same scale and show potential challenges along the drives. In addition to the plan and profile drawings, as many drawings as necessary should be prepared to convey the following information:

- Survey control points for project layout;
- Limits of work areas at all shaft locations;
- Ground improvement details;
- Additional work and staging areas;
- Rights-of-way and/or easements;
- Geotechnical instrumentation details and schedules, with tabular locations;
- Minimum requirements and design criteria for temporary shafts;
- Pipe details;
- Traffic management plans (vehicular and pedestrian);
- Work necessary for protection of adjacent structures;
- Utility relocations;
- Standard details per the appropriate agency (e.g., manholes, manhole lids, driveways, curbs and gutters); and
- Erosion control measures and details.

Appropriate notes are needed on the plans to advise the contractor of the need to locate utilities and confirm all crossing and adjacent utilities within the tunnel zone of influence and protect all utilities from damage. Additional notes should convey information necessary to guide the contractor in the execution of its work.

Protection of critical adjacent structures and utilities should not be left entirely to the contractor. Where special underpinning, grouting, or other measures have been included by the engineer, sufficient detail should be given so that the contractor can bid and construct the designed system.

15.6 GEOTECHNICAL INVESTIGATIONS AND DATA

Preliminary geotechnical investigation requirements are described in Chapter 8. Detailed geotechnical investigation requirements are described in Chapter 11.

All geotechnical data collected during the various phases of the project should be compiled and provided to all bidders on the project.

The owner of the project should request the engineer of record to write a GBR, depending on the level of complexity and risk. This report or letter report should follow the table of contents outlined in the ASCE/UTRC/AIME *Geotechnical Baseline Reports for Construction: Suggested Guidelines* (2007). The GBR should be provided to all bidders on the project.

Commentary. Some agencies compile and provide the geotechnical data on a searchable compact disc. All parties on the design team, including the owner, should participate in the preparation of the GBR.

15.7 MICROTUNNELING CONTRACTOR QUALIFICATIONS

Prequalifying a contractor and/or subcontractor to conduct microtunneling work should be considered. It is acknowledged

that technical prequalification is not always possible because of public bidding laws. When prequalification is not feasible, proof of minimum experience requirements for the project manager, MTBM operators, and the superintendents should still be stipulated as required submittals in the contract documents.

Commentary. Whereas the experience of the design team and a good set of contract documents are important to the success of a microtunneling project, the experience of the contractor is critical. Ideally, it is desirable to prequalify specialty microtunneling contractors or subcontractors on the basis of technical and experience criteria. Theoretically, this prequalification would ensure that the work would be done by a contractor or subcontractor whose key on-site personnel are familiar with the specific challenges and potential problems inherent in microtunneling.

The process of prequalification is challenged by some as unnecessary or as too restrictive in keeping new contractors out of microtunneling practice. However, it is not the intent of prequalification to preclude contractors from entering the microtunneling arena. Rather, the intent is to prevent unqualified, inexperienced contractors from gaining experience at the expense of the owner, and to the detriment of the completed project.

The specific mechanism for prequalification must be established on a case-by-case basis, depending on the risks associated with the project. The following criteria might be considered for prequalifying a microtunneling contractor or subcontractor:

- Proof of bonding capability;
- For the project manager and the superintendent, a minimum of five construction projects involving microtunneling in similar ground and groundwater conditions and with similar drive lengths and diameter range;
- For qualified personnel operating microtunneling equipment, a minimum of 1,000 m (3,000 ft) and/or five drives, with equipment similar to that proposed for the particular work; and
- Safety performance.

Technical prequalification is not always possible because of public bidding laws. When prequalification is not feasible, proof of minimum experience requirements should still be stipulated as required submittals in the contract documents for the contractor and its project manager, the MTBM operators, and the superintendents. See Section 21.1 for a discussion of the contractor's qualification submittals.

Some owners and engineers consider the prequalification of a bidder and its workforce for microtunneling, shaft design and construction, and associated ground improvement work to be a prerequisite for the owner to open the bidder's bid, for one or a combination of the following reasons:

- It is unfair for inexperienced contractors to learn or experiment using public funds, particularly on specialized or difficult work.
- Underground construction work is inherently risky and can be dangerous, particularly for contractors and workforces with insufficient experience.
- In the past, there have been problems such as project delays, damage to public and private properties, and exposure to third-party liabilities because of ground subsidence related to microtunneling errors, problematic shaft design and construction, inadequate ground improvement for MTBM ingress and egress, inadequate shaft perimeter support, and unsafe and messy worksites.
- In the past, there have been project problems related to estimators' errors during bidding, including inadequate

estimates of shaft sizes, and incomplete costing of shaft support.

- In the past, there have been microtunneling problems caused by contractors' use of inexperienced personnel in laying out shaft locations, and these have resulted in MTBM ingress problems and associated severe flows of ground into the shaft.
- Substantially higher inspection and construction management costs are associated with an inexperienced contractor or workforce.
- With experienced and qualified contractors and workforces, many projects have been successfully completed in difficult ground and groundwater conditions.

Here is an example of a city's prequalification process for bidders, in use since 1998:

- Owner requires that all contractors requested to submit and interested in submitting a bid for this project submit the qualification statement by the date specified in the Notice to Contractors.
- Prospective bidders must provide all of the requested information and answer all questions fully and explicitly.
- Only qualified contractors that have all the required prequalified workforce members and track records, or contractors that have prequalified subcontractors, shall submit a bid. Bids submitted by contractors that have not been prequalified are returned.
- Prospective bidders are encouraged to list more than one person for each position requiring prequalifying experience. With the exception of microtunneling operators and operators of or specialists in microtunneling automatic navigation systems, individuals must be currently employed by the bidding general contractor or by the bidding general contractor's subcontractor at the time when bids open. Only those individuals specified on the prequalification form, and whose names are submitted before the deadline for submitting the Bidder's Statement of Qualification set forth in the Notice to Contractors, are considered for review.
- Each prospective bidder submitting a Bidder's Statement of Qualification will be notified by the owner, either by mail, e-mail, or fax, before the opening of bidding, to indicate which individuals are deemed "qualified" and "unqualified." The bidding contractor may then select from those individuals deemed "qualified" by the owner of the project, to assemble a "workforce" for the three specialty areas of microtunneling (including operation of automatic navigation systems), shaft construction, and jet grouting work.
- Only work experience obtained before the Bidder's Statement of Qualification deadline date may be used as part of an individual's qualifying work experience. The owner, at its discretion, may ask for clarification or additional supporting data after this deadline.
- The general contractor may submit qualifications for as many subcontractors as the general contractor wishes. However, the general contractor must designate the subcontractors it intends to use in its bid submittal. Subcontractor qualifications must be submitted by a general contractor. Subcontractors may not submit qualifications directly to the owner.
- The final decision, based on the information provided in the Bidder's Statement of Qualification, to accept or reject a contractor, a contractor's workforce member, or a subcontractor lies solely with the owner.

- Fulfillment of the requirements of the Bidder's Statement of Qualification does not waive the requirements of the contract documents.

15.8 DISPUTE RESOLUTION

A method of resolving disputes between the owner and the contractor should be included in the contract documents.

Commentary. Even with the best engineering effort and exploration program, tunneling work, whether conventional or microtunneling, is often fraught with unforeseen underground conditions. Many owners and contractors do not fully understand the unpredictable nature of subsurface conditions and the impossibility of accounting for all of the conditions possibly encountered. It is imperative that any underground contract contain a "differing site conditions" (DSC) clause to allow the contractor to be compensated for extra costs involved without being forced into a breach of contract. This clause is required of federal contracts and should also be included for nonfederal government work. Typically, the federal clause should be used because it has been tested in court and found to be fair and enforceable.

Adversarial relationships should be avoided whenever possible. It is in everyone's best interest to resolve conflicts quickly, fairly, and equitably. The contract documents should provide for mediation or arbitration before litigation. The following are generally considered positive steps:

Clear, unambiguous contract documents. Owners should realize that spending a little extra time and money during the design phase for complete explorations and design tends to pay large dividends in decreased change orders and fewer extra costs during construction.

DSC clause and GBR: To help clarify exactly what constitutes a DSC on a particular job, tunnel designers should adopt the GBR. This document is a part of the contract documents and establishes baseline values for the conditions to be expected during construction, such as the number of cobbles, boulders, or groundwater inflows to be expected during construction. The joint ASCE/AIME publication, *Geotechnical baseline reports for construction: Suggested guidelines* (2007), describes GBRs in great detail. Readers are strongly urged to refer to this document for more information on GBR history, rationale, and preparation. Please note that the guidelines are written for large-diameter tunnels, and some provisions may need to be modified for small-diameter tunnels.

Fair and timely dispute resolution plan: The time for response between the engineer and the contractor on requests for information (RFIs) and notices of dispute should be specified so that turnaround is kept to a practical minimum. The quick resolution of disputes may reduce or eliminate the scheduling effects of a long-drawn-out or untimely, nonresponsive process.

Alternative dispute resolution (ADR): Once the preliminary dispute-resolution measures have contractually been exhausted, contracts should allow for some ADR procedure before the long and costly litigation process. Much good work done for the tunneling industry by the ASCE, the UTRC, and others can also be applied to the microtunneling industry. Such elements as dispute review boards (DRBs) and escrow bid documents might be considered, depending on the project's cost, time frame, and complexity. The reader is directed to an excellent reference published by ASCE in 1991, titled *Avoiding and resolving disputes during construction*. This document, the forerunner of the 2007 ASCE/AIME publication referenced above, discusses GBRs (then called *geotechnical design summary reports*, or GDSRs), escrow bid documents, and DRBs. The use of DRBs has been further

fine tuned and explained since that 1991 publication. The reader is directed to the Dispute Resolution Board Foundation website at www.drb.org.

The daily cost of construction is expensive for the contractor and the owner. Use of a DRB can reduce the cost of nonproductive work and the effect of disputes on work progress. When a dispute delays or stops work, the cost to all parties increases. Use of a DRB is a good method for helping to resolve a dispute quickly and keep the project moving forward. The benefit of the DRB is

that it brings in objective individuals who are familiar with micro-tunneling and shaft construction methods, and with the project itself, to provide input and expert opinion on the cause(s) of the dispute. It is also recommended that the DRB meet on a regular basis (at least quarterly, given the typically short duration of a microtunneling project) to remain informed about the project's status, even if there are no disputes; knowing that the DRB is staying current with project conditions helps the owner and the contractor keep an open conversation about the project.

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PART IV

PIPE MATERIAL ASPECTS

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CHAPTER 16

PIPE CHARACTERISTICS

16.1 GENERAL REQUIREMENTS

Microtunneling pipe should meet the following general requirements:

- Circular shape with a flush outside surface (including at the joints);
- Strength sufficient to withstand both the installation loads and the in-place, long-term service loads;
- Dimensional tolerances on length, straightness, roundness, end squareness, and allowable angular deflection;
- Durability for the service exposure (e.g., internal and external corrosion resistance, resistance to internal abrasion); and
- Joints capable of watertight performance for internal operating pressures (including working and transient pressures), as stated in the purchaser's documents, and external head pressure resulting from groundwater, slurry pressures, and lubrication. The joint should also be capable of efficiently transferring jacking forces from the hydraulic jacking frame, through the pipe wall and joint, to the MTBM face.

Commentary. These general requirements are applicable to all of the pipe types identified in Section 16.2. Any time multiple pipe material types are listed in these guidelines, they are listed alphabetically.

16.2 MATERIAL TYPES

There are six general types of pipe materials currently (2015) used in microtunneling (listed alphabetically):

- Ductile iron;
- Fiberglass-reinforced polymer mortar;
- Polymer concrete;
- Reinforced concrete;
- Steel; and
- Vitrified clay.

Commentary. Microtunneling is a trenchless method for constructing underground piping systems and is generally used in installations when one or more of the following demanding conditions exist:

- Urban setting, with limited access;
- Deep installations, with or without groundwater;
- Critical grade control; and
- Poor soils, with or without groundwater.

Microtunneling is most widely used for sewer and storm water applications. The process involves the direct jacking of pipes to advance the tunnel boring machine through the in situ ground

conditions. Low-head or gravity pipes most often jacked in these nonpressure applications include polymer concrete pipe (PCP) reinforced concrete pipe (RCP), fiberglass-reinforced polymer mortar pipe (FRPMP), and vitrified clay pipe (VCP). All of these pipe materials have a substantial microtunnel installation history in sewer applications.

The use of microtunneling for direct installation of pressure pipes has been limited because such installation typically follows the general topography along the pipeline alignment, with a minimum cover. However, when microtunneling is used for pressure pipe installation, suitable pipe materials include ductile iron pipe (DIP), reinforced concrete cylinder pipe (RCCP), FRPMP, and steel pipe.

Microtunneling is routinely used to install steel pipe as casing. Microtunnel installation of steel pipe for both gravity and pressure systems is a viable installation methodology, especially as new types of joints evolve and applicable coating and lining technologies are used.

16.3 MATERIAL SELECTION

Material selection should be based on many factors, including the following:

- Pipeline operating conditions (pressure: operating, test, transient, and vacuum);
- Pipeline service environment (fluid, temperature, and corrosivity);
- External loads (earth loads, surface live loads, and groundwater head);
- Inside diameter of the pipe required by the owner;
- Compatibility with the MTBM (outside diameter) and the jacking system;
- Anticipated jacking forces and drive lengths;
- Pipe deformation and rebound (during jacking) for plastic or elastic materials;
- Pipe hydraulic characteristics;
- Pipe availability, reliability, and durability; and
- Lifecycle cost.

Commentary. See the commentary in Section 13.2 and Chapter 17 regarding the design and selection of pipe materials.

When external loads are estimated, arching of the ground should be considered. Trench load factors calculated with such approaches as Marston's equation should not be used to estimate external loads on the pipe. As stated in ASCE MOP 60/WEF FD-5 (2nd ed.), when Marston's equation is modified to include cohesion, Marston's equation may be used to determine the gravity soil loads on jacked sewer pipe through undisturbed soil. See Section 13.3 for more discussion on earth loads.

Table 16-1. Standards for Different Jacking Pipe Materials

Material Type	Standards	
	Nonpressure	Pressure
DIP	AWWA C150/C151 ASTM A716 ASTM A746	AWWA C150/C151
PCP	DIN 54815-1 & 2 ASTM D6783 DIN EN 14636-1	N/A ³
RCP	ASTM C76 ASCE 27-00	ASTM C361 AWWA C300/ C302
FRPMP	ASTM D3262	ASTM D3517 ASTM D3754 AWWA C950
Steel Pipe	ASTM A36 ASTM A139 Grade B ¹ ASTM A515 ASTM A572 API 2B ²	AWWA C200 API 2B ²
VCP	ASTM C1208 EN 295-7	N/A ³

¹ No hydrostatic test required.² Dimensional tolerances only.³ Not applicable.**Table 16-2. Size Ranges for Different Jacking Pipe Materials**

Material Type	Range of Nominal Diameters, in. (mm)	Operating Condition	
		Nonpressure	Pressure
DIP	12–64 (305–1,625)	Yes	Yes
PCP	12–102 (305–2,591)	Yes	N/A*
RCP	12–144+ (305–3,658+)	Yes	Yes
FRPMP	18–126 (457–3,200)	Yes	Yes
Steel Pipe	12–144+ (305–3,658+)	Yes	Yes
VCP	15–48 (381–1,219)	Yes	N/A*

* Not applicable.

16.4 REFERENCE PRODUCT STANDARDS

Table 16-1 provides standards for different jacking pipe materials.

Commentary. See Section 16.2 for a list of jacking pipe material types.

The reference standard for reinforced concrete pipe is ASTM C76. ASTM C76 is written for concrete pipe used for open-trench construction, and the tolerances and joint design are not appropriate for jacking pipe used in microtunneling.

See Section 17.3 for recommended tolerances for all jacking pipe.

See Section 17.5 for recommended pipe joints.

16.5 SCOPE AND RANGE

Table 16-2 provides size ranges for different jacking pipe materials.

Table 16-3. Typical Jacking Pipe Lengths

Material Type	Standard Length(s)	Available Length(s)
DIP	19.5 ft	As required
PCP	8 ft and 10 ft and 1 m, 2 m, and 3 m	3 ft and 6 ft
RCP	7.5–24 ft	Variable
FRPMP	10 ft/20 ft	4 ft, 5 ft, 10 ft, and 20 ft
Steel Pipe	8 ft, 10 ft, 20 ft, and 40 ft	Any
VCP	4 ft, 6 ft, 8 ft, 10 ft and 1 m, 2 m, and 3 m	>2 ft

Commentary. Sizes of pipes are presented in English units because they are manufactured products with dimensions not actually converted to SI units.

Actual sizes applicable to microtunneling depend on the current state of the art. Currently (in 2015), jacking pipes that have an outside diameter (OD) between 16 and 144 inches can be installed with MTBMs. Actual internal diameters may vary from the nominal size. Consult with the various pipe manufacturers for applicable diameters and wall thicknesses.

16.6 SECTION LENGTHS

Table 16-3 provides typical lengths for pipe sections for the different jacking pipe materials.

Commentary. Lengths of pipe sections are presented in English units, where applicable, because they are manufactured products with dimensions not actually converted to SI units.

16.7 MECHANICAL PROPERTIES

Except in pressure applications, the most critical material property is normally compressive strength. For the jacking pipe materials covered in these guidelines, the compressive strengths range from approximately 35 to 414 MPa (5,000 to 60,000 psi), yet all of the different pipes can be used successfully in microtunneling and pipe jacking.

Commentary. During microtunneling through cobbly and bouldery ground or in rock, rock pieces can enter into the annular space and point load the jacking pipe. The flexural bending and tensile strength of the pipe material is an important mechanical property to evaluate when selecting the pipe for these complex ground conditions.

16.8 MANUFACTURING AND QUALITY CONTROL

The pipe manufacturing process and quality control must ensure that the required characteristics are achieved, including those pertaining to the following areas:

- General workmanship;
- Strengths and load capacities;
- Dimensions and tolerances;
- Joint integrity; and
- Surface consistency (smoothness and linings or coatings, if any).

Specific procedures and test methods vary with jacking pipe materials.

Commentary. None.

CHAPTER 17

PIPE DESIGN

17.1 IN-USE LOADS

The pipes and joints, once installed, must be capable of safely withstanding, for the project design life, all in-use service loads, including those associated with the following factors:

- Internal operating, transient, and test pressures;
- Earth overburden (tunnel construction may result in significantly lower loads than those experienced with open-cut installations);
- Surface loads (i.e., surcharge, vehicle, railroad, and/or aircraft, when present); and
- External static water head and vacuum (if present).

Commentary. Design procedures for these conditions vary somewhat by pipe material and are detailed in many other documents and standards. Therefore, they are not addressed further in these guidelines. See Section 13.3 for more discussion on earth loads.

17.2 INSTALLATION LOADS

Installation loads include the following:

- Jacking forces; and
- External pressure from groundwater, earth loads, surface loads, and injection of lubrication into the annular space.

Commentary. *Jacking forces:* Microtunneling pipe is subject to large transient axial loads, called *jacking forces*, applied during the installation process to advance the pipe and MTBM in the horizontal direction. See Section 13.4 for more discussion on jacking forces. Jacking forces are largest on sections of pipe nearest the jacking shafts or just in front of intermediate jacking stations. These forces are rarely distributed evenly around the pipe's end circumference because the squareness and mating of joints and the pipe alignment are seldom perfect. Some eccentricity of the axial load typically occurs in the field. The result is force concentrations in portions of the pipe ends (joints).

The maximum jacking stress at any point in the pipe is at least as great as the maximum jacking force recorded at the jacks (minus any friction losses along the drive between the jacking shaft and the point in question) divided by the effective minimum cross-sectional area of the pipe wall.

The pipe and joints must be able to withstand these stresses without cracking, breaking, or suffering other damage. Frequently, pipe ends of segmented pipes are separated with compressible packers (also called *compression rings* or *cushion rings*). These compression rings are normally used to distribute jacking forces between the pipe ends more evenly. The cushion material selected and its performance under repeated loads may cause hardening of the material during the drive. Most

compression ring materials selected have a low Poisson's ratio. Alternate methods of providing cushions include hydraulic and laminated designs of differing materials. Steel casings are often welded or have rigid mechanical designs that do not accommodate compression rings.

As the pipes are jacked through the ground behind the MTBM, the pipe and joint exterior surfaces experience skin friction from the surrounding ground. The pipes and joints must have sufficient durability and toughness to withstand this phenomenon without significant abrasion, loss of joint seal, damage, loss of corrosion resistance, or failure. Past successful experience in similar conditions is typically the best evidence of satisfactory capabilities. Adequate overcut, annular thickness, and lubrication can significantly reduce skin friction. See Sections 19.6, 20.1, and 20.2 for more discussion on lubrication, annular thickness, and overcut, respectively.

The factor of safety for jacking forces is the ratio of the pipe's ultimate (or yield) axial load capacity divided by actual applied load at the jacks. The allowable load calculated using the factor of safety may need to be reduced because of eccentric loading or end squareness tolerances. If the pipe is used for a curved microtunnel alignment, an additional reduction is needed. Consideration of special joint configurations may be necessary.

Different pipe materials require different safety factors to allow for various material characteristics.

External pressure: External pressure is exerted by groundwater and earth loads. Lubrication for the annular space is injected under pressure from the MTBM and/or through ports in the pipe walls and can impose an additional external pressure. The pipes and joints must not leak, be damaged, or fail from applications of these pressures. See Sections 13.3 and 19.6 for more discussions on earth loads and lubrication, respectively.

17.3 DIMENSIONAL CRITERIA

The engineer shall provide tolerances for each of the following dimensional parameters, where applicable, for the type of pipe and pipe material included in the contract documents:

- Circumferential uniformity;
- Exterior roundness;
- End squareness or planeness;
- Section straightness;
- Pipe length; and
- Allowable angular joint deflection tolerance.

Commentary. In microtunneling, it is extremely important that the owner, engineer, and contractor have a thorough understanding of dimensional tolerances for jacking pipes and the effect that variation may have on the process or the long-term performance of the pipe. Tolerances vary considerably by material

composition and method of manufacture and, in general, lower or tighter dimensional tolerances are ideal. However, it is also important to note that pipes with higher variations in dimensional tolerances have been successfully installed with microtunneling. This has been the case because experienced installing contractors can make certain strategic or functional changes to equipment or processes to compensate for less conservative tolerances.

End squareness, straightness, eccentricity, maximum diameter variation, and ovality are five tolerance parameters that can have a significant effect on the microtunneling process and pipe performance.

End squareness is the geometric relationship between the barrel of the pipe and the pushing surfaces of the pipe joint. If the pipe joint face is not perpendicular to the axis of the pipe, the jacking forces are not evenly distributed around the joint, and the result is areas of much higher stress concentrations or point loads. To compensate for some variations in end squareness, microtunneling contractors install compression rings or cushion rings that can more effectively redistribute the jacking loads and avoid potential joint damage.

The straightness deviation is often stated in standards as a deviation from a straight edge placed longitudinally above the pipe on the concave side. This deviation is reflected on the opposite side of the pipe and causes the effect of the pipe to be larger than the maximum diameter.

Eccentricity from the pipe axis of pipe joints affects the design of the shield cut and the required bore of the tunnel. This eccentricity should be included in the tunnel bore calculation, to provide adequate annulus.

The maximum outside diameter, the diameter tolerance, ovality, pipe straightness, and pipe eccentricity are all elements that affect the most critical dimension used in the microtunneling process. These values are used by the contractor to determine the required bore diameter and the associated annulus.

Pipe materials vary widely in their resistance to corrosive effects, both internally and externally. The pipe selection should evaluate the need to resist corrosion and the desired life of the installation for the owner. In some materials, internal or external sacrificial material may be added to the wall to extend the life of the material. This material should be allowed for in the design and installation annulus calculations.

The standards of manufacture of pipes vary widely with respect to measurement methods, use of appropriate statistical analysis, and dimensional limits. The engineer and installer must evaluate these variables for successful installation design and pipe selection.

Microtunneling is an installation method that is limited with respect to the amount of overcut and shield cut. It requires precise confinement of the MTBM and pipe string and is sensitive to fluctuation in the aforementioned dimensional parameters. The following critical dimensional parameters may have an effect on several aspects of the jacking process.

Jacking loads. As variations in circumference, roundness, end squareness or planeness, and section straightness increase, so do the jacking loads, sometimes substantially. This increase can result in decreases in the distance for safe jacking drives and/or in an increased need for intermediate jacking stations and ultimately, in severe cases, can cause pipe failures.

Steering and alignment. Poor control of end squareness or planeness and section straightness results in concentrated loads on the pipe ends and increased steering of the MTBM. When steering becomes excessive, jacking loads typically tend to increase significantly, which can result (as noted above) in shorter achievable drive lengths, more intermediate jacking sta-

tions, and/or increased risk of pipe failure. The same effects occur when the MTBM is intentionally steered to follow a curved alignment.

Load transfer. Square, plane pipe ends and straight sections improve the uniformity of jacking load distribution on the pipe ends. As deviations in straightness, squareness, and planeness increase, uneven loading on the pipe ends also increases. These load concentrations and point loads, when severe enough, may cause pipe damage or failure, sometimes even at forces below the calculated allowed load on the pipe.

Equipment. Some microtunnel equipment, particularly the spoils removal and transfer system, has auger or slurry circulation pipe lengths of uniform and discrete sizes. The microtunneling pipe's length must be consistent with these lengths, or adjustments may be required periodically.

In summary, although no pipes have perfect dimensional control, tolerances on these characteristics vary significantly from one pipe material to another, and from one manufacturer to another within a given type of pipe material. The dimensional tolerances required for successful performance vary according to several factors, including the following:

- Drive distance;
- Magnitude of jacking forces versus the ultimate pipe load capacity;
- Operational control, such as amount of overcut, effectiveness of lubrication, or degree of steering (note that allowance for wear of the cutting tools should be considered in determining required overcut);
- Compounding effects of several dimensional criteria;
- Ground conditions; and
- Type of pipe material.

In spite of the above-noted influences, pipe users generally agree that satisfactory jacking performance at a reasonable cost is achieved when the pipe dimensional tolerances are maintained within certain limits. The desirable tolerances for all microtunnel pipe products are shown in Table 17-1. Typical manufacturers' current achievable tolerances are summarized in Table 17-2.

Tolerances outside the desirable limits do not automatically ensure that pipe cannot perform successfully; however, it may be necessary to increase the jacking load safety factor to achieve acceptable reliability, increase the annulus, and/or increase the pipe load rating (capacity) in order to safely complete the required drive distances.

17.4 LUBRICATION PORTS

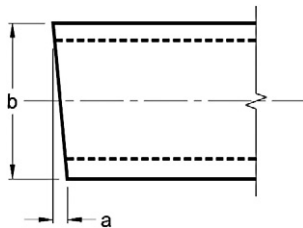
Pipes with diameters large enough to permit personnel entry are frequently equipped with lubrication ports (fittings) in the wall, to permit injection of a lubricant during jacking, or placement

Table 17-1. Desirable Dimensional Tolerances for All Material Types

Dimensional Characteristic	Desirable Tolerance
Outside diameter	Within 0.1%
Exterior roundness	Within 0.5%
End squareness or planeness	
Diameter $\leq$ 48 in.	Within 1/16 in.
Diameter $>$ 48 in.	Within 1/8 in.
Straightness	Within 1/8 in.
Joint length	Within 1/4 in.

Table 17-2. Manufacturers' Dimensional Tolerances by Material Types

Dimensional Characteristic	DIP, Concrete Coated	PCP	RCP ¹	FRPMP	Steel Pipe	VCP, ASTM	VCP, EN 295-7
Outside Diameter	±3/8" max	1%	±1/4" for ID ≤ 24"; ±1% with ±3/8" max. for ID > 24"	0.1%	0.1%	±0.20" ≤ 16" ±0.24" for 18" ±0.30" for 21" ±0.36" for 24"–30" ±0.48" for 33"–36" ±0.60" for 39"–48"	±0.20" ≤ 16" ±0.24" for 18" ±0.30" for 21" ±0.36" for 24"–30" ±0.48" for 33"–36" ±0.60" for 39"–48"
Exterior Roundness	²	0.5%	1%	0.1%	0.5%	2%	Included in External Diameter Tolerance
End Squareness or Planeness	1/8" ≤ 24" > 1/4"	±0.06" ≤ 39" ±0.12" for 42"–108" ±0.2" for 108"–144"	³ $a/b \leq 0.004$ for 12"–42" ID; ≤ 0.0035 for 42"–60" ID; ≤ 0.003 for ID > 60"	±0.06"	1/16"	±0.1"	±0.04"
Straightness	²	±0.33" ≤ 39" ±0.39" for 42"–108" ±0.46" for 108"–144"	±1/4" for ID ≤ 24"; ±1/8"/ft with ±3/8" max. for ID > 24"	±0.12"	1/8"	±0.4" at 8 ft	±0.2"
Joint Length	1/8" ≤ 24" > 1/4"	±1.97"	NA	±0.24"	1/4"	NA	±0.08"



¹ Tolerances are per ASCE 27-00; may vary per project requirements.

² Tolerances for concrete coating on Fastite Push-Bar joint ductile iron pipe are per the requirements of ASTM C76 and controls the following tolerances: Exterior Roundness, Outside Diameter, and Straightness. All other tolerances are per the requirements of AWWA C151 and the manufacturers' proprietary machining tolerance for fabrication of the Fastite Push-Bar joint.

³ The value a/b is the ratio between the two dimensions detailed in the object on the left.

of grout after jacking to fill any residual annular space. Pipes should be equipped with these fittings at the frequency determined by the contractor or required by the engineer.

Where drives are short and the ground friction is expected to be low, with resulting low jacking loads, the engineer should consider whether to exclude the lubrication ports.

Commentary. Typical lubrication port fittings are shown as either a flapper check valve (Fig. 17-1) or a witch's hat check valve (Fig. 17-2), and a photograph is provided showing check valves of different sizes (Fig. 17-3).

Lubrication port fittings are typically created by using a threaded bushing made of either steel or PVC, which can be cast in place during pipe manufacturing (Option A), or by a hole drilled into the pipe whereby the bushing is grouted in place with epoxy after pipe manufacturing (Option B). With a threaded bushing in place, the check valve can be added and positioned toward the external surface of the jacking pipe. A steel or PVC threaded plug is added from the internal side. The plug can have a hole to allow the insertion of a lubrication hose during jacking or a grout hose for contact grouting after jacking. The plug can also be solid, to serve as a permanent sealing plug. If the bushing and plug are steel, they can be welded to be watertight. The sealing plug can also be tapered, to form a seal. If the threaded bushing and plug are recessed, the internal surface of the jacking pipe can be filled with a patching material to form a smooth internal surface (if required by the owner).

17.5 JOINTS

All microtunneling pipe joints shall meet the following performance requirements:

- Flush or slightly recessed in relation to the pipe outside diameter (OD);
- Capable of watertight performance for internal operating pressures (including working and transient pressures), as stated in the contract documents, or external head pressure caused by high groundwater table;
- Capable of efficiently transferring jacking loads from the hydraulic jacking frame, through the pipe wall and joint to the machine face; and
- Capable of modification to allow the use of intermediate jacking stations (IJSs).

Commentary. *Types.* Most microtunnel pipe joints are gasket-sealed to facilitate rapid assembly and to provide the required rotational capabilities and connection-sealing performance. Steel pipes are also available with either interlocking push-together rigid joints or butt-welded rigid joints. Articulating steering joints may be needed when rigid pipes and joints (interlocking push-together or butt-welded) are used.

Configuration. There are several gasket-sealed joint configurations. A typical joint configuration is shown in Fig. 17-4.

Compression rings. Compression rings (packers or cushions) made of plywood, particle board, or other low-modulus

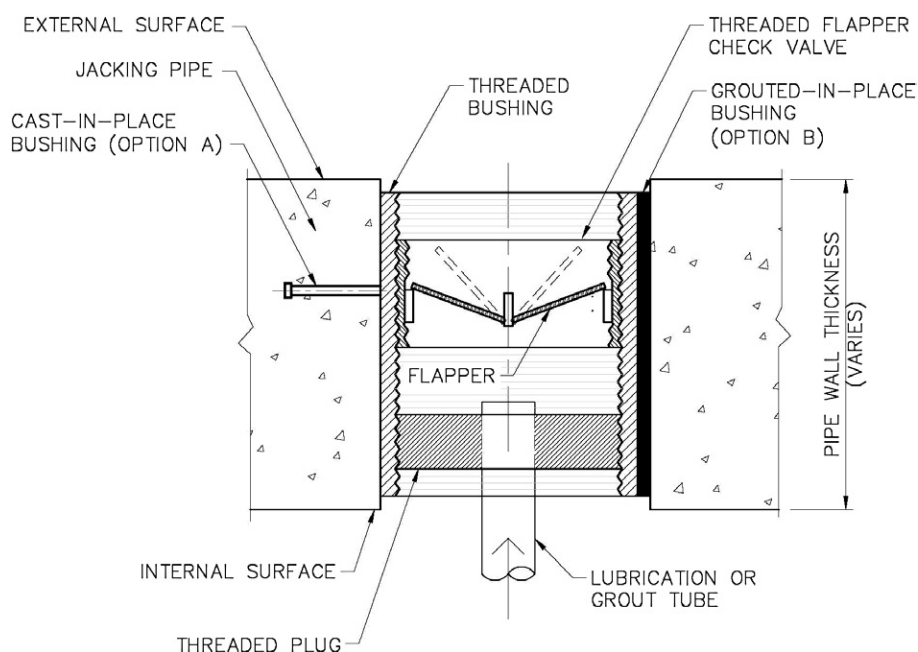


FIGURE 17-1. Typical flapper check valve (detail)

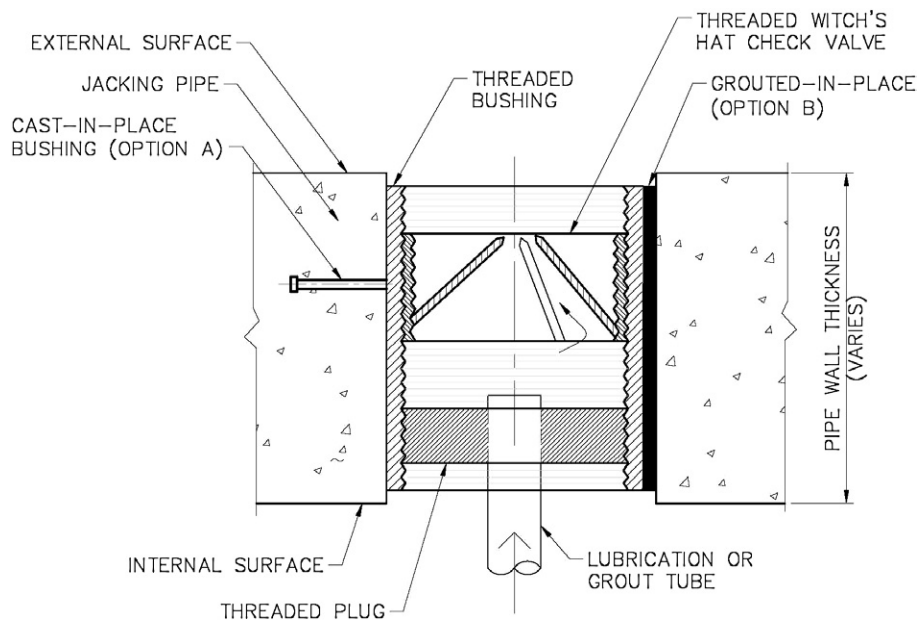


FIGURE 17-2. Typical witch's hat check valve (detail)

(compression resistance) material(s) should be placed between the pipe ends (when required) to reduce load concentrations on the pipes caused by nonplane or nonsquare ends, nonstraight sections, or angles between the axes of adjacent pipes (caused by steering). Compression rings should be as uniform as possible in compression resistance so as not to be the cause of load concentrations.

Intermediate jacking station (IJS) joint configuration. When additional jacking force capacity is required, an IJS may be inserted between jacking pipes at desired locations. A typical configuration is shown in Figs. 17-5 and 17-6.

17.6 RESISTANCE TO THE ENVIRONMENT

Pipe material and appurtenances and their resistance to environmental factors should be considered in the selection process. The use of coatings and linings is a function both of the pipe material selected and of the environment.

If a pipe is to be installed and it must transition through an area of known contaminated ground or an area subject to contamination, consult with the pipe manufacturer regarding the potential negative effects of the contaminate on the pipe, coating, and gasket materials before selecting the pipe material for use in that area.



FIGURE 17-3. Typical check valves for lubrication or grouting: Flapper and witch's hat check valves of different sizes
Source: Courtesy of Timothy R. Coss, Microtunneling, Inc.; reproduced with permission.

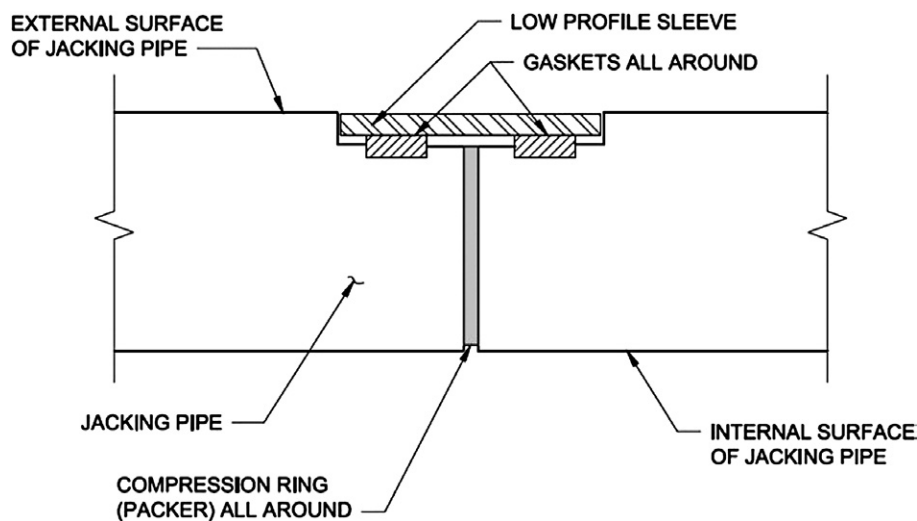


FIGURE 17-4. Typical joint configuration for jacking pipe

Commentary. In the long term, the microtunnel pipes and joints are exposed internally to the fluids and/or gases conveyed and externally to groundwater, chemicals in the ground and, occasionally, stray electrical currents and hydrocarbon contamination.

Much information is available from other sources regarding corrosive or galvanic attack of the various materials covered in these guidelines and the methods available for protection, if needed. However, exterior pipe friction during the jacking drive and extensive activity inside the pipes throughout the operation are unique to microtunneling installation. These events may have a damaging effect on the durability of the materials as well as on any coatings, linings, or other protective measures used.

Pipe installed by direct jacking, including steel casings, typically does not have a coating. The coating is damaged by the

jacking process as the exterior of the pipe is pushed through the ground.

Use of linings varies with the type of pipe used. PCP, FRPMP, and VCP are made of inert materials and are not typically subject to corrosion. DIP, RCP, and steel can be prone to corrosion in some environments, and addition of a lining can increase the pipe's effective life. A lining can be added before or after jacking. Installing a lining before jacking runs the risk of damage to the lining.

A casing installed by jacking and then sliplined with a carrier pipe is a typical means of installing a pipe that needs a coating and/or lining. The space between the carrier pipe and the casing is filled with a backfill material that can also improve and provide protection.

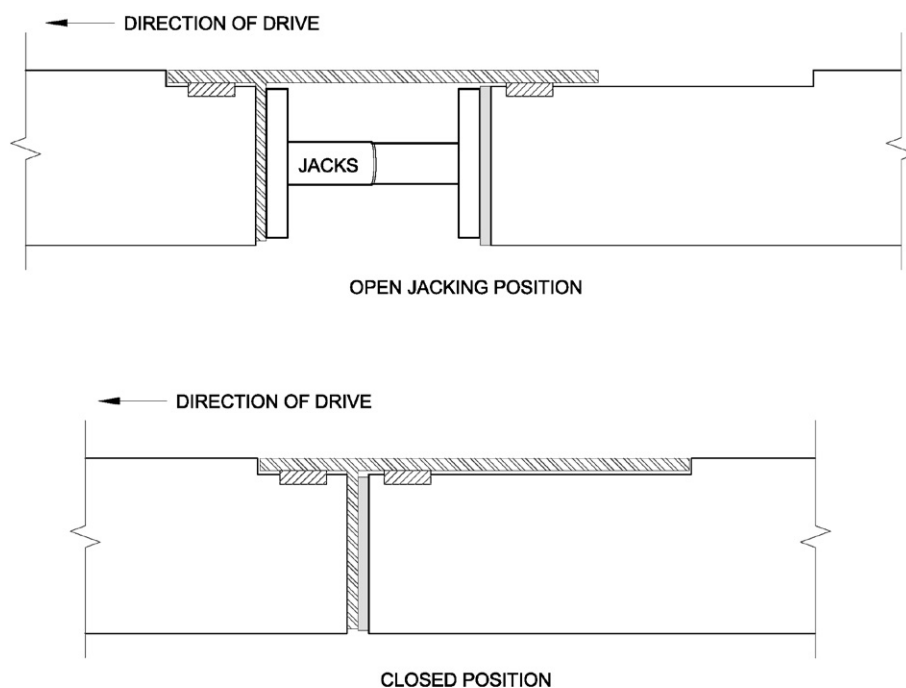


FIGURE 17-5. Typical IJS joint configurations (open position during active jacking, and closed position after jacks are removed)



FIGURE 17-6. Typical IJS assembly installed within the pipe string (jacks are recovered from the IJS after pipe installation; IJS steel can remains in the ground when the pipe is brought to a closed position)

Since joint bell sleeves may be of a different material than that used for the pipe (e.g., steel sleeves are frequently used with PCP, RCP, and VCP), separate consideration of resistance to the environment is prudent.

Consult with the various manufacturers of the pipe materials for applicable means of resisting the environmental factors

anticipated for the project. With all of the pipe materials, the engineer should use a risk-based protocol to determine the best means and methods of addressing how to resist the environment in which the pipe string is to be installed. Giving the contractor as many pipe materials and installation options as possible can be cost-effective for the project.

17.7 ANSI/NSF INTERNATIONAL 61 CERTIFICATION

All pipe materials that will be in contact with potable water should be certified to meet ANSI/NSF International 61.

Commentary. Pipes used for sanitary sewer and storm drain installations do not need ANSI/NSF International 61 certification.

For potable water applications, some of the pipe materials identified in Section 16.2 have ANSI/NSF International 61 certification. Check with the specific pipe manufacturer to determine if the proposed water pipe material has gone through and received ANSI/NSF International 61 certification. Certification can take months to obtain and is expensive. Check with the pipe manufacturers early if considering the use of a new pipe material for potable water applications. Certification of a pipe material is dependent on the final product. Pipe materials made from individual components from different sources, like sand and binder, may need recertification if the source of the component changes.

To avoid the need and expense of pipe certification, some potable water installations use a two-pass installation method. A casing is installed using microtunneling, and then the ANSI/NSF International 61 certified potable water pipe material is inserted into the casing. The space between the carrier pipe and casing is typically filled with backfill grout.

CHAPTER 18

FIELD ASPECTS

18.1 PREINSTALLATION INSPECTION

As soon as practical after their arrival at the project site, all pipes, fittings, and appurtenances should be thoroughly checked for compliance with the contract documents and accepted submittals by the contractor. The owner's representative and the microtunneling subcontractor should be given the opportunity to participate in this inspection. Any pipe, fitting, or appurtenance not in compliance should be immediately removed from the project site.

Commentary. Normally, a routine inspection of the pipes after delivery is sufficient to ascertain their condition and detect any shipping damage. Dimensional consistency should be verified at this time; of particular interest are diameter, length, squareness, minimum wall thickness, joint end surface defects, and joint seal defects or damage. On some projects, the general contractor orders the pipe. The microtunneling subcontractor should also inspect and accept the pipe before use.

Some materials, because of their characteristics, may require more extensive inspection and/or testing to verify their suitability for on-site use. The pipe manufacturer should be consulted for any specific suggestions.

Where standards of manufacture specify requirements for inspection upon delivery, these requirements should be followed, unless otherwise specified in the contract.

Additional inspections are typically done at other times, including immediately before lowering the pipe section into the shaft.

Records of all pipe inspections and acceptances should be documented by the contractor.

18.2 POSTINSTALLATION INSPECTION

The pipe should be inspected after installation to determine if any damage has occurred.

Commentary. Typically, a video or a visual inspection of the installed pipe is conducted to carry out the following activities:

- Ensure that proper joint assembly has been maintained;
- Review the general pipe condition and alignment; and
- Locate damage visible from the inside of the pipeline that may have been caused by the jacking operation or ground conditions.

The type of inspection is a function of the pipe size. Pipes smaller than 1,200 mm (48 in.) are typically video-inspected. Pipes with larger diameters typically allow for visual inspection.

Internal inspection may not reveal damage to some pipe materials. Calculation of maximum stresses at the exterior surface should be conducted as an indicator of whether spalling has occurred. If the pipe material contains steel reinforcement, spalling or cracking could cause exposure of the rebar and accelerate corrosion.

In addition, pressure testing, appropriate for the system design, may be done to confirm the watertightness of the pipe and joints.

18.3 DAMAGE CORRECTION

The method of correcting damaged pipe segments should be in accordance with the manufacturer's recommendations and should be agreed to by the owner before work begins. The contract specifications should describe methods acceptable to the owner.

Commentary. Damage to pipes and/or joints can occur during microtunneling. Because of the depth and possible surface obstructions, excavation to repair or replace the affected pipes is normally undesirable. Fortunately, there are several options for correcting some occurrences with little or no excavation, although some pipe damage is not repairable:

Pull back: MTBM and pipe sections may be pulled back to the jacking shaft with appropriate ground support for the excavated bore.

Push through: Frequently, it may be possible to push a damaged pipe through to the receiving shaft and remove it.

In-place repair: When the diameter is sufficiently large and access permits safe entry, some types of damage or leakage may be repaired in place, internally. Such repairs may be made with the following:

- Mechanical devices;
- Welding or bonding methods;
- Removal and closure techniques;
- Linings (sealing or structural reinforcement of damaged or defective areas may also be corrected with cured-in-place or form-fit linings or by sliplining with another pipe); or
- External grouting in combination with other pipe repairs.

Point excavation: Isolated defects can also usually be corrected by gaining access via construction of a small point excavation shaft. The damaged or deficient part can then be removed and replaced or repaired.

A plan must be developed and implemented to monitor and repair damage that occurs during jacking so that the drive can be completed.

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PART V

CONSTRUCTION ASPECTS

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CHAPTER 19

MICROTUNNELING OPERATION

19.1 SHAFTS

The shaft should provide safe working conditions and fully support the ground. Jacking and receiving shaft support system and sizes are typically determined by the contractor. See Chapter 14 for the design of shafts. The shaft needs to be constructed in accordance with the design.

Commentary. The shaft design should be submitted to the owner for review and acceptance. The submittal should be complete and should include earth, groundwater, thrust, and surcharge loads; wall structure; groundwater control; invert slab; wall penetrations; wall seals; and safety equipment (e.g., railings, baseboards, ladders or stairs, ventilation, lighting, and sump pumps).

In some cases the prime contractor or other shaft subcontractor builds the shaft for the microtunneling subcontractor. Close coordination is recommended between the shaft contractor or subcontractor and the microtunneling subcontractor to ensure that the shaft meets the needs of the microtunneling subcontractor.

Geotechnical instrumentation should be installed, as required in the contract documents, with sufficient time to obtain a baseline survey and data.

During shaft construction and excavation, the geotechnical engineer should visit the site to verify that the ground conditions encountered are similar to those in the design.

Any significant changes to the shaft design during construction should be reviewed and approved by the professional engineer of record.

Jacking shafts are typically circular or rectangular and aligned in the direction of the microtunnel. The ground at the shafts may require stabilization.

Circular shafts are more efficient for resisting earth pressures and may not need internal bracing as the ground loads place the support elements into ring compression. Rectangular shafts are exposed to nonuniform loading conditions and need internal bracing and struts for stability.

Base slab construction may need to be tied to the shaft support system.

Contingency plans should be developed to address the following:

- Infiltration of groundwater and/or ground;
- Excessive wall deflections; and
- Bottom heave or boiling.

The shaft support system should be constructed or modified to accommodate the breakout and reception of the machine without compromising the lateral internal support of the shaft. Breakout framing can be used for reinforcement of support systems.

A launch and retrieval seal suitable for the shaft and ground conditions should be used to control inflow of water and earth into the shafts. See Sections 14.4 and 19.2 for more discussion on machine launches.

The thrust block should be designed and installed to transfer the anticipated jacking forces to the ground mass behind the shaft through the shaft support system.

Ground treatment around the shaft may be required to provide adequate capacity for the anticipated thrust. Ground treatment may also be required outside of the shaft to control running or flowing ground conditions and groundwater inflow into the shaft during cutting of the shaft wall during launch and retrieval. The ground treatment also provides a suitable foundation for the cantilever load created by the MTBM during breakout and recovery. The weight of the MTBM can overload the ground, leading to additional grade deviation.

The guidance control instrument should be properly positioned to provide the design line and grade. The instrument should be set up in the shaft to be independent and isolated from anything that can move (i.e., shaft support system, jacking frame, and thrust blocks). Monitor the position of the instrument for any movement during the entire jacking process. See Section 19.5 for more discussion on guidance systems.

A pump should be provided in the base of the shaft to remove excess water.

Microtunneling should be completed before open-cut construction is begun near a jacking or receiving shaft.

19.2 MTBM LAUNCH

An MTBM should be positioned to launch from the jacking shaft or portal at the desired line and grade. Launch seals should be designed and used to prevent the entry of material from outside the shaft. See Section 14.4 for more discussion on seals and launch.

Commentary. Launching of the machine is the one element of the work most prone to problems. When the machine penetrates the shaft wall, ground and groundwater under pressure are trying to enter the shaft. The amount of pressure is a function of the depth of the launch and the nature of the ground conditions. Drilling fluid used to stabilize the face and lubricants pumped into the annular space are under pressure as well and also try to return into the shaft. Launch seals must be designed for the ground and load conditions for each project. Simply reusing a launch seal because it worked on a previous project is not enough.

An appropriate launch seal should be fitted before the launch of the MTBM. If the seal is damaged, it must be repaired, replicated, or replaced.

Launch seals include single or double rubber donut gaskets mounted to the shaft wall, with slide plates to prevent inversion.

Inflatable seals have been used with injectable grease or brushes. Ground improvement, in the form of jet grouting, can be used to improve the ground's bearing capacity and stand-up time during launch.

If the launch seal fails and material enters into the shaft, many problems can result. A loss of grade with the machine or pipe string can occur. Voids or sinkholes outside the shaft can form. Surface settlements can result. Injuries to workers in the shaft can occur. Loss of shaft support can occur.

On completion of setup of the jacking frame in the shaft to correct line and grade, the MTBM should be prepared for launch. All functions, from steering, face support, spoils removal, and spoils handling and disposal to the lubrication system and supplies, should be checked immediately before the start of microtunneling.

The MTBM should be positioned correctly on line and grade with reference to the guidance system and should enter the ground through the shaft wall. Sometimes the MTBM cuts its own entrance window in special circumstances, such as through a predesigned unreinforced concrete window (eye) or through a preweakened launch eye, depending on the shaft construction method used. Other times, the shaft supports are cut out in advance or simply pulled up to expose the ground conditions when steel sheeting that is not excessively long or toed in is used.

Once the MTBM exits the jacking shaft, it becomes supported by the ground. The bearing capacity of the ground directly outside the shaft is critical in providing the necessary MTBM support. Furthermore, these ground conditions may be disturbed by the shaft support installation and shaft excavation as well as by removal of the shaft supports allowing the MTBM to launch. Diving or settlement of the machine can occur. This problem can be aggravated by low-strength ground conditions and/or by the amount of cantilever of the MTBM as its center of gravity moves beyond the jacking frame rails and through the entrance eye. The machine operator should be prepared for these occurrences. A launch plan should be prepared and followed.

When extending the MTBM through the launch seal, make sure to consider and factor in the cutter wheel tooling used to create overcut as well as sharp weld spatter, fins, rough ends, articulation joints, and other features that could cut, damage, or affect the profile of the sealing system.

MTBMs are being launched from deep shafts with more than 30 m (100 ft) of groundwater head. Special seal designs are needed in areas where there is high groundwater. Also, excessive groundwater pressures push the MTBM and pipe string back into the shaft when the jacks are retracted. To resist these pressures, contractors should use pipe clamps or brakes applied to the jacking pipe. If the pipe string is allowed to move backwards, the launch seals can invert, allowing material and fluids from outside to enter into the shaft. This is an important consideration because losing the seal with such high groundwater pressure can lead to catastrophic failure from rapid shaft flooding and ground loss.

19.3 MICROTUNNELING PROCESS

The microtunneling process begins with the jacking forward of the MTBM steering head. When the MTBM has been advanced sufficiently, the first pipe is placed in the jacking frame, and the pipe joint and service lines are connected. Microtunneling proceeds by jacking the MTBM and successive jacking pipes forward. The spoil transport system (auger or slurry) provides earth pressure balance (EPB). EPB is controlled by the rate of removal of the spoils in relation to the rate of MTBM advance. This process is used to keep settlement or heave within the

required tolerances. For slurry machines, an additional constraint is that the circulation of slurry must be maintained at a velocity great enough to ensure that spoils are transported to the surface without plugging.

A microtunneling system balances the earth and groundwater pressures by adjusting the slurry pressure in the head and the face pressure of the machine to equal the pressures measured. See Section 19.4 for more detailed discussion on drilling fluid mix design and the properties needed to counterbalance pressures and transport excavated material.

The MTBM is actively steered by reference to a laser beam or other guidance system, and steering corrections are made to maintain line and grade within acceptable tolerances. See Section 19.5 for more discussion on the guidance system.

Enough pipe to complete the drive should be available, inspected, and ready for use before launching. Once the pipe has been pushed into the ground, the cutter wheel is stopped and the muck removal system is stopped or closed, to isolate and maintain the pressure balance. Service lines are disconnected, and the jacking frame is retracted. A new pipe is placed in the jacking frame, and the service lines are reconnected. The pipe's leading edge is pushed to mate with the joint of the previously jacked pipe. Slurry circulation is started, and tunneling continues.

Lubrication is injected into the annular space between the excavated ground and the outside surface of the jacking pipe. Lubrication typically commences once the first pipe is completely through the launch seal and into the ground. Lubricant can also be injected into the launch eye to lubricate the pipe as it passes the seal. See Section 19.6 for more discussion on lubrication mix design and the properties needed to reduce friction and jacking forces.

Commentary. The actual ground conditions dictate the operation of the MTBM. The balancing of earth and groundwater pressures is achieved by monitoring the face pressure (thrust against the steering cylinders), the cutter head torque and speed, the slurry chamber inlet and outlet pressures, and the slurry flows. From this information, the operator analyzes the ground conditions and adjusts the excavation and slurry balancing to advance efficiently without over- or underexcavating. The operator is responsible for maintaining constant face support.

The MTBM allows the operator to implement a variety of slurry path flows (bypasses). The slurry inlet valve may be closed, which opens the slurry bypass. The bypass mode is used to flush the slurry circulation system and during short periods of shutdown. For longer stoppages, the slurry pumps are turned off, and the valves in the bore pit are also closed. Some MTBMs allow drilling fluid to be injected in front of or behind the MTBM's cutting chamber. This injection can have a significant effect on the advance rate and on the ability of the MTBM to support the ground. Other MTBMs rely on cutter wheel configurations instead of on the drilling fluid injection point to control the rate of excavation and the amount of face support along with the resulting slurry pressure.

The operation of the MTBM should be recorded, either with an automated system or manually on a paper form. Either way, appropriately selected operational functions should be recorded. The frequency with which the functions are recorded should be spelled out in terms of elapsed minutes or number of times per pipe joint. The minimum number of recorded data points per pipe joint of 10 ft or less is three. The record should also include reference information, such as project name, time, date, pipe number, and weather conditions. The operator should also include breakdowns and work stoppages, along with the reason

for the breakdown or stoppage, in addition to observations of settlement, unusual events, and guidance system adjustments.

It is prudent to launch the MTBM only after the receiving shaft has been successfully constructed. If the machine is launched with the receiving shaft still under construction, the project runs the risk that the machine may arrive outside the receiving shaft and have to wait for the completion of the shaft. Letting the machine sit in the ground can be problematic and should be avoided.

19.4 DRILLING FLUID USE DURING CONSTRUCTION

The drilling fluid used in the MTBM's slurry circulation system should be designed to accomplish the following tasks:

- Mix with and transport the excavated material to the surface for separation;
- In combination with mechanical pressure, counterbalance the earth and groundwater pressures at the face of the MTBM; and
- In certain ground conditions, accommodate the addition of additives to enhance performance of the excavation process.

Commentary. See Section 13.6 for a discussion on drilling fluid design.

The slurry circulation system pumps drilling fluid to the face of the MTBM. The drilling fluid is mixed with the spoils and is then pumped back to the surface for separation as a slurry.

During the circulation process, the MTBM operator counterbalances face pressure with a combination of adjustments affecting the following factors:

- Advancement rate;
- Rotational speed of the cutter wheel (if a fixed-rpm MTBM is not being used); and
- Drilling fluid pressure in the slurry circulation system.

Different ground conditions require different settings and/or pressures for the aforementioned functions. Yet in all cases the total applied face pressure should equal the total realized pressure being exerted against the face of the MTBM from a combination of the hydrostatic and mechanical pressures. The total face pressure exerted can be calculated by dividing the total steering cylinder thrust at rest by the total face area.

To launch, the MTBM operator selects an initial slurry path configuration (i.e., drilling fluid to face, or to the chamber if the MTBM is equipped for multiple drilling fluid injection location points) and slurry velocity. The operator observes the feed and return line pressures that the initial configuration generates when the slurry circuit is opened to the face. The operator adjusts and maintains a drilling fluid pressure by changing the pump speed on the feed side or the return side of the slurry circuit, or both. The operator attempts to counterbalance the estimated groundwater pressure but may not know the exact groundwater pressure because of the variability of the ground. The operator may try to measure the groundwater pressure by cleaning out the slurry circuit, shutting the pumps off, closing the feed valve, opening the return valve, and observing the pressure sensor reading in the drilling fluid chamber. The feed side pressure of the slurry circuit should slightly exceed (typically 2 to 6 psi) the return side (groundwater) pressure of the slurry circuit. The slurry feed and return flow rates must also be approximately balanced; otherwise the slurry can wash out the spoils, leading to settlement. The key is that the total applied pressure must be equal to or slightly

exceed the actual pressure exerted on the MTBM face by the combination of earth and groundwater pressures.

Another function of the drilling fluid is to mix and transport the excavated material for removal. The drilling fluid is circulated and pumped at a velocity great enough to ensure that the spoils are transported to the surface without plugging the slurry return line.

The composition of the drilling fluid mix varies according to the anticipated ground conditions. It may be water or it may be an engineered water-based fluid mixture that may contain bentonite, polymers, or other additives to enhance desirable properties and/or to aid in the separation of solids.

If a drilling fluid containing bentonite, polymers, wetting agents, and/or additives is used in the microtunneling system, it should be engineered to the specific ground conditions encountered during the mining process. Specific testing procedures to monitor the drilling fluid are recommended. Typically these include testing the drilling fluid that is being sent to the face and testing the slurry returned to the recycling plant. Periodic testing of the drilling fluid and slurry is a good practice. Tests could include density (with a mud balance), viscosity (Marsh funnel test; see ASTM D6910), sand content (sand content kit), pH (pH test strips) and available calcium (hardness). In brackish water or saltwater, chloride testing (Cl^- test kit) is highly recommended. Test results should be logged, and copies should be forwarded to the appropriate personnel. A complete history of the testing results should be kept on site for reference. Trends observed from testing results may help in predicting potential problems and mitigating them.

In noncohesive soils (i.e., sands, gravels, cobbles, and boulders), drilling fluid with a bentonite base is recommended. Viscosity should be closely monitored to minimize the effect on the slurry separation plant. It is important to have enough bentonite to effectively control the face during the mining operation. Face control minimizes over excavation and loss of drilling fluid to the formation. To minimize fluid loss into the formation, it is recommended that the drilling fluid be designed to include the ability to form a filter cake.

In soils with greater than 15 to 20 percent fines (i.e., minus No. 200 sieve), a bentonite-based drilling fluid may not be effective and is not recommended. As the viscosity increases from the fines being mined, the pump pressures increase, and the ability to transport the spoils decreases. Adding a centrifuge and using a polymer additive can help to maintain the viscosity of the drilling fluid and the slurry.

In clay soils, jetting the face may be permitted, to prevent clay balling or clogging of the cutter head. Jetting may be done on a periodic basis. Water, foam, surfactants, and dispersants are commonly used in the jetting system. The advance rate should increase from jetting with an appropriate fluid. Large volumes of jetting fluids increase the slurry volume and are likely to change the composition of the slurry. The contractor should plan for additional disposal costs when jetting is used. Any polymeric additive used in jetting must be compatible with the slurry separation plant.

19.5 GUIDANCE SYSTEM

The MTBM is actively steered by reference to the laser beam or other guidance system, and steering corrections are made to maintain line and grade within the required tolerances.

The instrument and guidance system should be properly installed and calibrated before launch of the MTBM. If using a laser, the laser beam's focal length should be at least half the length of the longest drive. If using a gyroscopic system, the

system should be calibrated. If using a theodolite system, the target positions should be confirmed. If using a water-level system, the operational requirements should be compatible with the design grade change. In all cases, the contractor's surveyor needs to ensure that the survey method and frequency are compatible with the guidance system and that the accuracy of the survey will provide a design within acceptable tolerances.

The line and grade readouts should be checked to ensure that the guidance system has not moved during the pipe connection process. The alignment of the guidance system should be checked at least once per shift to minimize directional deviations.

The operator should maintain a log sheet that documents all before-and-after readings for each adjustment, and the survey results including the date, the time, and the location as a pipe number. All offsets should be recorded with equal care, and all readings from that offset should be properly noted.

Commentary. Before the MTBM is launched, the guidance system should be calibrated and set up according to the manufacturer's written directions. The MTBM operator and the project personnel should be familiar with the proper operation and reading of the instrument as well as with the survey requirements. Some guidance systems use plane surveying (simple systems using a pipe laser), whereas others use geodetic surveys, which allow for greater survey accuracy and account for the Earth's spheroid shape. Some guidance systems provide a direct reading, with a positive reading indicating that the MTBM is high, and other systems provide an inverse reading, with a positive reading indicating that the MTBM is low. An MTBM operator using a new system may want to use two systems in tandem, to verify proper operation. If the systems are not consistently producing identical readings, the issue should be resolved before continuing.

If using a laser, the MTBM operator should set up the laser, and a second competent person should confirm the line and slope readings before launch. The second person should confirm the line and slope readings each time the laser is reset. The operator should also confirm that the vertical control is reading in the proper units (degrees, radians, grade, or slope).

The guidance control laser or other instrument should be positioned and isolated from the shaft so that it is not affected by any movement induced by the jacking force. If the jacking force causes movement, it is normally first detected in the shaft walls, the thrust block, the jacking frame, and the concrete base slab and is typified by a rising laser as forces increase and by a dropping laser as forces decrease. In extremely soft or sensitive soils, the entire shaft may move with each jack thrust and may deflect the alignment of the laser as well. This movement may not be detectable with lasers equipped with automatic alignment checks, so if such conditions are a possibility, daily surveys of the shaft and tunnel may be warranted. Some lasers have the beneficial feature of shooting an isolated target to identify laser movement. The contractor should verify the line and grade of the tunnel and provide marks at each end of the shaft to allow a string line to be set up with vertical "plumb bobs" to align the laser beam reference accurately. The owner's surveyor should confirm that line and grade requirements in the contract documents are being met. Although laser guidance is the most commonly used system, other systems should be installed and used with similar precautions to ensure accurate guidance (Boyce and Bradshaw 2015).

Once the guidance system is set in place in the shaft and correctly aligned to the planned line and grade, the instrument and support need to be protected from accidental movement.

The guidance system needs to be checked against a reference, daily and at any other time when it has been moved. At least

once a week the entire alignment should be checked against a reference that is removed from the construction activity.

All reference checks and alignment changes should be recorded, with the effect on the alignment noted.

Currently (in 2015), the laser is the most common instrument used for providing the design line and grade reference during construction. The laser is typically limited to 185 to 245 m (600 to 800 ft) unless a specially manufactured long-distance laser is used. These lasers are typically limited to 365 m (1,200 ft). Other guidance systems that are currently available include water-level systems for grade only, stationary robotic theodolite systems, moving robotic theodolite systems, and gyroscopic systems. The moving robotic theodolite system and the gyroscopic system are both designed for curves.

Lasers are beams of high-intensity light. Light beams can be deflected with temperature variations. When a laser is used, the pipe string should be ventilated at all times, to maintain an air flow that helps to circulate the air and avoid hot and cold spots with the pipe string. Pumps, booster pumps, the MTBM motors, hydraulic transmission lines, and hydraulic power pack motors are sources of heat generation and can deflect the laser's position on the target. In cold-weather environments, and especially when temperatures fall below freezing, the atmospheric distortion of the laser can be much greater. Covering and/or heating the shaft may be necessary to minimize this temperature gradient. Total station survey instruments rely on lasers as well, so they are also subject to the same atmospheric distortions as a shaft laser unless the surveyor takes the total station into the tunnel and closes the survey loop to null out the distortion. The guidance system should be protected from direct exposure to the sun and heating up of the unit over the course of the day.

The contractor should select a surveyor who is experienced in underground surveying. Underground surveying is becoming a lost art because of the use of GPS surveying instruments, which do not work underground. And, as stated above, the surveyor may be required to enter the tunnel to overcome atmospheric distortion, depending on the survey equipment being used.

Additional surveying is needed near the end of the drive to help ensure that the MTBM will hit the receiving shaft and entry seal. This surveying should take place well within the distance of the receiving shaft, to allow steering corrections of no more than 1:300.

Underground surveying has several aspects that are unique to the underground environment. Some of these aspects include the use of a GPS for surveying. The transfer of line into a shaft cannot be performed using a GPS because of the limited space in the shaft. Transfer of line is a special skill that is limited to underground surveying and is not practiced regularly by other surveyors. A GPS does not work from within a shaft or underground because the instrument cannot establish access to the satellites required for a reading. Underground surveying uses a very short back sight, which limits the accuracy of a survey. For these reasons, the owner, the engineer, and the contractor must use an experienced underground surveyor. Once the tunnel is completed, a closed-loop survey needs to be completed to properly locate the tunnel and ensure that the tunnel meets the design requirements.

The following are several issues that may affect line and grade control during tunneling, or that may affect the ability to verify the as-built condition:

- *Bumping lasers:* The laser and support should be protected from contact with people and equipment.
- *Shaft walls and other structures:* Lasers should be independent of the shaft walls, the jacking frame, and any other

structure that is likely to move when jacking forces are applied. A laser can be mounted on an isolated support that is anchored to the base slab. In extremely soft soils, however, and when the required jacking thrust is high, even shaft slabs can move and deflect the laser. A separate survey may be required to verify the stability and integrity of the shaft wall.

- *Removing the laser at the end of the day:* This procedure may cause a delay in the morning if line and grade controls are not reestablished. It may also cause error if the laser is improperly reset. The procedure introduces one more source of error.
- *Using different personnel to sight the line:* Each person may sight the plumb bob lines differently, which can lead to a deviating line. An untrained person may use the wrong measurement or unit of measurement to set the grade.
- *Heat within the tunnel:* Laser light may become distorted because of inversion layers within the tunnel. The tunnel should be ventilated to remove heat from the tunnel. This problem is worse if jacking extends more than 600 ft (183 m), depending on the jacking pipe's diameter (smaller diameters are more problematic), whether the temperature is below freezing, and whether a hydraulic power pack is used in the tunnel, regardless of the amount of ventilation applied.
- *Machine off line and/or grade:* The guidance laser may need to be offset vertically or horizontally in order to see the MTBM target and thereby provide guidance. These offsets need to be recorded, and all future readings need to clearly note such offsets.

All line and grade readings must be recorded immediately before and after adjustments are made to the guidance system and to the survey control points.

19.6 LUBRICANT SYSTEM

Lubricants should be used in the microtunneling and pipe jacking process. The pipe lubricant should be designed to accomplish the following tasks:

- Reduce friction between the pipe and excavated ground as the pipe is jacked forward;
- Maintain the annular space between the pipe and ground;
- Form a filter cake to limit the amount of fluid loss into the ground;
- Maintain its composition when pumped under pressure into the annular space;
- Maintain its composition and properties during long periods of inactivity; and
- Resist the loss of integrity when exposed to chemical contamination.

A lubricant should be injected from the rear of the MTBM and, if size allows, through ports in the pipes at regular intervals to reduce friction and fill any annular space remaining from the overcut of the cutter wheel.

Commentary. The lubricant is water-based material with bentonite and/or polymers. Lubricant is pumped into the annular space created by the overcut. Lubricant blends with ground in the annular space. The lubricant should have the following characteristics:

- Viscosity geared to the ground conditions being tunneled through; and

- Enough thinness to flow around the pipe, but enough thickness not to be lost to the formation.

In sand or gravel, the working viscosity should typically be 80 to 100 seconds per quart (s/qt), using a Marsh funnel test (ASTM D6910). In clays, the lubricant should contain polymers and the viscosity is lower.

Fluid loss should be tightly controlled and in amounts less than 12 mL (0.4 oz), using a standard API fluid loss test.

The original concept for lubrication fluids was taken from the horizontal directional drilling (HDD) industry. This approach is not correct. Good HDD fluids have the ability to carry a lot of cuttings because they exhibit high yield points and high gel strengths. Lubrication fluids used in microtunneling do not need to have these capabilities. Typical HDD projects are completed in days, whereas microtunneling projects may be completed in weeks and months. Nonproductive times are common with microtunneling. Jacking frames need to be reset in order to drive a full pipe. Power cables, slurry lines, and lubrication lines must be disconnected before another pipe can be placed in the jacking frame. Lowering and aligning the pipe and reconnecting the lines result in the loss of production time. During these needed operations, the lubrication fluid is static, with the force of the surrounding ground acting to close the annular space. Designing the fluid with a high gel strength and yield point when pumped could cause ground erosion and would increase jacking forces if it is allowed to sit (and gel) in the annular space. Lubricants with high gel strengths and highly progressive gels should be avoided, if possible.

When designing the lubricant, consider whether the lubrication fluid will become contaminated during the jacking process. Contamination can change the chemistry of the fluid and neutralize the lubricant's properties. Contamination can include high water hardness, saltwater inclusion, concrete used for the shaft support, and jet grouting used for launching and receiving seals. The effect of mining through grout, flowable fill, and contaminated areas should be calculated when making the lubricant. Treatment plans to minimize these effects should be put into place. Effects of groundwater should also be calculated, and treatment plans should be initiated.

Once the annular space has been calculated, the amount of lubricant to pump can be determined. Sandy and gravelly ground conditions typically benefit more from lubrication and may require more lubricant to be pumped into the annular space than clayey ground conditions. A method of tracking the volume of lubricant pumped per pipe is recommended. Automatic injection systems are highly recommended for controlling specific amounts and for placement of the lubricating fluid. The automated system opens the appropriate valve and allows the lubricant to be injected at a predetermined pressure and duration. The duration is either time- or volume-based. The cycle continues as new valves are added as the pipe string is extended. At a bare minimum, the annular space must be completely filled. Sandy or gravelly ground conditions can often take 2 to 3 times the amount of the calculated annular space. Firm clay soils often take 1 to 1.5 times the calculated annular space.

In most projects, all of the materials added to the lubrication should be ANSI/NSF International 60 Clean Water approved. Soda ash is not ANSI/NSF International 60 Clean Water approved but may be required to control calcium and increase the pH. Saltwater additives are not ANSI/NSF International 60 Clean Water approved because saltwater is not potable water. Commonly used liquid PAC (polyanionic cellulose) is suspended in a proprietary mixture containing food-grade mineral oil. These additives are used to control fluid loss. In all cases, consideration

should be given to the amount of hydrocarbons included in the additives. Hydrocarbons are particularly common in drilling fluids for oil wells and should be eliminated or minimized, if possible.

19.7 WORK HOURS

If possible, work hours should not be restricted during microtunneling operations. See Section 7.7.

Commentary. The typical microtunneling project does not have to be a 24-hour operation. However, microtunneling is a time-sensitive operation and cannot be shut down for extended periods of time without the risk of tunneling failure. The contractor performing the microtunneling is frequently not local to the project, and therefore the contractor needs to be notified of any restrictions on work hours and informed about the application process for acquiring a waiver. In addition, the contractor should not start a drive that cannot be completed before one of these extended shutdown periods. A typical example of an extended shutdown would include Thanksgiving Day (a Thursday) plus the following Friday, Saturday, and Sunday, making a four-day weekend.

Noise and traffic ordinances may impose requirements that prohibit certain operations during certain time periods, and the contractor needs to be informed of these restrictions and about how to obtain a variance.

Multiple shifts should be permitted if noise does not exceed the maximum levels contained in local ordinances. In particular, continuous jacking should be permitted under the following circumstances:

- Where squeezing soils are encountered;
- Where the actual jacking force that is required approaches the capacity of the jacking system or the designed jacking capacity of the pipe; and
- Where drive lengths are long.

Restrictions on truck traffic may stop a project if spoils cannot be removed from the site in a timely manner.

An extended-hours operation requires at least two operators and crews, which may introduce additional coordination and operational challenges. These challenges may include fatigue; differences in operator styles; difficulties in performing surveys; and difficulties in performing maintenance, deliveries, and support services.

19.8 MTBM RETRIEVAL AT DRIVE COMPLETION

An MTBM is retrieved at drive completion from a receiving shaft installed before the beginning of the microtunnel drive. In some cases, the MTBM can be recovered from an open trench or portal on land or underwater. Care must be taken to prevent excessive amounts of ground and groundwater from entering the receiving shaft as the MTBM is jacked into it. The contractor should provide the means and methods for the MTBM recovery on the basis of the type of receiving shaft supports and ground and groundwater conditions.

Before the MTBM enters the receiving shaft, the ground at the point of retrieval may require stabilization. The receiving shaft wall should not be removed until the MTBM has arrived at the shaft wall.

Underwater retrievals of the MTBM may be required for intake and outfall projects; see Section 19.9 for more discussion.

Commentary. The receiving of the MTBM is a multistep process. It must be determined on the basis of the ground

conditions in the area outside the receiving shaft wall at the point of MTBM retrieval and depends on the shaft supports used. The goal is to prevent excessive amounts of ground and groundwater from entering as the MTBM is jacked into the receiving shaft.

The following are typical receiving shaft systems used in microtunneling:

- *Steel sheet piling:* When steel sheeting is used for the receiving shaft supports, there are two procedures, depending on the ground conditions. First, if the soil is stable and groundwater is controlled, the steel sheeting must be cut out of the way of the MTBM once its location is determined. Second, if the soil is not stable and there is groundwater pressure to contend with, a concrete headwall must be constructed with an exit eye and seal at the projected point of MTBM entry. The exit eye must be filled through the concrete headwall with material to create a “soft eye” that will support the soil and hydrostatic water pressure when the steel sheeting is pulled up to allow MTBM entry. The soft eye material must allow the MTBM to mine through to enter the receiving shaft. The exit seal minimizes inflow of materials around the MTBM as it enters. Alternatively, a top hat filled with soft eye material can be used to confine and minimize the inflow of material if the headwall soft eye and seal are judged inadequate for the anticipated soil and groundwater pressures.
- *Fixed shaft supports:* If the shaft supports, such as liner plate, soldier pile and lagging, and even steel sheeting, cannot be pulled up, they must be removed at the point of MTBM reception once the MTBM is at the shaft. The ground conditions must be stable to allow this removal without excessive inflow of material. Stabilizing the soils outside the shaft at the MTBM point of entry may be required and in some cases can be accomplished by back grouting of the shaft supports or by locally injecting grout into the soils from inside the shaft. Concrete headwalls and exit seals may not be necessary if the soils are stabilized and dewatered. Ground stabilization of the soils from the surface, outside the shaft at the point of MTBM entry, may be required before the shaft supports can be removed. Ground stabilization should be installed in advance, to allow proper curing time.
- *Minable shaft supports:* Secant piles and slurry walls using fiberglass reinforcing at the exit eye can serve as the soft eye, and the MTBM can mine through them. A concrete headwall with a shaft seal may be required to control material inflows around the MTBM.
- *Open trench or portals:* Occasionally, MTBMs are recovered by open-trench construction or jacking into a portal. Exit headwalls and seals are not used. The MTBM is simply jacked to a station, the trench above it is excavated, and the MTBM is lifted out. For portals, the MTBM is simply jacked into the open portal. This method is often used in underwater retrievals for raw water intakes and other such utilities. See Section 19.9 for underwater retrievals.

Once the receiving shaft has been constructed, the area outside the shaft wall at the point of retrieval may need to be stabilized to prevent ground and groundwater inflow. The ground improvement material must have sufficient time to cure before the MTBM arrives so that the MTBM does not become lodged within the improved ground. The MTBM must be able to mine through the material even if the MTBM's progress is slower or faster than anticipated.

The first step in receiving an MTBM is to locate the center of the MTBM before mounting an exit seal. Once the actual or projected center of the MTBM is located, the receiving process commences with mounting the exit seal to the shaft wall, bracing the shaft wall as required by the design, removing the shaft wall, jacking the MTBM into the receiving shaft, disconnecting the MTBM utility services (including slurry lines), removing the MTBM from the pipe string, and then lifting the MTBM and any trailing cans to the surface as well as removing the utility services (including slurry lines) from the installed pipe string and cleaning the installed pipe string.

Accurate surveying of the receiving shaft exit eye and the MTBM is necessary so that the receiving shaft can be prepared for entry. In self-supporting dewatering soils, pulling or removing the supports is done only when the MTBM is at the shaft and its center is located. This method allows accurate removal of the shaft supports and minimizes material inflow.

In projects with hydrostatic heads that are not compatible with ordinary seals, grouting outside the shaft reception wall and high-pressure seals may be required. Grouting may include jet grouting, chemical grouting, soil mixing, or another ground stabilization method to prevent flowing ground from flooding the shaft. High-pressure exit seals include double seals, inflatable bladder seals, and top hat seals. Double seals require two elastic seals mounted to the steel ring. An inflatable seal has an air bladder that is inflated to increase the hydrostatic pressure that the seal can resist when jacking stops. A top hat is a seal that is mounted to the shaft wall and backfilled with weak grout. The MTBM then tunnels through the shaft wall and into the top hat. Once the MTBM is in the top hat, the front steel plate is removed and the MTBM is pushed through the seal and into the shaft.

The recovery process consists of the following steps in general:

- Drive the MTBM near or against the shaft supports.
- In stable or stabilized soils, locate the MTBM and neatly remove the shaft supports and mine into the shaft.
- In unstable soils, pull the steel sheeting, mine through the soft eye or into the top hat, and engage the seal.
- In minable shaft supports, the MTBM mines through the shaft wall and engages the seal, if necessary.
- Jack the MTBM and trailing cans into the receiving shaft on steel rails or bedding stone, or attach them to a crane hoist as they enter so that they can be hoisted to the surface.
- Recover all the MTBM utilities from the pipe string by back pulling to the jacking shaft.
- Perform contact grouting at the entrance and exit seals to minimize inflows of materials and movement of the jacking pipe. Also perform contact grouting along the jacked pipe if voids are suspected or if contact grouting is required by the permitting agency.
- Clean the pipe string, and test accordingly.

19.9 WET RETRIEVAL OF MTBM

Underwater retrieval of the MTBM may be required for intake and outfall projects. Specialized procedures are needed in underwater retrievals to ensure safety.

Commentary. The retrieval of an MTBM at the bottom of a lake or reservoir can be done but requires special procedures. These are a few of the special procedures to be included:

- Install a watertight (water-resistant) bulkhead at the rear of the MTBM, and install a second watertight bulkhead inside the pipe between the jacking shaft and the MTBM, with air and water fittings attached before the launch of the MTBM.
- Mobilize a heavy all-terrain crane and work barge for bench construction and retrieval support.
- Prepare a bench in the lake or reservoir bottom for MTBM exit.
- Excavate a microtunnel from the jacking shaft onto the prepared bench in the lake or reservoir.
- When the MTBM daylight in a body of water, stop rotation of the MTBM cutter wheel. Dead-push the MTBM, without rotating the cutter wheel, to ensure that no soil or rock remains at the face to obstruct retrieval. Use divers to confirm that the face is free.
- Remove slurry lines, control cables, and other equipment from within the pipe string.
- Jack the MTBM forward. Attach lift slings and flotation devices to the MTBM, using divers.
- Flood the pipe between the bulkheads. Pressurize with air, and push the MTBM off the pipe.
- Slowly float the MTBM to the surface, using divers and flotation devices.
- Retrieve the MTBM with the barge and the crane.
- Load the MTBM onto the barge and transport it back to shore.
- Lift the MTBM from the barge onto land.
- Install a valve in shaft in the dry. Flood the pipe between the valve and pipe bulkhead. (The pipe will also be filled with water from the pipe bulkhead to open-end in the lake or reservoir, balancing water pressures on each side of the bulkhead).
- Use divers to remove the interior bulkhead. The bulkhead may have to be cut into pieces and removed through a valve.
- Use divers to install a trash and fish screen on the end of the pipe to complete the intake pipeline.

For outfall, a diffuser may be installed on the open end of the pipe.

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CHAPTER 20 MACHINE PERFORMANCE

20.1 ANNULAR THICKNESS, ANNULUS, AND ANNULAR SPACE

The annular thickness is the radial distance between the largest outside radius allowed in the tolerance of the jacking pipe and the outside diameter of the excavation as created by the gauge cutter. The annular thickness is equal to the overcut plus the shield cut. The annular thickness is normally between 10 and 25 mm (0.4 and 1 in.) for soil conditions and 20 and 40 mm (0.8 and 1.6 in.) for rock conditions. If the annular thickness is greater than 20 mm (0.8 in.) and the internal diameter of the jacked pipe is of sufficient size for personnel entry (typically 42 in. (1.07 m) or larger), or if voids are suspected, contact grouting is often required by the permitting agency. The use of an annular thickness is required principally to reduce pipe friction and not to facilitate steering. The amount of the annular thickness should be based on the following factors:

- Heave and settlement requirements;
- Jacking forces;
- Pipe diameter;
- Ground conditions;
- Contractor's experience in similar conditions;
- Relative size of the MTBM and pipe;
- Type of lubrication (if used); and
- Horizontal and vertical alignment requirements.

The annulus is the theoretical area between the gauge cut and the outside diameter of the jacking pipe. The annulus is calculated by assuming that the pipe remains centered within the bored excavation. The actual annular thickness varies around and along the bored excavation.

The annular space is the theoretical volume and is equal to the annulus times the length of the microtunnel drive. The annular space is used to calculate the settlement trough, lubrication quantities, and potential quantities of contact grouting.

Commentary. Fig. 20-1 illustrates the various terms associated with the annulus and annular thickness. See Section 20.2 for a discussion of overcut.

The amount of annular thickness may be included in the specifications.

The annular thickness, which is used to calculate the annular space, is critical to the determination of ground settlement for soils. The engineer must assume the minimum and maximum annular space to evaluate the settlement risk. The contractor uses the minimum and maximum annular space to determine lubrication quantities. The assumptions used in the calculations should be confirmed in the field. The specifications should allow the contractor flexibility in selecting an annular thickness compatible with the contractor's means and methods and with the evaluation of settlement risk. If the annular thickness and the resulting annular space in soils are too large, the following problems may occur:

- Excessive surface subsidence;
- Misalignment of the pipe segments, creating differential thrusts and possible point loads at pipe joints;
- Difficulty in steering;
- Groundwater and slurry movement along the external surface of the pipe string; and
- Reduction in the effective steering range of the MTBM.

If the annular thickness and the resulting annular space in soils are too small, the following problems may occur:

- High jacking forces from increased skin friction;
- Pipe breakage caused by increased skin friction; and
- Peeling of the pipe joint collar.

For working in rock, the contractor should be allowed flexibility regarding the amount of annular thickness, since subsidence is not an issue. Contact grouting, if needed, should be performed within the contract-specified time after completion of a drive.

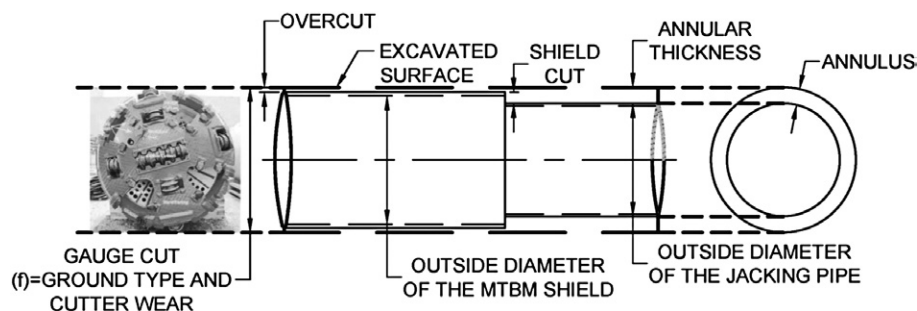


FIGURE 20-1. Diagram illustrating terms associated with the annulus and annular thickness

If the annular thickness and the resulting annular space in rock are too large, the following problems may occur in addition to those listed above:

- Flotation, resulting in failure to achieve vertical alignment tolerances; and
- Misalignment or deflection of the pipe segments, creating differential thrusts and possible point loads.

If the annular thickness and the resulting annular space in rock are too small, the following problems may occur in addition to those listed above:

- High skin friction caused by buildup of cuttings under the MTBM and the pipe string;
- Spikes in the jacking forces as rock fragments protrude into the annulus; and
- Peeling of the joint collar (once the collar is moved, the internal gaskets no longer seal the joint, and the pipe is subject to leakage).

Each of the above problems may result in failure to complete the project or drive.

Groundwater moving along the pipe exterior can erode naturally occurring fines, leading to settlement, or the groundwater or slurry can deposit fines along the pipeline, leading to high jacking forces. This is not normally an issue if slurry pressures are properly controlled, if inadvertent slurry returns to the surface have not occurred, or if the entrance seal is functioning properly.

Contact grouting of the annular space is usually required when the annular thickness is greater than 20 mm (0.8 in.) or when the ground is noncohesive. However, the annulus is usually full of lubricant or slurry, or the annular space has significantly diminished with ground relaxation, making contact grouting of minimal value. Contact grouting pressures must be carefully determined and closely controlled to avoid damaging the jacked pipe and the pipe joints, causing unexpected returns to the surface and into nearby utilities, or disturbing other features.

20.2 OVERCUT

The overcut is obtained by using gauge cutter tools placed at the periphery of the cutting wheel. These tools create an excavation larger than the body of the MTBM, trailing cans, and trailing pipe string. The use of an overcut is required to reduce skin friction between the ground and the MTBM body and to allow steering. The overcut creates the outside cut, leading to the formation of an annular thickness between the excavated ground and the outside radius of the trailing pipe string (see Fig. 20-1). The difference between the radius of the MTBM shield and the outside radius of the jacking pipe is called the *shield cut*, and it contributes to the annular thickness and the resulting annular space. The amount of the overcut should be based on the following factors:

- Ground types;
- Ground behaviors;
- Contractor's experience in similar conditions;
- Type of lubrication (if used);
- Heave and settlement requirements; and
- Horizontal and vertical alignment requirements.

Commentary. Overcuts generally range from 3 to 10 mm (0.1 to 0.4 in.) on the radius in soil conditions, and from 15 to 40 mm (0.6 to 1.6 in.) on the radius in rock conditions. The overcut is developed by the use of gauge cutters on the cutter wheel. Wear

on the gauge cutters during the drive reduces the overcut as the drive progresses. Wear on the gauge cutters can increase the risk of not completing the drive and is often the controlling factor in determining the drive length if the MTBM does not have cutter chamber access so that the gauge cutting tools can be changed during the drive.

A minimum overcut is required to steer the machine.

If the overcut is too large, the following problems may occur:

- Rolling of the MTBM;
- Poor steering response;
- Surface subsidence; and
- Slurry and groundwater movement along the external surface of the MTBM.

If the overcut is too small, the following problems may occur:

- Inability to steer;
- High jacking forces; and
- Wedging of the MTBM.

If the overcut in rock is too small, the following problems may occur in addition to those listed above:

- Seizing of the MTBM in abrasive rock if the gauge cutters wear down before the drive is completed and cannot be changed through the MTBM; and
- Difficulty in steering.

Each of the above problems may result in failure to complete the project or drive.

The overcut should be documented both by the engineer in the specifications and by the contractor in the submittals.

A larger overcut and its associated annular space are required for curved jacking alignments, to allow the trailing components to turn or curve. The sharper the curve, the larger the overcut and the shorter the pipe segments need to be to complete the curve.

20.3 SLURRY OR AUGER SYSTEM

The contractor should propose the type of machine suitable for the ground conditions.

Commentary. There are two types of MTBM. The first and most common is the slurry-type MTBM. Slurry-type MTBMs were designed to install pipelines in ground, below the groundwater table, to line and grade, and they potentially reduce surface subsidence and heave. The later generation of MTBM is the auger-type MTBM. Auger-type MTBMs were designed to excavate cohesive soils while eliminating slurry and reducing overall project costs. The new pilot tube method, front steer systems, and vacuum systems are based on the auger-type MTBM.

The slurry microtunneling machines are designed with a chamber at the face to counterbalance the groundwater and earth pressures. This counterbalance is created by a combination of mechanical and drilling fluid pressure. The presence of the drilling fluid prevents over excavation when appropriate drilling fluid pressures are maintained in conjunction with the advance rate. The slurry system consists of a feed line, which carries drilling fluid to the suspension chamber located behind the cutter wheel, and a return line, which carries the spoil-laden slurry away and back to the surface at the jacking shaft for processing. The use of a drilling fluid with bentonite or polymer additives helps carry the excavated spoils in suspension, reducing the pumping velocities. Slurry-type machines are used in variable ground conditions and for variable pipe diameters. Slurry machines are owned or leased by various contractors.

Auger-type machines are designed with an auger to remove excavated ground from the tunnel zone. Counterbalance is created by maintaining an earth plug (which is a result of using a screw auger); controlling the advance rate of the machine, auger speeds, and material discharge; soil conditioning; and volumetric monitoring of excavated spoils. Auger-type machines are limited by the type of ground, groundwater table height above the machine, and pipe diameter. Auger-type machines are still owned by various contractors and may still be used on some projects. Many of the auger-type machines have been superseded by the pilot tube method. One modification to these machines has been the use of a visual light guidance system in place of laser guidance. A second modification has been replacement of the auger itself with a vacuum system for spoils removal.

See Section 12.2 for more discussion on the types of MTBMs available.

20.4 CUTTER WHEEL DESIGN AND SELECTION

The contractor should be responsible for selecting a cutter wheel suitable for the ground conditions anticipated along each drive. Different cutter wheels may be required on a project, depending on the variability of the ground conditions.

Commentary. Cutter wheels are typically designed by the manufacturer and should be designed to excavate through the anticipated ground conditions. They differ from one another by their cutting tools and by the geometry of the wheel, particularly the size of openings enabling the passage of the ground through the cutter wheel face. Documentation of the ground conditions is important and essential. The anticipated ground conditions should be documented in a Geotechnical Baseline Report (GBR) or geotechnical design memorandum (GDM) prepared by the engineer.

Cutter wheels can be equipped with

- *Blades:* Steel tools used to excavate clay and other soils with high N-values.
- *Buckets:* Fabricated steel components used to pull material into the crusher chamber.
- *Button bits or strawberry bits:* Rolling tools used to excavate soft rock (buttons or strawberries are steel or carbide inserts placed into the roller to offer better resistance to wear).
- *Cone-shaped disc cutters (strawberry and multikerf bits):* Rolling tools used to excavate rock (conical shape along the shaft of the cutter has different circumferences on each end and is designed to minimize skipping and skidding as the cutter rotates).
- *Conical picks or conical bullet teeth:* Steel with carbide tools that revolve within its block, used to excavate rock.
- *Crushers:* Mechanical components used to reduce excavated material before it enters the slurry transport system.
- *Drag teeth or chisel points:* Steel with carbide tools used to excavate soils.
- *Disc cutters (single and multikerf):* Rolling tools used to excavate rock. *Disc* refers to the number of kerfs on a single shaft and bearing assembly that engage the rock for cutting; a single disc has one cutting surface, and a multikerf disc has more than one cutting surface.
- *Grizzly bars:* Steel bars welded over the cutter wheel openings to control the size of material entering the cutter chamber.
- *Hard facing:* Abrasion material that is applied to steel using welding equipment.

- *Injection nozzle:* Nozzle used to inject slurry water into the crusher chamber.
- *Rippers:* Steel tools used to fracture and excavate rock.
- *Slots:* Holes within the cutter wheel that allow material to pass.
- *Water jets:* Jet used to inject clean water to clear cohesive materials from the cutter wheel and crushing chamber.

The size of the opening in the crusher chamber controls the size of the excavated material to be transported. Smaller openings in the cutter wheel keep material in front of the machine longer, allowing the tooling to further reduce the size of the excavated material. Smaller openings are sometimes used to keep cobbles, boulders, and rock in front of the machine to allow the disc cutters and other tooling to work. Allowing too many cobbles or boulders into the crushing chamber can cause the cutter wheel to lock up. Larger openings in the cutter wheel allow larger material to enter the chamber for size reduction. Machines with a stone and rock crusher built into the front compartment might use larger openings. The size of the opening in the crushing chamber controls the size of the excavated material to be transported by the slurry. Some contractors have considered and used nonsymmetrical openings to help control the intake of larger material into the crushing chamber. Care should be taken in limiting cutter wheel opening sizes if cohesive soils may be encountered that will block the smaller openings.

In cohesive ground, the cutter wheels may be equipped with high-pressure or medium-pressure water jetting nozzles. The jetting nozzles use water and may contain additives to help flush and wash excavated material sticking to the cutter wheel that is not otherwise entering the slurry transportation system.

Though not intended for soils with boulders, typical soft-ground cutter wheels ingest boulders between 25 and 30 percent of the MTBM diameter and do not contain disc cutter tooling. On soft-ground cutter wheels the rock, cobbles, and/or boulders must be processed in the crushing chamber to be further reduced to particle sizes small enough to pass through the slurry system.

Mixed-face cutter wheels are generally outfitted with a combination of disc cutters, drag teeth, and conical bullet teeth. The mixed-face cutter wheel openings are reduced, to allow rock and cobble to be processed by the disc cutters before entering the crushing chamber.

The rotating cutter wheel on all disc cutters should be manufactured to create the properly sized excavation and should not be skinned up like the MTBM body. Care should be taken in upsizing cutter heads to provide proper overcut for the MTBM shield body. Upsizing beyond the next size MTBM should be done with input from the manufacturer. Upsized cutter wheels can force the contractor to advance the MTBM at a slower rate to force the excavated material into the chamber so that the MTBM does not move upward on the undigested material. However, the slower advance rate may be the economical choice versus using a new cutter wheel.

Cutter wheels are uniquely designed by almost all manufacturers and some contractors and thus provide varying amounts of overcut between the cutter wheel gauge tools and the MTBM shield body. The perimeter gauge tools wear away as the MTBM advances during the drive. If the MTBM does not have face access so that the perimeter gauge tools can be changed during the drive, then the MTBM may become wedged in place before completing the drive. This is a particular challenge when microtunneling through abrasive metamorphic and igneous rock.

20.5 SLURRY SEPARATION PLANT

The contractor should select the type of slurry separation equipment needed for each drive based on the geotechnical information available and other project constraints.

Commentary. Methods to separate slurry into the drilling fluid and muck include the following:

- Settlement basins or tanks;
- Shale shakers and vibrating screens;
- Desanding cones;
- Desilting cones;
- Mud cleaners;
- Centrifuges;
- Polymers; and
- Vertical clarifiers.

The separation system should be designed to meet the needs of the project. Any additives used should have ANSI/NSF International 60 approval and should not negatively affect the slurry separation plant.

Each type of separation method works best on different soil particle sizes. Contractors should work with qualified suppliers to design the slurry separation plant to handle the full range of material anticipated to be encountered on the project. The designer should provide documentation of gradation of the anticipated excavated material down to 1 micron. Some contractors, for example, do not expend the effort to remove the finest grain particles from the slurry. Over time, the slurry can become heavy with additional native clays. The properties of the slurry change as the microtunnel face advances. The slurry can become less effective in counterbalancing face pressures. See Section 19.4 for more discussion on the slurry design. The slurry is also harder to pump, both to the face and back to the jacking shaft. The cost of lost production can equal or exceed the cost of a well-designed slurry separation plant. In addition, heavier slurry requires higher pumping pressures to move slurry to and from the face of the MTBM. The higher pressures can result in heave of the ground, movement of slurry into other utilities above the alignment, and even inadvertent return (frac-out) of slurry to the surface.

Excavated material removed from the slurry may have a wet component. If necessary, dump trucks used to transport the excavated material from the jacking shafts should be sealed to prevent loss of any fluids from the excavated materials during transport.

Typical flow rates of the slurry system must maintain turbulent flow in order to suspend the excavated material. Slurry separation equipment must be able to accommodate these flows. The typical slurry velocity required, regardless of slurry line diameter, is 3 to 4.5 m/s (10 to 15 ft/s).

The slurry separation plant generates noise and vibrations because of moving parts. Sound walls are normally required to control noise because the equipment cannot be contained or designed for silent operation. These environmental effects should be considered during project layout and operations. See Section 7.7 for special considerations.

Removal of the fine particles from the slurry may require the use of one or more centrifuges. The centrifuge is typically run on a circuit that allows continuous operation, even between pipe changeovers, in order to achieve desirable results.

The following is a description of slurry separation components, with technical data, that are used with slurry microtunnel boring machines. The concept of separation is to remove the spoils, excavated material, with as little energy as possible.

Adding energy, whether from pumps or slurry processing components, to the slurry allows the excavated material to break down into the smallest particles and become suspended in the slurry. Clay-removal screens and primary shakers are placed in the return slurry line flow and process while tunneling is advancing. The mud conditioners, centrifuges, and clarifiers are continuously fed by reprocessing the slurry that is contained on the surface during normal shutdown cycles.

- A. Clay-removal screens, the gumbo box, the prefeeder, and the rotating belt are designed to scalp large clay balls that are being transported from the MTBM in the slurry lines. These units are capable of processing, on average, 600–800 gallons per minute (gpm) and produce 20 tons of material per hour (tph). When these units are used, they are the first processing component.
- B. Primary shakers are designed to scalp sand and other coarse-grained materials. As a general rule, screen sizes on the primary shakers range from 10 mesh to 70 mesh. These units typically contain three or four screen units that operate between 5 and 7 g's. Each shaker generally processes between 100 and 1,000 gpm, depending on screen size and viscosity of the slurry. These units produce 10–15 tph of sand per unit.
- C. Mud conditioners are designed to remove the majority of the sand entrained in the slurry. The solids are concentrated through various sizes of hydrocyclones and then deposited in fines screens of up to 325 mesh. The high-frequency fines shaker screens operate at between 5 and 8 g's and dewater the material. There are different types of hydrocyclones available, and all have their advantages in different soil applications. With all cones, more head pressure is required as the viscosity and density of the slurry increase in order to achieve satisfactory results. Here are the differences among some types of cones.
 1. *Hydrovacuum cyclones:* These are the largest cones. One 15-in. hydrovacuum cyclone can process 650 gpm and can produce 35 tph of sand. These cones make a D_{50} cut point of 55 microns, or 325 mesh. These work well in any circumstance where viscosity of slurry is below a 70 s/qt Marsh funnel viscosity. They work on relatively low head pressure and do not degrade the finer and softer clays that are encountered in the formation. Head pressure for the hydrovacuum cones is typically 8–12 psi.
 2. *Desander cones:* One 10- or 12-in. desander cone can process 500 gpm and can produce 6 tph of sand. Desander cones remove a D_{50} cut point of 50 microns. These cones need a minimum of 30 psi for proper separation.
 3. *Midrange cones:* One 6-in. cone can process 150 gpm and produce 2.5 tph of sand. These need to operate at 75 feet per hydraulic head (ft/hd). The cut point on these cones is a D_{50} of 35 microns.
 4. *Desilter cones:* One 4-in. desilter cone can process 50 gpm and produce 1 to 1.2 tph, with an average cut point of 20 microns. These also operate at 75 ft/hd.
 5. *2-in. cyclones:* These are used primarily for clay removal. One 2-in. cyclone can process 15 gpm and produces 0.3 tph with a D_{50} cut point of 7 microns. These are also run at 75 ft/hd. The only time that these cones would be beneficial is if the heavy underflow from these cones were collected and fed straight to a centrifuge for dewatering.
- D. The final cleaning stage uses centrifuges, belt presses or filter presses, and clarifiers.

Centrifuges: These are available in basically three sizes. The diameter of the tunnel and the amount of clay expected dictate the size centrifuge needed.

1. *Big bowl centrifuges:* Bowls are typically 21 in. in diameter and approximately 72 in. long. These centrifuges require large horsepower (hp) motors to operate and process large amounts of volume. They typically come with a 150-hp main drive and a 40–50-hp back drive. They are rated at 500 gpm and can produce up to 25 tph of solids. These machines are generally fully adjustable and operate at between 2,000 rpm and 3,500 rpm.
2. *Midrange centrifuges:* Bowls are typically 18 to 24 in. in diameter and are between 38 and 50 in. long. They run on one 50-hp motor and can be adapted to have a variable-speed back drive run by a 20–25-hp motor. These units are rated for 150–200 gpm and produce 6–8 tph of solids.
3. *Small bowl centrifuges:* Bowls are generally 14 in. in diameter and range from 38 to 50 in. long. They usually run on a 40- or 50-hp motor and are capable of running up to 3,500 rpm. These units are generally offered with fully adjustable bowl speeds and back drives. Back drive motors are either 15 or 20 hp. These units are rated for 125 gpm and 4–6 tph.

Centrifuges are good pieces of equipment and are a necessary solids-removal tool. However, centrifuges on their own only make a D_{50} cut point of between 5 and 7 microns, no matter how fast they can be spun. In some cases, 95 percent of the material in the slurry can be under 5 microns. Every jobsite should have a polymer injection and delivery system. This system allows the particles under 5 microns to be made artificially larger and lets the centrifuge remove these ultrafine particles. Through the use of polymers, clear water can be achieved for disposal or dilution back into slurry.

Belt presses or filter presses: Belt presses are typically used for permanent installations, such as water treatment plants or sand and gravel applications. It is difficult to achieve the desired results in the field. Belt presses require a large footprint and can be time-consuming, requiring a lot of personnel. Belt presses suitable for the microtunneling industry would range in size between 1.5 m and 3.0 m. Under ideal circumstances, a 1.5-m press can produce 5–7 tph of solids, and a 3-m press can produce up to 25 tph. All solids need to be put through a thickener to achieve the correct amount of solids before being sent to the belt press.

Filter presses are gaining popularity in the sand and gravel business but need to be in a stable environment and require a large footprint to be effective.

Clarifiers: These units are composed of a tank that is 8 ft in diameter and 20 to 30 ft high. Clarifiers are rated at 600–800 gpm and can produce up to 8 tph of solids. Polymer is injected into the slurry on the way to the clarifier, to separate the solids and water. The solids drop down to the bottom for removal, and the clean water is recovered off the top. This unit is generally used to concentrate fine particles so that the centrifuge can be limited out on solids tonnage rather than liquid flow. This is most advantageous in mixed-soil areas where only pockets of clay are encountered.

Managing the slurry separation plant requires planning, customizing of equipment, monitoring, and continually revising as the excavated soils change. The planning starts

by reviewing the soils report and the geotechnical baseline report and determining the soil types, fractions, and locations. Once the soils are understood, the contractor needs to select the proper components and additives for the bid. These items help determine operating costs and disposal costs. The contractor should review the plan with the preferred suppliers and adjust the plan as needed. Once the project is won, the contractor needs to select and provide the equipment, accessories, supplies, and items that are required, basing these selections on contingency plans or past experience. The contractor should then review these plans along with monitoring requirements with the person who implements the slurry separation plan in the field.

20.6 BURIED OBJECTS AND OBSTRUCTIONS

Materials and buried objects identified in the design site investigations by the engineer should be factored into the type of cutter wheel and microtunnel equipment to be deployed by the contractor. They can also be used to determine the measurable properties that would define an obstruction.

Commentary. An obstruction is defined as any object or feature that lies completely or partially within the cross section of the microtunnel face and impedes continued forward progress along the design path and within allowable tolerance. Potential buried objects should be identified during the site investigation phase and design stage. Buried objects that cannot be overcome by the tunneling equipment should be avoided or removed before tunneling commences. Plans should be developed for dealing with potential buried objects physically and contractually. The contract documents should require submittals for dealing with potential buried objects, and procedures should be established to remove or bypass the buried objects if encountered.

Methods that may be available include

- Sinking a rescue shaft;
- Hand mining from the receiving shaft or a rescue shaft;
- Removing the buried object by accessing the face through an access door in the MTBM, provided that the MTBM is 1,500 mm (60 in.) or larger and the MTBM is so equipped, and that the ground is stable or prevented from movement with some type of stabilization method;
- Drilling down on the object and breaking it up, allowing the MTBM to progress through the object; and
- Installing a drilled shaft in front of the alignment and the obstruction deeper than the diameter of the microtunnel, and pushing the object with the microtunnel machine into the void left by the drilled shaft.

If work is to take place in ground known to contain significant buried objects, it may be necessary to consider alternatives to using a remote-controlled MTBM. In stable ground, an open-face shield can be equipped with a boom cutter or backhoe to provide the flexibility needed to deal with a variety of conditions. For ground that is both unstable and likely to contain buried objects, selecting and designing a machine with a compressed air chamber may be an option. Typically, a machine must be 1,800 mm (72 in.) or larger to house a compressed air chamber in the tunnel; however, the air lock can be situated in the drive shaft. Work in compressed-air environments can be subject to many health and safety regulations that should be considered and can increase the cost of the project. There have been many interventions to remove obstructions where the only other viable option would have been to excavate a rescue shaft. The ability

to access the tunnel face from within the machine may be of great benefit. Having face access does not guarantee that face access is possible or that face access allows removal of the obstruction. If an intervention is envisioned, the owner should obtain the necessary approvals from state and federal regulators before bidding the work and should consider including a bid item and quantity for the intervention(s).

Some buried objects, like wood piles, timber shoring, bricks, concrete blocks, and construction debris, can usually be excavated, but advance rates are reduced. Buried objects such as wood piles and timber shoring may not have a significant effect on the cutter wheel and crushing chamber; however, these buried objects may have a detrimental effect on the slurry circuit by causing fibrous plugs inside the slurry lines and pump impellers. Compressible wood like root balls is difficult to excavate because the material compresses, absorbing energy, and rebounds once the force is released.

Since an obstruction is defined as a buried object that might deflect the microtunnel drive off the design line and grade, consideration should be given to using a two-pass tunneling approach to allow for this occurrence. The two-pass approach should always be considered when the installed line and grade are critical for the final pipeline's operation and function. A rescue bid item and quantity should be included when obstructions are perceived as a risk factor. A rescue shaft must be designed and sized as a launch shaft if the pipe string freezes in place while the rescue shaft is installed and the obstruction is removed. This decision must be made before the size of the buried object is determined. See Section 27.6 for additional discussion of rescue bid items.

When dealing with obstructions, some owners might consider using an equitable approach. The owner agrees to pay the contractor for downtime and test pits. By digging a test pit, both parties have an opportunity for discovery and determination of the buried object that is impeding forward progress. The two parties are then able to negotiate a final cost that is based on what is discovered and on whether equipment is damaged.

20.7 STEERING

Steering accuracy should be maintained to within the required line and grade of the design.

Commentary. The criterion of a good guidance system is that it gives accurate information about the current position and attitude of the shield so that steering actions can be minimized.

Excessive steering should be avoided because it increases jacking loads and could overstress the pipe at the joints. Steering corrections to return to the design line and grade should be done over the remaining portion of the drive. If that is not possible, then the rate should not exceed 1 in 300. Returning to line and grade and then continuing the drive may actually increase the risk of failure because of oversteering, increased skin friction, and jacking pipe point load failures. The rate of return is affected by the type of pipe used. The steering system of the machine should be able to steer in all directions. Typically, machines have a minimum of ± 3 degrees of steering capability for straight alignments. For curved alignments, additional steering capability as well as passive steering joints may be required to negotiate the alignment.

For pipe jacking in a straight line, the conventional theodolite survey and laser are all that are typically used. For drives more than 1,200 ft (366 m) and curved drives, the owner should consider requiring a fully automated theodolite or gyroscope system (Camp et al. 2013).

A skilled operator should be employed on the basis of flexibility and experience to control the steering and other adjustments.

Curved microtunneling requires the MTBM to negotiate the curve and the trailing pipe string to also negotiate the curve. In a simple curved alignment there would be a single short radius curve with a short arc located at the receiving shaft end of the drive. This alignment configuration limits the number of pipes that pass through the curve and requires the least amount of steering and guidance changes. A complex curved project would include multiple curves in many different directions (left and right, up and down, and possibly compound curves), short radii, long arcs as compared to the radius, very short pipe joints, and intermediate jacking stations (IJSs) that pass through multiple curves. The guidance system needs to be selected on the basis of the project requirements, and the survey method needs to be selected on the basis of the guidance system requirements. Owners, engineers, and contractors need to work together to build a curved microtunnel because the owner is buying the curve into its system.

See Section 19.5 for more information on the guidance system to be used.

CHAPTER 21 SUBMITTALS

The contractor should submit the items listed in the specifications for review and acceptance by the engineer. The submittals are generally divided into *preconstruction* and *postconstruction* submittals.

Commentary. Presented in the following subsections is a general summary of typical submittal requirements for microtunneling projects. This summary is not intended to be all-inclusive, and specific submittal requirements should be evaluated on a project-by-project basis. Typical submittals may include the following elements:

Preconstruction

- Contractor's qualifications
- Microtunneling equipment, including IJSs;
 - Method statement (general description of how system works and operates);
 - Jacking system;
 - Intermediate jacking stations (IJSs), as applicable;
 - Slurry design, pressure balancing control;
 - Water jetting operation plan, if allowed;
 - Lubrication design, pressure, and volume;
 - Guidance and steering system; and
 - Earth pressure balancing control (how measured and operated);
- Shop drawings;
 - Project layout;
 - Shaft layout;
 - Jacking pipe with joint design; and
 - Geotechnical instrumentation;
- Schedule;
- Calculations;
 - Jacking forces;
 - Settlement or heave; and
 - Shaft design;
- Ground stabilization (if required);
- Contingency plans;
- Preconstruction surveys;
- Safety plan (submitted for information only and not approved);
- Daily construction records;
 - Production (daily or weekly);
 - Operation (daily or weekly);
 - Photographs (weekly, or as specified);
 - Schedule updates (weekly, or as specified);
 - Survey measurements and interpretation (weekly, or as specified); and
 - Geotechnical instrumentation readings and interpretations (weekly, or as specified).

Postconstruction

- Drawing of as-built pipeline based on closed-loop survey showing design;
- Ground movement data and summary table;
- Combined guidance and steering with the line and grade survey control record; and
- As-built schedule.

The contractor should assemble a package that includes all of the submittals identified in the contract specifications to minimize the number of submittals and resubmittals. Coordination with submittals required by other specification sections and with those prepared by subcontractors is necessary.

21.1 MICROTUNNELING

The contractor should submit the following items:

- Manufacturer's literature for the MTBM, mixing equipment, and slurry separation plant;
- Proposed type and size of equipment, guidance systems, and method of grade and alignment adjustments;
- Description of similar projects (including names, addresses, and phone numbers of owners' representatives) on which this similar system by the same manufacturer has been successfully used;
- Calculations of the anticipated jacking loads;
- Calculations for the shafts and associated shoring, including the ability to handle the anticipated jacking forces;
- A work plan that includes the following elements:
 - Description of the slurry mix to be used along the alignment, including safety data sheets (SDSs);
 - Description of the lubrication and grouting systems, including SDSs, grout mix, proportions, slump, and design strength as well as equipment and injection pressure;
 - Description of the method to remove and dispose of spoils;
 - Layout of equipment at the shafts;
 - Schedule and sequence of construction;
 - Plan for dewatering; and
 - Details of survey and guidance systems, including mounting;
- Daily log sheets detailing machine performance and operational data; and
- Daily log sheet detailing settlement monitoring data.

Commentary. All calculations related to jacking forces and the shaft must be signed and stamped by a professional engineer licensed in the state in which the work is to be conducted.

21.2 QUALIFICATIONS OF PERSONNEL

Documentation summarizing the qualifications of the project superintendent and machine operators should be requested by the owner in the contract documents and provided by the contractor.

Commentary. Microtunneling is a trenchless installation methodology that requires a contractor who can demonstrate technical prowess in microtunneling through a prebid and prequalification process. See Section 15.7 for recommended prequalification requirements. If the owner and engineer are reluctant to require prequalification of bidding contractors or microtunneling subcontractors, it is of paramount importance that they include the same requirement for postbid qualifications. This way, the owner and engineer can determine the competency of the microtunneling contractor before the start of the project and adjust inspection of the project to an appropriate level.

21.3 MICROTUNNELING WORK PLAN

The contractor should submit a plan showing the proposed method of construction and operation, including design and construction of the shafts, thrust block, pipe lubrication procedure, location and design of intermediate jacking stations, arrangement and position of main jacks, sequence of operation, and other elements complete and in assembled position. The contractor should submit, in a format acceptable to the owner and engineer, a daily log of the jacking operations at intervals of not more than one pipe length, including the following information for each pipe jacked:

- The position of the tunneling machine in relation to the design line and grade;
- The jacking forces exerted on the pipe at each jacking station;
- The date, starting time, and finish time of the jacking operation;
- Inclination;
- Cutterhead torque;
- Slurry flow rates in both the feed (charge) and return (discharge) lines;
- Slurry chamber pressure;
- Position of steering jacks;
- Amount of lubricant used (in gallons) and its viscosity and pumping pressure;
- Volume of spoils removed from the site;
- Number of pipes installed or distance tunneled;
- Important observations, e.g., settlement; and
- Detailed plans for installation and monitoring of instruments as well as a plan for baseline surveys and reports.

In addition, the contractor's proposed sequence of operation and schedule should be submitted.

Commentary. None.

21.4 PIPE SELECTION

The contractor should submit a certificate of compliance that verifies that the pipe complies with the project specifications, including pipe design data, calculations, and tolerances.

Commentary. This submittal should include literature describing the microtunneling pipe and indicating allowable safe jacking loads with the designated factor of safety. Details of pipe compression rings (when required at the joints to safely distribute

jacking forces) and instructions on storage, handling, transporting, and installation requirements should also be provided.

21.5 CONTINGENCY PLANS

Contingency plans should be prepared by the contractor for review by the owner and engineer for potential conditions, such as:

- Damage to pipe structural integrity and repair;
- Mixed faces;
- Mixed ground;
- Obstructions;
- Steering corrections;
- Loss of ground;
- Inflows at the shafts;
- Slurry migration to the ground surface or to other subsurface structure(s);
- Excessive thrust wall deflections;
- Excessive settlement or heave;
- Contaminated ground; and
- Traffic control plan.

Commentary. Damage to pipe structural integrity and repair: The owner and engineer should be notified of any pipe damage that occurs before or during construction. The documentation should note the location of the damage, whether in the barrel of the pipe or at the joint; the location and number of the pipe; and the severity of pipe damage. The damage should be defined into one of three classes:

- Class 1: The least critical is minor damage, which can be corrected after the pipe jacking is completed.
- Class 2: The next level is damage that threatens completion.
- Class 3: The most severe is a failure that prevents completion.

All repairs should be performed with acceptable means and methods that are established by or agreed to by the original pipe manufacturer. A Class 3 failure might require using open-cut methods to remove and replace pieces of pipe.

Mixed faces: Contingency plans should be developed for how the MTBM is to be operated if mixed-face conditions are encountered during a drive. These plans could include faster or slower penetration rates, or increased or decreased revolutions per minute (rpm).

Mixed ground: Contingency plans should be developed for how the MTBM is to be operated if mixed ground conditions are encountered during a drive. These plans could include faster or slower penetration rates, or increased or decreased revolutions per minute (rpm).

Obstructions: Contingency plans vary depending on the location of the encountered obstruction. The accessibility from the surface can be limited or restricted by the presence of railroad tracks, private rights-of-way, bodies of water, highways, or runways. Obstructions may have to be dealt with by pulling back the machine and/or driving a second oversized tunnel from the receiving shaft. These procedures depend on the selected size of the MTBM. Larger diameter machines may have an accessible face or may be configured with an air lock allowing use of compressed air for direct access at the face. If those methods are not feasible or are unsuccessful, obstruction removal or machine retrieval using a rescue shaft may be needed.

Steering corrections: The tolerances for line and grade should be established in the specifications. A contingency plan should

require the work to stop when the line or grade exceeds a certain percentage of the specified tolerances. The contractor should determine the cause of the steering difficulties and correct the identified problem. The corrections in line or grade should be made for the remaining portion of the drive. The contractor should calculate a new bearing or slope so that the remaining portion of the drive hits the final intended target. Corrections should be minimized because they increase the time to complete the drive. Curves require special consideration and can introduce additional problems without additional planning.

Loss of ground: This occurs when the microtunnel is inadvertently over excavating the ground without maintaining machine advance (i.e., when excavated volumes exceed calculated volumes). When loss of ground occurs, adjustments may need to be made to the machine advancement rate, cutter wheel rotation speed, slurry pressures, slurry separation plant, the slurry mix, and/or the pipe string integrity. Loss of ground should be defined with consistent units, such as half cubic yards for earth material.

Inflows at the shaft: Contingency plans should be made for inflows of fluids into a shaft, whether through or at shaft penetration points. Inflows should be defined with consistent units, such as half cubic yards for earth material and gallons per minute (gpm) for water. The contractor may need to adjust the ground support methods to create a watertight shoring system (inflow <5 gpm) if inflows hamper the tunneling efforts. Plans may also include the use of ground improvements, such as jet grouting, the use of submersible pumps, and/or imposing fines. Additionally, the contractor may need to improve the entrance and exit seals or ensure that the jacking loads are acceptable to the shaft designer.

Slurry migration to the ground surface or to other subsurface structure(s): Slurry is usually injected from the crown (lubrication from the invert) and controlled with pressure and volume. When an inadvertent return occurs, the contractor should be prepared to carry out the following action, in accordance with the contingency plan: shut down the tunneling operations, let the ground rest (generally for 24 hours), and then resume pressure. The specifications should address allowable pressure, which in most typical cases is 1 psi for every 2 ft of cover. Vertical and horizontal clearances should be considered when examining potential migration issues.

Excessive thrust wall deflections: The shaft designer should provide the maximum allowable deflections and measuring methods to the contractor. Communication between the designer and the contractor should take place to ensure that the jacking loads are implemented in the thrust wall design. Thrust wall deflection should be monitored throughout the jacking process. A telephone list with the designer's contact information should be maintained by the contractor.

Excessive settlement or heave: The owner or engineer should provide the maximum allowable vertical movement for each desired item. Measurement points should be in locations that are independent of the ground surface. A telephone list with the owner's contact information should be maintained by the contractor.

Contaminated ground: The crew needs to be trained and prepared to handle contaminated material if it is anticipated. Projects that are located in urban environments should anticipate encountering contaminated ground. The 40-hour training course

must be OSHA-compliant in hazardous waste operations and emergency response (HAZWOPER). The crew should be trained before the start of the project, and all necessary recertifications should be obtained. A hazardous disposal landfill should be appropriately located before the start of the project. Information about the amount of expected contaminated material should be provided to the contractor, in terms of volume or weight. Additionally, the contractor should have an appropriate plan to deal with contaminated groundwater when contaminated ground is expected.

Traffic control plan: The contractor should have a detailed plan to facilitate the smooth flow of traffic around the project area in accordance with local ordinances.

21.6 DEWATERING PLAN

The contractor should be responsible for the design, construction, operation, and maintenance of a dewatering system that meets the needs of the construction method.

Commentary. Pumped or diverted water should be controlled by surface structures as necessary to comply with environmental runoff regulations. The dewatering plan should comply with federal, state, and local environmental regulations.

The owner of the project is responsible for establishing the constraints placed on the dewatering program. These constraints may include not allowing any dewatering, imposing environmental testing, and limiting discharge quantities.

If dewatering is allowed, the engineer should estimate the effects of the dewatering plan on adjacent structures and the potential for ground subsidence.

The contractor needs to obtain discharge permits and may be limited in the amount of water that can be disposed of in the local sewer. These requirements would also apply to the collection and discharge of construction water.

Alternatives to dewatering include building watertight shaft support systems, excavating the shafts in the wet, and placing a tremie slab at the shaft invert as a seal.

Caution should be used in efforts to use standard dewatering specifications and lowering water below the excavation bottom. Most dewatering specifications are written for open-trench work and are not appropriate for shaft construction.

21.7 ADDITIONAL SHOP DRAWINGS

The contractor should submit all shop drawings and plans for the excavation method and the excavation support system to the engineer for review before the construction operations begin and within the time limits specified in the contract.

Commentary. The engineer's or the contracting officer's approval of such plans should not be construed in any way as relieving the contractor of his or her responsibilities for the excavation method or the excavation support system but are primarily for the purpose of ascertaining compliance with the requirements of the contract.

Such requirements include issues related to overloading or misuse of existing structures, adequacy and safety of any such works, and potential damage to or undermining of existing structures.

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CHAPTER 22

SHAFTS AND TEMPORARY STRUCTURES

22.1 SHAFT DESIGN

Shafts should be of a size to accommodate the equipment and pipe selected and to allow for safe working practices. If the shaft must accommodate the permanent works, it is the contractor's responsibility to include these requirements in the shaft design. Permanent works are designed by the engineer and specified in the contract documents.

Commentary. With review and acceptance by the engineer, the contractor should be able to relocate shafts to better suit the capabilities of the microtunneling method proposed.

When possible, shaft locations should be kept clear of road intersections and within a single traffic lane to minimize disruption to the flow of traffic. If shafts are located in intersections or other areas of high effects, the location should be considered for a receiving shaft because of its normally smaller footprint and lower use. The design of shafts should ensure safe exit from the jacking shaft and entry into the receiving shaft of the MTBM. Special consideration should be given to avoiding interference with existing utilities, especially overhead electrical and communication lines.

22.2 EXCAVATION TECHNIQUE AND SUPPORT SYSTEMS

Selection of an excavation technique, excavation support system, and dewatering system is the contractor's responsibility, within the framework established by the contract documents.

Commentary. The contractor is to choose the method of excavation and support to be used, be responsible for design of any appropriate system, and ensure that it is operated correctly.

The contractor can support the shafts with a number of techniques. The different support systems may include the following elements (listed alphabetically):

- Caisson (i.e., cast-in-place concrete or precast segments);
- Cutter soil mixing (CSM) method;
- Drilled shaft with corrugated metal pipe (CMP) or liner plate;
- Ground freezing;
- Interlocking steel sheet piling;
- Liner plates;
- Secant piles;
- Shotcreting and rock bolts;
- Slide rail system;
- Slurry walls;
- Soil nails;
- Soldier piles and steel plates or timber lagging;
- Steel ribs and timber lagging;

- Tangent piles; and
- Trench boxes.

See Table 14-1 for information about the various support systems and the limits on size, shape, and effective depth.

22.3 PERMITS

The owner is required to obtain all necessary permits, licenses, and rights-of-way, unless another party is required to do so as part of its contract of services.

Commentary. The owner needs to provide copies of all permits, licenses, and rights-of-way obtained for the project and listed in the contract documents. The contractor often is required to obtain construction-related permits for its work, which could include blasting, storm and construction water discharge, and traffic control.

Some jurisdictions require a tunnel classification permit issued by OSHA. The permit classifies the microtunnel as one of the following (with or without special conditions):

- Nongassy;
- Potentially gassy;
- Gassy; or
- Extra hazardous.

If a compressed-air intervention is anticipated, a compressed-air variance from OSHA is required. The contractor, in coordination with the owner, needs to comply with the various requirements.

The contractor needs to comply with all permit requirements. Copies of all contractor-obtained permits should be provided to the owner. Copies of all the permits should be maintained on site for the duration of the project.

22.4 RESPONSIBILITY FOR DESIGN ADEQUACY

The contractor's design for the shaft should include working drawings and calculations stamped and signed by a professional engineer licensed in the state where the project is located. The review and approval of contractor submittals, such as shop drawings, as required by the contract, is only for the limited purpose of checking for conformance with the design concept and the information expressed in the contract documents.

Commentary. The contractor should provide accepted submittal information to the construction crew on site and provide timely notice if the contractor elects to change the submitted information. The owner should provide accepted submittal information to its field staff before that work commences. The design should include all construction details and the construction sequence.

22.5 CONTRACTOR SUPERVISION

Construction administration quality control by the contractor should include continuous on-site supervision during all phases of the construction by one or more competent, technically qualified, and experienced supervisor(s).

Commentary. The contractor's competent person (e.g., the superintendent) should be on site during all construction activities.

22.6 PUBLIC SAFETY

The contractor should provide fencing, plates, and other barriers to maintain public safety at shaft locations and in the working area. The contractor should provide provisions for the following:

- Shaft location and working areas;
- Construction traffic and traffic management;
- Noise and vibration;
- Dust control; and
- Project water runoff.

Commentary. Shaft location and working areas: The contractor should provide fencing, plates, and other barriers to maintain public safety. The shaft and construction working areas should be isolated or fenced off from public entry.

Construction traffic and traffic management: An approved proper traffic management plan according to department of transportation (DOT) and local government regulations needs to be maintained by the contractor. In some cases, traffic diversions and lane closures may be necessary. These should be planned well in advance, and it is recommended that local businesses and residents be informed of diversions and closures.

Noise and vibration: Noise and vibration should be kept to a minimum at all times and below the OSHA and local allowable standards. In residential areas, around hospitals and nursing

homes, and in other sensitive areas, careful consideration should be given to working hours and days. Where possible, silenced or containerized generators, machinery, and other equipment should be used. Barrier walls should be considered, to minimize the noise per building code. A change of methods may be required to minimize vibration in sensitive areas, such as eliminating sheet piling by using an alternative shaft construction method and by changing the layout of equipment.

Dust control: Roads should be cleaned regularly and sprinkled with the water truck to suppress dust. For slurry and cement mixing, barriers or dust sheets should be erected around the slurry and cement mixing areas, and care should be taken to reduce dust, specifically in sensitive areas. The spoils should be stored so as not to become an environmental nuisance or should be taken away by truck before they become a nuisance to the environment. The spoils runoff at the separation plant should be contained and not allowed to enter into surface drains or sewers.

Project water runoff: The excess water from storms or the shaft or tunnel should be disposed of in the proper manner. The quality of water should be checked, and permission should be obtained from the local authority before disposal into the sewer system.

22.7 VENTILATION REQUIREMENTS

Shafts and the microtunneled pipe should be adequately ventilated. All OSHA confined-space requirements should be met. The contractor should also provide ventilation into the pipe string during the microtunneling operation to remove heat pockets that could deflect the laser during microtunneling.

Commentary. Ventilation is required to remove heat and moisture in the pipe string and the interior of the MTBM. The ventilation helps create a uniform environment to maintain a more stable laser position.

CHAPTER 23

CONSTRUCTION SUPPORT SERVICES

23.1 PROTECTION OF ADJACENT STRUCTURES

A preconstruction survey of existing structures within the anticipated construction zone of influence should be conducted before any construction begins. A plan for structures that need protection should be prepared before construction begins. A postconstruction survey should be conducted. Any major discrepancy should be reported.

Commentary. A preconstruction survey is the best tool to identify possible problems with nearby structures. Where possible, shafts should be located away from critical structures and utilities. If shafts must be located near structures and utilities, suitable excavation support should be used to protect these structures. Still photos and videos are the most common media used to provide records for the survey.

Temporary support of utilities should be handled with extreme care. Utility companies should be consulted. If the utilities cross the shafts and are to remain in place, necessary arrangements should be made.

Before beginning construction for the shafts, the contractor should adequately protect existing structures, utilities, trees, shrubs, and other permanent objects that are visible or shown on the drawings.

A building and facilities assessment plan should be developed that includes documenting the existing conditions before construction commences. The documentation should include photographs of sufficient detail to document the conditions. One should consider photographing with a scale cracks and other features that would indicate movement. Poles and other vertical structures should include measurements that include their vertical nature. Locations should be surveyed so as to provide reproducible results.

Unless otherwise specified, all existing surface improvements damaged or removed as a result of microtunneling operations should be restored to their original conditions.

23.2 INSPECTION OF THE WORK

The owner should furnish or hire a full-time inspector for all work. The inspector should be knowledgeable about the microtunneling process, shaft construction, and pipe material. The inspector should monitor and document that all the requirements of the contract and specifications are being met and that all workmanship and construction practices equal or exceed the standards called for in the construction contract documents and/or in the contractor's approved shop drawings.

Commentary. The inspector typically has no authority to relieve the contractor of any obligations under the contract. The owner,

inspector, or engineer should not direct, supervise, or assume control over the means, methods, techniques, sequences, or procedures of the construction except as specifically called for in the contract specifications. The owner's inspector should prepare a record of work activities performed on all shifts. The data should be sufficiently detailed that a project time line associated with construction can be easily constructed. The records should include the following elements:

General inspection and photographic documentation of the equipment, per the submittals:

- MTBM;
- Jacking system, including intermediate jacking stations (IJSs);
- Guidance system;
- Wall seals;
- Pumps;
- Separation plant;
- Generators;
- Mixers; and
- Safety equipment.

Pipe inspections (see Section 18.1).

Personnel and equipment log:

- Date, day of week, sequential construction day;
- Start of work: day, swing, and graveyard shifts;
- Weather and temperature conditions;
- Labor classifications: daily count for each; start and end hours;
- Equipment on site with usage codes for hourly activity (down, idle, active); and
- Inspection of the IJS and documentation of its location.

Microtunnel activity log:

- Date;
- Start and end times for jacking each pipe;
- Cycle time for pipe placement and reconnects;
- Running time log of commentary and remarks on microtunnel activities, and reporting and tracking of other microtunnel activities, such as lubrication, separation plant cuttings, slurry checks, slurry modifications, slurry test results (e.g., viscosity, sand content, density), maintenance, repair, replacement of equipment, changes or unusual events; and
- Summary of tunnel progress;
 - Tunnel length at start of shift;
 - Feet mined during shift;
 - Tunnel length at end of shift; and
 - Tunnel length remaining.

Microtunnel mining and advance log:

- Start and finish times for each pipe;
- Pipe number;
- Time for mining each pipe;
- Downtime and associated reason;
- Movement of pipe with removal of jacking forces;
- Face pressure while not mining;
- Slurry face pressure during mining;
- Horizontal and vertical alignment;
- Pitch, roll, and yaw;
- Jacking forces, including IJSs;
- Slurry feed and charge (psi and gpm);
- Slurry return and discharge (psi and gpm);
- Steering jacks (psi);
- Cutterhead torque/rpm; and
- Cumulative length.

The data above should be collected at least once for each pipe cycle over the course of installation.

The inspector should also collect samples of the cuttings off the spoils piles at regular intervals during the jacking process. The samples should be contemporaneously logged onto a chain-of-custody form in the event that sample possession needs to be relinquished. The samples should be grabbed and stored in sealable bags. A marker should be used to label the bags with the

date, time, project number, and station or location of the MTBM at the time the sample is collected. There is a time lag between the excavation of the ground and the arrival of the grab sample on the surface. Though the stationing does not match, collecting samples at a regular interval allows for the documentation of ground conditions as the jacking pipe is advanced. Additionally, the spoils piles should be routinely checked during advances for noticeable and visually apparent percentages of the various soil fractions being screened.

The collection of data allows the plotting of information—specifically, jacking forces and advance rates. The jacking forces should be compared to the theoretical jacking force plot provided as part of the submittal process. Typically, the other data collected and grab samples are used only to help resolve construction issues or claims.

Access to the control cab to collect jacking data by the owner's inspector can be an issue for the contractor. If the contractor does not want to allow access during active jacking, then a remote monitor or webcam should be provided.

The on-site inspector should visually monitor and inspect the shafts daily for signs of movement and stability. In critical locations, the specifications may require monitoring of the shafts for movement by optical measurements or other devices, such as inclinometers. See Chapter 25 for a discussion on instrumentation and monitoring.

CHAPTER 24 REGULATIONS

24.1 SAFETY REGULATIONS

The contractor should meet federal, state, and local regulations. The contractor is responsible for jobsite safety.

Commentary. Procedures to meet applicable OSHA and other safety requirements should be submitted for information only and are not subject to approval by the engineer. At a minimum, the contractor should provide the following:

- Job-specific safety plan job hazard analyses (JHAs) of shaft construction, including invert slabs, head walls, and thrust blocks;
- Form drawings for head walls and thrust blocks;
- JHAs of microtunneling construction;
- Protection against flooding, earth instability, and all inflows;
- Adherence to all applicable regulations regarding shaft ingress and egress, including provisions for ladders, stairs, walkways, and hoists with personnel cages;
- An accounting system for tracking all people underground, such as a tag in/tag out board;
- Adherence to all applicable hoisting and rigging regulations, to include any required training and/or certification of hoisting equipment operators, riggers, and signal workers;
- Adherence to all applicable regulations regarding gas testing and ventilation that may be required for shafts and tunnels. (Note that some states, such as California, require the owner to obtain an Underground Classification before any work may begin; this classification is commonly obtained during the latter stages of the design phase);
- A written emergency action or evacuation plan, to include a rescue plan;
- Worksite and tunnel lighting;
- Coordination with the public response teams and local fire department for shaft and tunnel entry and evacuation;
- Protection of the jobsite and all shafts (including traffic control devices), and prevention of accidental or unauthorized entry. (The contractor should protect unoccupied shafts from unauthorized entry and access; methods of protection may include fencing and plating over the shaft);
- Supervision of all aspects of safety on the jobsite;
- A written fire response plan, to include a map of the location of all fire extinguishers on the jobsite; and
- A site-specific pollution prevention plan designed to prevent the accidental release of any pollutant into the environment.

Microtunneling projects do become points of interest for visitors who want to see the construction process. Visitors to the construction site must sign in with the contractor, be oriented,

be equipped with personal protective equipment (PPE), and be escorted while on site.

24.2 POLLUTION CONTROL

The contractor should monitor all worksites for pollution and take corrective actions. Pollution control measures may include the following:

- Disposal of sediment and runoff to streams and sewers;
- Fueling areas with containment, and spill kits for remedying release of greases, oils, and/or fuel from equipment;
- Proper storage and containment of products used for construction, such as polymers, paints, glues, grouts, cements, or solvents; and
- Proper ducting including air scrubbers for foul air from operating equipment such as generators, welders, and batch plants.

Commentary. See the EPA website as well as those of local jurisdictions for the latest requirements and regulations pertaining to pollution control measures and documentation.

Pollution control documentation may include the following:

- Proper documentation of the products on site and the measures necessary to have proper control of those products;
- Procedures for containment or cleanup of material used for spills or accidental rupture of tanks;
- Fueling areas, with spill kits;
- Grouting, and slurry mix areas with containment;
- Baker tanks to treat water runoff, along with products to neutralize expected chemicals or allow for quick settling of bentonites or cements;
- Safety sections on safety data sheets (SDSs) or material safety data sheets (MSDSs) containing health precautions for products used in construction;
- Hazard waste transportation plan; and
- Waste determination results, to determine recycle, reuse, and disposal options.

See Section 24.3 for erosion measures.

24.3 EROSION CONTROL

The contractor must implement erosion control measures on all worksites of a project. These erosion control measures are mandated by local laws. Basic erosion control measures include the following:

- Entrance and exit wheel washes; broom-off areas to remove muck, latent concrete, and bentonite from the tires of trucks and vehicles entering or exiting the jacking shaft area or shaft construction areas;

- Dam and inlet protection and other best practices devices around drainage facilities to prevent slurry and waterborne soil from entering storm sewer systems and waterways;
- Sediment fences around work areas to prevent runoff from leaving the site, and settling tanks or other devices to settle out solids so that discharge water from the site complies with EPA or local jurisdiction requirements under the Clean Water Act;
- Truck bed liners and truck covers as necessary to prevent spillage of wet soils, muck, or waste slurry;
- Sweeper trucks and vacuum trucks available to clean up spillage from excavation, truck tracking, and slurry not contained at the project site; and

- Suitable erosion control logs describing measures in place, maintenance dates, and turbidity of runoff water.

Commentary. Microtunneling generates waste soils mixed with bentonite clay; shaft construction generates cement-mixed soils and waste that when mixed with water is easily transported. The results can be a violation of EPA or local department of environmental quality (DEQ) permit requirements. Microtunneling can generate large quantities of partially separated soils and waste slurries that contain bentonite (which does not easily settle or separate from water). Measures in place are required to contain, clean up, and limit the extent of water-transported soils.

CHAPTER 25

INSTRUMENTATION AND MONITORING

A detailed plan for monitoring possible ground surface movements caused by the microtunneling operations should be prepared. The plan is typically prepared by the engineer in consultation with potentially affected facility owners, as described in Section 13.5. The plan should address the equipment, method, and frequency of survey measurements. The engineer should identify special monitoring requirements. A baseline survey needs to be conducted before the start of construction sufficiently early to establish preexisting conditions. In addition to baseline surveys, before construction the contractor should document existing conditions of potentially affected structures, utilities, and facilities. The documentation of existing conditions establishes a baseline against which damage claims can be evaluated.

Before the installation of instrumentation, the contractor should meet with the engineer to discuss frequency and formatting of reports as well as action levels and contingency plans.

Commentary. Instrumentation to measure ground movements includes the following elements:

- Inclinometers;
- Settlement casings;
- Multiple-point borehole extensometers;
- Surface settlement points;
- Surface reference points; and
- Utility monitoring points

Section 13.5 describes the basics of required instrumentation and monitoring plans, including threshold and action levels.

Typically, required actions are described in the contract documents, should these levels be reached. Regardless of whether they are reached, the contractor should meet with the engineer before construction to discuss required actions.

These are some typical actions that might be required if the threshold levels are reached during construction:

- Schedule a meeting with the contractor, engineer, and owner.
- Increase the frequency of monitoring of instrumentation.
- Perform third-party confirmation surveys.
- Review contractor means and methods and revise as required. Evaluate such things as torque, lubrication practices, drilling fluid and slurry pressures and flow rates, and face pressures.

These are some typical actions that might be required if the action threshold levels are reached during construction:

- Inform stakeholders if settlements are confirmed.
- Review contractor means and methods and revise as required. Evaluate such things as torque, lubrication practices, drilling fluid, slurry pressures, flow rates, and face pressures.
- Reevaluate critical structures in the area of concern for damage. Install additional instrumentation if warranted.
- Implement contingency plans outlined in the contract.

When tunneling adjacent to the sensitive structures, consider 24-hour operation and the placement of vibration monitoring devices in addition to other instrumentation.

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CHAPTER 26

RESTORATION AND CLEANUP

The contractor should be responsible for site restoration and keeping the construction site clean, neat, and orderly at all times.

Commentary. Acceptance of site restoration and final cleanup should be a condition for payment. Continual good housekeeping and slurry containment can reduce the amount and cost of cleanup required at the end of a project. Removal of grease, slurry, and debris from the carrier pipe must be a part of the overall cleanup. Prevention of off-site drainage is critical. Simple housekeeping tools and equipment, such as pails, brooms, straw bales, shovels, silt fences, sandbags, and vacuum trucks and trailers must be on site or available at short notice.

Haul routes should be policed daily to make sure that truck washes and other measures of site containment are effective.

Cleanup should be continuous to minimize traffic hazards or obstructions along the street or to driveways. Trenching, backfill, pavement repair (as necessary), and cleanup should be coordinated with the local jurisdiction.

Unless otherwise specified, all existing surface improvements damaged or removed as a result of microtunneling operations should be restored to their original conditions. These items may include such elements as landscaping, curbs and gutters, sidewalks, and road surfaces.

CHAPTER 27

MEASUREMENT AND PAYMENT

The owner shall establish a bid schedule to obtain pricing for all work associated with the microtunneling installation. This bid schedule provides the bidders the opportunity to establish their lump sums or unit prices for individual work items. These bid items become part of the contract.

Commentary. Bid items can be as a lump sum or as a unit price, depending on the owner's preferences. The bid schedule should include the following bid items, at a minimum:

- Mobilization, which includes MTBM system preparation, delivery, and setup;
- Jacking and receiving shafts;
- Ground improvement;
- Pipe;
- Microtunneling installation;
- Rescue shafts;
- Manhole and access installations;
- Site restoration; and
- Demobilization.

The lump sum or unit price should include full compensation for bid items. Lump sum amounts are typically further subdivided by a "schedule of values" established by the contractor during the submittal process.

Incidental work items, such as dewatering of shafts, providing access to the shafts or portals, backfill and replacement of shafts, or other improvements, and furnishing and installing the pipe connections, should be included in the bid items above.

The bid item for the rescue shaft should include recovery of the MTBM, standby time, and the resumption of tunneling. In some cases, a back tunnel may be required and should be included in the bid item. This item is paid only if an obstruction is determined to be the cause.

27.1 MOBILIZATION

Mobilization expenses should include transportation costs of equipment, temporary facilities, and setup of equipment at the worksites.

Commentary. A separate bid item for mobilization should be included because the contractor tends to have many up-front costs before actually installing pipe in the ground. These up-front costs are experienced by both the prime contractor and any subcontractor and should be compensated. Mobilization is normally a lump sum item. Most owners allow a percentage of the total contract amount; 2 to 5 percent of the contract amount is typically used. Microtunneling projects are equipment dependent, and this percentage may need to be higher.

Mobilization of additional equipment after start-up as a result of change conditions should be included as part of the new work.

27.2 JACKING AND RECEIVING SHAFTS

Jacking and receiving shaft costs should include preparation of shop drawings and construction of shafts in accordance with accepted drawings and contract documents.

Commentary. Bid items for jacking and receiving shafts should be included because the work can be completed by the prime contractor or a subcontractor. Having bid items allows for easier definition of work. Some owners include a bid item for each shaft as a lump sum.

Intermediate shafts are sometimes included by the owner as optional bid items for the contractor to use. These intermediate shafts should be included in the bid.

Items such as installation of the entry and exit seals, thrust blocks, invert leveling slab, setting of rails, and setting of jacks are part of the microtunneling operations and should be paid for under the microtunneling bid item.

Construction of rescue shafts should be included as separate bid items. See Section 27.6 for a discussion of these shafts.

27.3 GROUND IMPROVEMENT

When ground improvement is identified during design as being needed, whether at the shafts or along the drive, this work should be included as a bid item in the contract.

Commentary. If ground improvement is envisioned or anticipated, then a bid item should be included in the contract. A contractor cannot afford to voluntarily include a bid price for ground improvement in the bid, for fear of being undercut by another bidder who does not include the ground improvement costs and seeks to be paid later via a change order. Ground improvement methods include soil-cement mixing, formational grouting, jet grouting, permeation grouting, and ground freezing, as examples.

It is important that any areas requiring ground improvement be designed by an engineer with experience in microtunneling and ground improvement to identify and specify the extent, properties, and strength of ground improvement zones. It is important to include as part of the bid item a testing program to ensure that the ground improvement is compatible with the project needs and intended purpose. The ground improvement zones should not become an obstacle for the microtunneled drive.

A separate bid item for ground improvement should be included because the work is typically completed by a specialty subcontractor.

27.4 PIPE

Pipe supplied and installed should be paid on the basis of distance from center of manhole to center of manhole.

Commentary. A separate bid item should be considered for paying for pipe material when it is delivered to the site. The pipe material is typically specially made for the contractor's machine and delivered well before installation. Typically, the contractor purchases additional pipe sections at various lengths for connections. Any additional pipe purchased by the contractor and not installed is considered incidental and not paid for by the owner.

27.5 MICROTUNNEL INSTALLATION

Payment for the microtunnel installation should be on the basis of distance from center of manhole to center of manhole for each pipe diameter.

Commentary. A separate bid item should be considered to pay for the microtunnel. The cost of pipe between the shaft and the manhole is typically considered incidental to the bid item.

27.6 RESCUE SHAFTS

Provisions should be established in the contract to cover the cost and time associated with rescue shafts. A rescue shaft may be needed for obstruction removal, differing site conditions, equipment failure, or excessive deviation from the design line and grade. The need and value of a rescue shaft allowance should be defined on the basis of anticipated site and ground conditions.

Commentary. Rescue shafts can be paid for as time and materials under a fixed price allowance for all bidders. The amount of the allowance should reflect the project risk. For most projects, the location and level of effort required to build a rescue shaft are unknown. An allowance item paid for by time and materials apportions risk fairly. Compensation for a rescue shaft should be based on the evaluation of cause. Compensation should include

the contractor's downtime costs and ownership or leasing costs of the microtunneling equipment.

27.7 MANHOLE AND ACCESS INSTALLATIONS

A bid item should be established for the installation of manholes and other access facilities in any of the shafts.

Commentary. A separate bid item should be established for the manhole and access construction. The work is typically completed by a different subcontractor than the contractor or subcontractor that did the microtunneling.

27.8 SITE RESTORATION

Site restoration expenses should include repaving the road surface at the shaft locations and bringing the work spaces back to their original conditions.

Commentary. A separate bid item for site restoration should be included for repaving the road surface where shafts were dug. Worksites must be returned to an acceptable condition, and final payment must be made to the contractor.

27.9 DEMOBILIZATION

Demobilization expenses should include removal of equipment, transportation costs of equipment off site, decommissioning of temporary facilities, and cleaning of worksites.

Commentary. A separate bid item for demobilization should be considered because the worksites must be returned to an acceptable condition, and final payment must be made to the contractor.

27.10 DIFFERING SITE CONDITIONS

If site conditions are different from those described in the contract documents, and if the contractor can show that the different conditions adversely affected the work, the contractor is entitled to some form of compensation. See Section 15.8.

Commentary. None.

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